**PHINMA UNIVERSITY OF ILOILO**

College of Information Technology Education

# CAPSTONE THESIS

## FlexSpace: Your Gateway to Co-Working Spaces in Iloilo City

*Co-Working Space Discovery and Reservation System*

A Comprehensive System Development and Documentation

In Partial Fulfillment of the Requirements for the Degree of

**Bachelor of Science in Information Technology**

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# CHAPTER 1

# INTRODUCTION AND BACKGROUND OF THE STUDY

## 1.1 Background of the Study

### 1.1.1 Global Emergence of Co-Working Spaces as Socio-Technical Ecosystems

The emergence of co-working spaces represents one of the most significant structural transformations in how contemporary societies organize productive labor. Far from being a transient trend or a marginal phenomenon confined to technology hubs, the proliferation of co-working infrastructure signals a fundamental reconfiguration of the relationship between work, space, and community. Globally, co-working hubs have evolved from small, community-driven initiatives—often rooted in the hacker spaces and maker culture of the early 2000s—into sophisticated, multi-layered ecosystems that simultaneously support entrepreneurship, foster innovation, catalyze academic productivity, and generate new forms of social capital.

In cities recognized as global innovation leaders, such as Singapore, Berlin, and San Francisco, co-working has become virtually synonymous with start-up culture, providing flexible, scalable infrastructure for digital nomads, freelancers, small enterprises, and even established corporations seeking to decentralize their operations. These hubs function not merely as physical containers for work activities but as dynamic socio-technical environments where digital technologies, urban planning strategies, and socio-economic policies converge to create sustainable, inclusive, and innovation-driven work environments. The co-working phenomenon thus transcends the simple provision of desks and internet connectivity; it encompasses the cultivation of professional communities, the facilitation of knowledge exchange, the reduction of barriers to market entry for entrepreneurs, and the democratization of access to high-quality work infrastructure.

The global co-working industry has demonstrated remarkable resilience and growth trajectories even in the face of significant disruptions. According to industry analyses conducted by the Global Co-working Unconference Conference (GCUC) and parallel academic studies, the number of co-working spaces worldwide exceeded 20,000 by the year 2023, with projections indicating continued expansion at compound annual growth rates exceeding 15% through 2030. This growth has been propelled by multiple convergent forces: the proliferation of remote and hybrid work

arrangements accelerated by the COVID-19 pandemic, the increasing prevalence of freelance and gig economy participation, the rising costs of traditional commercial real estate in urban centers, and a generational shift in workforce preferences toward flexibility, autonomy, and work-life balance.

The socio-economic significance of co-working spaces extends beyond their immediate users to encompass broader urban development dynamics. In Singapore, co-working hubs are strategically integrated into the government's Smart Nation initiative, with spaces located near transport nodes, innovation districts, and educational institutions to maximize accessibility and foster cross-sectoral collaboration. In Berlin, co-working infrastructure has played a pivotal role in the city's transformation into a European start-up capital, with hubs such as Betahaus and Factory Berlin serving as incubators for ventures that have subsequently achieved international scale. In San Francisco, the co-working ecosystem—despite experiencing consolidation following the WeWork restructuring—continues to serve as the backbone of the Bay Area's entrepreneurial culture, with spaces like Galvanize and Runway Innovation Hub providing integrated programs that combine workspace with mentorship, funding access, and technical training.

The academic study of co-working has evolved in parallel with the industry's growth, generating a substantial body of scholarly literature that examines these spaces through multiple disciplinary lenses. Urban planners analyze co-working hubs as instruments of spatial restructuring and neighborhood revitalization. Sociologists investigate the community dynamics, identity formation, and social network effects that emerge within co-working environments. Information systems researchers evaluate the technological infrastructures that enable space management, reservation optimization, and user experience enhancement. Organization theorists examine how co-working spaces function as alternative institutional forms that challenge traditional corporate structures. This multi-disciplinary scholarly attention underscores the intellectual significance of co-working as a phenomenon worthy of sustained academic inquiry.

The Philippine context presents a distinctive case within the global co-working landscape. As a developing economy with a rapidly growing services sector, a large and youthful population, and widespread English language proficiency, the Philippines has emerged as a significant market for co-working expansion. Metro Manila and Cebu City have led this growth, driven by the country's business process outsourcing (BPO) industry, the rise of freelancing platforms, and government initiatives to promote digital entrepreneurship. However, the co-working ecosystem in the Philippines remains unevenly developed, with substantial variation in digital maturity, service

quality, and accessibility across different cities and regions. This uneven development creates both challenges and opportunities for localized interventions that can accelerate the maturation of co-working infrastructure in emerging digital hubs such as Iloilo City.

### 1.1.2 Smart City Frameworks: Comparative Analysis of Singapore, Berlin, and San Francisco

The concept of the Smart City has emerged as a dominant paradigm for urban development in the twenty-first century, emphasizing the integration of digital technologies, data analytics, and citizen-centric governance to enhance urban efficiency, sustainability, and quality of life. Smart City frameworks provide the policy, technological, and institutional foundations within which co-working ecosystems can thrive. By examining the Smart City initiatives of Singapore, Berlin, and San Francisco, this study establishes a comparative benchmark against which to evaluate Iloilo City's trajectory and to identify the specific contributions that a co-working discovery and reservation system can make to the city's digital transformation agenda.

Singapore's Smart Nation initiative, launched in 2014 and continuously refined since, represents one of the world's most comprehensive and systematically implemented Smart City frameworks. The initiative operates across three pillars: Digital Economy, Digital Government, and Digital Society. Under the Digital Economy pillar, Singapore has invested heavily in digital infrastructure, including nationwide high-speed broadband, 5G networks, and sensor deployments for environmental monitoring and traffic management. The government's co-working strategy aligns directly with these investments: the Jurong Innovation District and one-north development zone integrate co-working spaces with research institutions, corporate innovation centers, and startup accelerators. The National Research Foundation provides funding for co-working operators that meet criteria for innovation support, community building, and accessibility. Singapore's co-working reservation platforms—such as those developed by JustCo and The Great Room—integrate with the city's broader digital services ecosystem, including the Singpass national digital identity system, the LifeSG government services portal, and the GoBusiness licensing platform. This integration ensures that co-working users can authenticate seamlessly, access government services, and comply with regulatory requirements through a unified digital experience.

Berlin's approach to Smart City development emphasizes open data, civic participation, and digital sovereignty. The Berlin Smart City Strategy, adopted in 2021, positions the city as a leader in ethical technology governance, with explicit commitments to data privacy, algorithmic transparency, and citizen control over personal information. Berlin's co-working ecosystem

benefits from the city's robust telecommunications infrastructure, affordable commercial real estate relative to other European capitals, and a vibrant creative and technology sector. The city's co-working spaces—numbering over 200 by 2024—are concentrated in districts such as Mitte, Kreuzberg, and Neukoelln, with emerging clusters in former industrial areas undergoing regeneration. Berlin's Smart City framework supports co-working through the Berlin Partner for Business and Technology, which provides market entry assistance, co-working vouchers for startups, and networking events that connect space operators with potential tenants. The city's open data portal, FIS-Broker, provides geospatial information that co-working platforms can leverage for location-based services, transit integration, and neighborhood analytics. Berlin's emphasis on digital commons and cooperative ownership models has also given rise to community-governed co-working spaces that operate as non-profit associations, offering an alternative to commercial operators.

San Francisco's Smart City initiatives reflect the city's position as the epicenter of global technology innovation while also addressing acute challenges of housing affordability, income inequality, and homelessness. The city's Digital Equity Strategy prioritizes universal internet access, digital skills training, and affordable technology services for underserved communities. San Francisco's co-working ecosystem is characterized by diversity, ranging from premium spaces serving venture-backed startups to community technology centers providing free or low-cost access for job seekers, students, and entrepreneurs from marginalized backgrounds. The San Francisco Municipal Transportation Agency (SFMTA) integrates co-working location data into its transit planning processes, using anonymized reservation patterns to optimize bus routes and frequency. The city's Office of Economic and Workforce Development (OEWD) operates the SF Shared Spaces program, which streamlines permitting for outdoor co-working and commercial activities, demonstrating how regulatory innovation can expand co-working beyond traditional indoor environments. San Francisco's experience illustrates both the potential and the limitations of Smart City frameworks: while digital infrastructure and policy support have enabled co-working ecosystem growth, persistent socio-economic inequalities mean that access remains uneven across demographic groups and neighborhoods.

Comparing these three frameworks reveals several dimensions relevant to Iloilo City's context. First, infrastructure investment is foundational: Singapore's fiber-to-the-home deployment, Berlin's 5G rollout, and San Francisco's municipal broadband initiatives all demonstrate that high-quality connectivity is a prerequisite for co-working ecosystem development. Second, institutional coordination matters: successful Smart City frameworks involve alignment across government

agencies, private sector operators, academic institutions, and civil society organizations. Third, user-centric design principles ensure that digital services—including co-working reservation platforms—are accessible, intuitive, and responsive to diverse user needs. Fourth, data governance frameworks must balance innovation with privacy protection, ensuring that user data collected through reservation systems is handled responsibly and transparently. Fifth, equity and inclusion considerations should be embedded in system design, preventing the emergence of digital divides that exclude vulnerable populations from co-working benefits. These comparative insights inform the design of the FlexSpace system, ensuring that it contributes meaningfully to Iloilo City's Smart City development while addressing the specific socio-economic conditions of the local context.

The Smart City Framework for Iloilo City must therefore be understood not as a direct replication of Singaporean, German, or American models, but as a localized adaptation that incorporates relevant best practices while responding to the city's unique demographic profile, economic structure, institutional capacity, and cultural context. Iloilo City, with a population of approximately 480,000 within the city proper and over 2 million in the metropolitan area, is classified as a highly urbanized city in Western Visayas. The city's economic base encompasses business process outsourcing, tourism, education, healthcare, and retail trade, with growing sectors in information technology, creative industries, and food processing. The Iloilo City Government, under the leadership of successive administrations, has articulated a vision of digital transformation that includes e-governance initiatives, public Wi-Fi deployment, smart traffic management, and support for the city's burgeoning start-up ecosystem. The FlexSpace system is designed to advance this vision by addressing a specific gap in the city's digital infrastructure: the absence of a centralized platform for discovering and reserving co-working spaces.

### 1.1.3 Iloilo City as an Emerging Digital Hub in Western Visayas

Iloilo City occupies a strategically significant position within the Philippine archipelago and the broader Western Visayas region. Located on the southeastern coast of Panay Island, the city serves as the regional center for commerce, education, governance, and culture, with a metropolitan economy that generates approximately PHP 120 billion in annual gross domestic product. The city's historical importance as a Spanish colonial center—evidenced by its well-preserved heritage district—combines with contemporary development dynamics to create a distinctive urban environment that blends tradition with modernization. This historical depth provides Iloilo City with cultural assets and urban character that differentiate it from newer, more uniformly planned

cities, but also presents challenges in terms of infrastructure modernization and spatial reconfiguration.

The city's designation as a highly urbanized city (HUC) in 1979 marked the beginning of accelerated urban development, with subsequent decades witnessing substantial investments in transportation infrastructure, educational institutions, healthcare facilities, and commercial real estate. The establishment of the Iloilo Business Park in Mandurriao district—developed by Megaworld Corporation—represents the most significant contemporary urban development project, transforming former airport land into a mixed-use central business district that includes office towers, retail complexes, residential condominiums, and hospitality establishments. The business park has attracted major BPO operators, including Accenture, Teleperformance, and TaskUs, creating employment opportunities that have drawn workers from across the region and stimulated demand for complementary services, including co-working spaces.

Iloilo City's educational landscape further reinforces its position as an emerging digital hub. The city hosts numerous higher education institutions, including Central Philippine University, University of San Agustin, West Visayas State University, John B. Lacson Foundation Maritime University, and PHINMA University of Iloilo, collectively enrolling over 80,000 students annually. These institutions produce graduates in information technology, business administration, engineering, and other fields relevant to the knowledge economy, creating a talent pipeline that supports the growth of technology-enabled industries. The concentration of academic institutions in districts such as Jaro and Molo generates substantial demand for study spaces, collaborative work environments, and internet-connected facilities, particularly during examination periods when campus libraries and study halls reach capacity.

The city's information and communications technology (ICT) infrastructure has improved significantly in recent years, though challenges remain. Major telecommunications providers—Globe Telecom, Smart Communications, and DITO Telecommunity—have expanded 4G LTE coverage across the city, with 5G deployment initiated in central business districts. The Iloilo City Government, in partnership with the Department of Information and Communications Technology (DICT), has deployed free public Wi-Fi hotspots in select public spaces, including plazas, government offices, and transport terminals. However, internet connectivity remains uneven across districts and barangays, with peripheral areas experiencing slower speeds and less reliable service. The cost of broadband subscriptions relative to average household incomes also creates

accessibility barriers for lower-income residents, a consideration that informs the design of FlexSpace's user interface and service delivery model.

Iloilo City's Smart City aspirations are articulated in the Iloilo City Comprehensive Development Plan and various sectoral strategies adopted by the city government. These documents identify digital transformation as a cross-cutting priority, with specific objectives including: (a) enhancement of e-governance capabilities and online service delivery; (b) improvement of digital infrastructure and connectivity; (c) promotion of the city's ICT and creative industries; (d) development of human capital for the digital economy; and (e) leveraging technology for environmental sustainability and disaster resilience. The Iloilo City ICT Council, established as a public-private partnership mechanism, coordinates implementation of these objectives, bringing together government agencies, private sector representatives, academic institutions, and civil society organizations. The FlexSpace system aligns with these Smart City priorities by digitizing a component of the city's service economy, generating data that can inform urban planning decisions, and demonstrating the viability of locally developed technology solutions.

The socio-economic context of Iloilo City presents both opportunities and constraints for co-working ecosystem development. The city's relatively lower cost of living compared to Metro Manila and Cebu City makes it attractive to freelancers, remote workers, and start-up entrepreneurs seeking to optimize operational costs. The presence of a young, educated, and English-proficient workforce provides a talent base for knowledge-based industries. The city's growing reputation as a tourism destination—particularly for heritage and gastronomic tourism—creates opportunities for co-working spaces that cater to digital nomads and location-independent professionals. However, the city's co-working market remains underdeveloped relative to its potential, with existing spaces operating at suboptimal efficiency due to fragmented discovery channels, manual reservation processes, and limited visibility to potential users. Addressing these inefficiencies through the FlexSpace system represents a targeted intervention that can unlock significant value for multiple stakeholder groups while contributing to the city's broader digital transformation agenda.

### 1.1.4 Growth of Co-Working Spaces Across Iloilo City Districts: A Spatial Analysis

The spatial distribution of co-working spaces in Iloilo City reflects the city's socio-economic structure, land use patterns, and transportation infrastructure. An analysis of existing and emerging co-working hubs reveals distinct clustering patterns across the city's seven districts—City Proper, Molo, Arevalo, Mandurriao, Lapuz, La Paz, and Jaro—each characterized by unique demographic

profiles, economic activities, and accessibility conditions. Understanding these spatial variations is essential for designing a discovery and reservation system that effectively serves users across the city's diverse urban landscape.

Mandurriao district has emerged as the primary concentration of co-working activity in Iloilo City, driven by the development of the Iloilo Business Park and the co-location of BPO companies, hotels, retail establishments, and residential developments. Co-working spaces in Mandurriao—including WorkSpace Iloilo, Gridpad, and shared office facilities within business process outsourcing complexes—cater primarily to professionals employed in the district's office towers, as well as to freelancers and entrepreneurs seeking proximity to the city's central business functions. The district's advantages include superior transportation connectivity via major arterial roads (Sen. Benigno Aquino Jr. Avenue, Mandurriao Road), access to commercial services and amenities, and the prestige associated with the business park address. However, the higher rental costs in Mandurriao translate to higher membership fees, creating affordability barriers for students and early-stage entrepreneurs. The district's co-working spaces also experience peak demand during standard business hours, with underutilization during evenings and weekends, suggesting opportunities for demand management through dynamic pricing and reservation optimization.

Jaro district, located north of the city center, presents a contrasting co-working ecology shaped by its concentration of educational institutions and residential neighborhoods. Home to Central Philippine University, West Visayas State University Medical Center, and numerous secondary schools, Jaro generates substantial demand for study-oriented co-working and shared learning spaces. The district's co-working offerings include academic-focused spaces such as StuPad and The Study Box, which provide quiet environments, extended operating hours, and student-friendly pricing. Jaro's co-working ecosystem also encompasses informal study spaces within cafes, bookstores, and student housing complexes, many of which are not formally marketed as co-working facilities but functionally serve similar purposes. The district's transportation infrastructure—including jeepney routes connecting to other districts and the nearby Iloilo International Airport—provides reasonable accessibility, though traffic congestion during peak hours affects travel times. For the FlexSpace system, Jaro represents a high-priority district due to the concentration of student users, the prevalence of price-sensitive demand, and the potential for partnerships with educational institutions.

Molo district, situated south of City Proper, combines residential, commercial, and institutional functions, with a growing presence of small and medium enterprises in retail, services, and food

processing. The district's co-working infrastructure is less developed than Mandurriao or Jaro, but emerging demand from local entrepreneurs, home-based professionals, and students from the nearby John B. Lacson Foundation Maritime University and St. Therese College of Iloilo suggests growth potential. Molo's co-working spaces tend to be smaller, more affordable, and community-oriented, often operated by local business owners as ancillary services. The district's accessibility via Molo Boulevard and General Luna Street provides connections to City Proper and other districts, though public transportation options are less frequent than on major corridors. FlexSpace's coverage of Molo district can contribute to spatial equity by making co-working options visible to users who might otherwise be unaware of facilities outside the central business district.

City Proper, encompassing the historic central business district and government center, hosts a limited number of formal co-working spaces but extensive informal work environments within cafes, restaurants, and hotel lobbies. The district's role as the administrative and commercial heart of Iloilo City generates demand for meeting spaces, temporary workstations, and professional venues for client presentations. However, the limited availability of dedicated co-working facilities, high commercial rents, and parking constraints restrict the district's co-working capacity. The heritage character of City Proper—designated as a heritage zone by the National Historical Commission—also imposes restrictions on building modifications that could accommodate co-working infrastructure. FlexSpace's integration of City Proper venues can capture demand from government workers, legal and financial professionals, and tourists seeking temporary workspaces, while also promoting the adaptive reuse of heritage buildings for contemporary economic functions.

Arevalo, Lapuz, and La Paz districts represent peripheral areas with nascent co-working demand. Arevalo's coastal location and tourism attractions create seasonal demand from digital nomads and location-independent workers, while Lapuz's industrial and transport functions generate demand from logistics and trade professionals. La Paz, known for its commercial markets and the La Paz Batchoy heritage food district, has limited formal co-working infrastructure but potential for development tied to the district's commercial vitality. The FlexSpace system's district-based filtering and mapping capabilities ensure that users can discover options across all city districts, not merely in the central business and academic zones, thereby promoting spatial equity and supporting the development of co-working ecosystems in underserved areas.

### 1.1.5 The Freelance Economy in the Philippines: Structure, Dynamics, and Implications for Co-Working Demand

The Philippines has emerged as one of the world's largest markets for freelance labor, with an estimated 1.5 million active freelancers contributing approximately USD 15 billion annually to the national economy. This freelance workforce spans diverse sectors including information technology, creative services, digital marketing, virtual assistance, content writing, graphic design, software development, and customer support. The country's competitive position in the global freelance market rests on several foundational advantages: widespread English language proficiency, cultural compatibility with Western business practices, relatively lower labor costs compared to developed economies, and a large, young, and technically capable workforce. These factors have enabled Filipino freelancers to secure contracts with clients in the United States, Europe, Australia, and other markets, creating a geographically dispersed but economically significant sector.

The rise of online freelancing platforms—Upwork, Fiverr, OnlineJobs.ph, Freelancer.com, and Toptal among them—has been instrumental in connecting Filipino professionals with international clients. These platforms function as digital labor markets, enabling workers to create profiles, showcase portfolios, bid on projects, receive payments, and build reputations through client feedback systems. For many Filipino freelancers, platform-based work represents a deliberate career choice that offers advantages over traditional employment: autonomy in selecting projects and clients, flexibility in work scheduling, elimination of commuting requirements, and earning potential that often exceeds local salary benchmarks. However, freelance work also entails risks and challenges: income volatility, absence of employment benefits such as health insurance and retirement contributions, isolation from professional communities, and the need for continuous self-directed skill development.

The infrastructure requirements of Filipino freelancers create direct demand for co-working spaces. Reliable, high-speed internet connectivity is the most critical requirement, as freelance work typically involves video conferencing, file sharing, cloud-based collaboration tools, and real-time communication with clients across multiple time zones. While home-based internet subscriptions have improved in recent years, many freelancers experience connectivity problems including service interruptions, bandwidth limitations, and latency issues that affect client interactions. Power supply reliability is another concern, as the Philippines experiences periodic electricity outages that can disrupt work schedules and damage professional relationships. Co-working spaces

address these infrastructure gaps by providing enterprise-grade internet connections, backup power systems, and technical support that individual freelancers cannot affordably replicate in home environments.

Beyond infrastructure, co-working spaces fulfill important social and professional needs for freelancers. Working from home, while convenient, can lead to social isolation, blurred boundaries between professional and personal life, and reduced opportunities for skill development through peer interaction. Co-working environments provide structured work settings that enhance productivity through the psychological effects of social presence and accountability. They also create opportunities for networking, collaboration, and knowledge sharing among professionals with complementary skills. For freelancers in Iloilo City, these social and professional benefits are particularly valuable given the relatively smaller size of the local freelance community compared to Metro Manila or Cebu, which makes spontaneous professional encounters less frequent.

The demographic profile of freelancers in Iloilo City encompasses diverse age groups, educational backgrounds, and experience levels. Young graduates entering the freelance market seek affordable, accessible workspaces that enable them to establish professional routines while building client portfolios. Mid-career professionals transitioning from corporate employment to freelance status require environments that signal credibility to clients and provide facilities for meetings and presentations. Experienced freelancers operating established practices need spaces that support team collaboration, project management, and business development activities. The FlexSpace system addresses this demographic diversity through its filtering capabilities, which allow users to identify spaces based on price points, amenities, atmospheres, and professional orientations that match their specific requirements.

The temporal patterns of freelance work in Iloilo City also inform system design. Many freelancers serving international clients work non-standard hours to accommodate time zone differences, creating demand for co-working spaces with extended or 24-hour operating schedules. The academic calendar affects demand patterns, as students engaged in freelance work increase their workspace needs during semester breaks and summer periods. Seasonal variations in client demand—often corresponding to fiscal year cycles in client countries—generate fluctuations in workspace utilization that co-working operators must manage. FlexSpace's real-time availability tracking and reservation management features provide the data infrastructure necessary to understand and respond to these temporal dynamics, enabling both users and space operators to optimize their planning and resource allocation.

### 1.1.6 Student Demand for Study Infrastructure: Academic Pressure, Space Scarcity, and the Co-Working Alternative

Students constitute a substantial and growing user segment for co-working and study-oriented spaces in Iloilo City. The city's higher education institutions—collectively enrolling over 80,000 students across undergraduate, graduate, and professional programs—generate demand for conducive study environments that exceeds the capacity of institutional facilities. This demand-supply imbalance is particularly acute during examination periods, when campus libraries, study halls, and computer laboratories reach capacity, forcing students to seek alternative spaces or compromise their study effectiveness in suboptimal environments.

The academic calendar of Philippine universities creates predictable cycles of space demand. The midterm examination period, typically occurring in the eighth to tenth week of each semester, generates the first major surge in study space requirements. Students engage in intensive review activities, group study sessions, and completion of pending assignments, all of which require access to quiet, well-lit, internet-connected environments. The final examination period, occurring in the sixteenth to eighteenth week, produces the peak demand cycle, as students undertake comprehensive review of entire semester curricula and prepare for high-stakes assessments. The weeks preceding major standardized examinations—such as the Philippine Licensure Examinations for nursing, engineering, education, and other professions—generate additional demand from reviewees who require sustained access to study facilities over extended periods.

The characteristics of effective study spaces differ from those of general co-working environments in important respects. While both require reliable internet connectivity and comfortable physical conditions, study spaces prioritize acoustic privacy, minimize distractions, and provide extended operating hours that accommodate diverse study schedules. Lighting quality affects reading comfort and cognitive performance, with natural light and adjustable artificial lighting preferred over harsh fluorescent illumination. Temperature control ensures that students can maintain focus without discomfort from excessive heat or cold. Access to power outlets for charging laptops and mobile devices is essential, as digital study materials and online research tools have largely replaced physical textbooks. Refreshment facilities—whether on-site cafes or proximity to food establishments—support sustained study sessions without requiring users to leave the premises.

The institutional study facilities of Iloilo City's universities face structural constraints that limit their ability to meet student demand. University libraries, while comprehensive in their collections, typically operate within fixed hours that may not accommodate students with evening or early

morning study preferences. The design of library spaces often emphasizes individual silent study, which may not support the collaborative learning approaches preferred by many students for certain subjects. Computer laboratories, while providing equipment access, are frequently reserved for scheduled classes, leaving limited open hours for general student use. The physical capacity of these facilities—determined by seat counts, floor areas, and fire safety regulations—creates hard constraints that cannot be easily expanded within existing buildings.

Co-working spaces offer a complementary alternative to institutional study facilities, addressing capacity limitations while providing superior amenities and greater flexibility. Study-oriented co-working spaces in Jaro district—such as StuPad, The Study Box, and similar establishments—have developed specialized service models that cater to student needs, including: affordable hourly and daily rates; extended operating hours, often until midnight or 24 hours; provision of study materials such as whiteboards, printers, and reference books; quiet zones for individual study alongside collaborative areas for group work; and academic support services such as tutoring, review sessions, and peer mentoring programs. These spaces function as para-academic institutions that extend the educational infrastructure beyond university campuses, contributing to student success and academic performance.

The economic accessibility of co-working spaces for students is a critical consideration. Student budgets are typically constrained by limited income sources—allowances from families, part-time employment, scholarships, and loans—and high sensitivity to price levels. Co-working operators that target student markets must balance service quality with affordability, often through differentiated pricing strategies: lower rates during off-peak hours, student membership packages with monthly fees, group discounts for study teams, and partnerships with universities that provide subsidized access. The FlexSpace system supports price-based filtering and comparison, enabling students to identify spaces that match their budget constraints while meeting their study environment requirements. Integration with simulated payment systems also allows students to plan and manage their workspace expenditures within their overall financial planning.

The COVID-19 pandemic has accelerated shifts in student study behaviors that increase the relevance of co-working alternatives. Extended periods of remote learning accustomed students to studying outside institutional environments, developing preferences for spaces that combine independence with structured settings. The proliferation of online learning platforms and digital study tools has reduced dependence on physical library collections, while increasing the importance of reliable internet connectivity and power access. Health and safety concerns have

heightened awareness of ventilation, spacing, and sanitation conditions in shared environments. These evolved preferences create opportunities for co-working spaces that can position themselves as safe, healthy, and technology-enabled alternatives to crowded campus facilities.

### 1.1.7 Digital Transformation of Local Small and Medium Enterprises

Small and medium enterprises (SMEs) form the backbone of Iloilo City's economy, comprising over 95% of registered businesses and employing the majority of the city's workforce. These enterprises span diverse sectors including retail trade, food service, professional services, construction, manufacturing, and agriculture-based processing. The digital transformation of these SMEs—the adoption of digital technologies to improve operational efficiency, expand market reach, and enhance customer engagement—represents both a significant economic opportunity and a substantial challenge for the city's business ecosystem. Co-working spaces play a multifaceted role in supporting SME digitalization, providing not merely physical infrastructure but also access to digital tools, technical expertise, and collaborative networks that accelerate technology adoption.

The digital transformation journey of Iloilo City's SMEs encompasses multiple dimensions. E-commerce adoption enables businesses to extend their market reach beyond physical storefronts, selling products through online marketplaces such as Lazada, Shopee, and Facebook Marketplace, as well as through proprietary websites and social media channels. Cloud-based business applications—including accounting software, customer relationship management systems, inventory management tools, and productivity suites—replace manual processes with automated, data-driven operations. Digital marketing through social media advertising, search engine optimization, and content creation allows SMEs to target customers with greater precision and measure campaign effectiveness. Online payment systems, including GCash, Maya, and bank transfer platforms, facilitate cashless transactions that improve convenience for customers and financial transparency for businesses.

Despite the evident benefits of digital transformation, Iloilo City's SMEs face significant barriers to adoption. Financial constraints limit investment in digital infrastructure, software subscriptions, and technical training. Skills gaps mean that business owners and employees lack the competencies required to effectively implement and utilize digital tools. Uncertainty about return on investment creates hesitation in committing limited resources to technology initiatives. Cybersecurity concerns generate resistance to storing business and customer data in digital systems. Infrastructure limitations, including unreliable internet connectivity and inconsistent power supply, undermine

the reliability of digital operations. These barriers are particularly pronounced among micro-enterprises and businesses in peripheral districts, creating a digital divide within the SME sector.

Co-working spaces address these barriers through multiple mechanisms. By providing enterprise-grade internet connectivity and reliable power with backup systems, co-working spaces eliminate infrastructure constraints that impede digital adoption in home and traditional office settings. Access to shared equipment—high-quality printers, scanners, projectors, video conferencing facilities—reduces the capital investment required for individual businesses to acquire these assets. The co-working environment exposes SME owners to peer learning opportunities, where they can observe digital practices implemented by other businesses, receive informal technical assistance, and develop collaborative relationships with digitally proficient professionals. Some co-working operators in Iloilo City have developed formal digital support programs, including training workshops, technology consulting services, and partnerships with digital service providers that offer member discounts.

The networking function of co-working spaces creates particular value for SME digitalization. Business owners operating from co-working environments develop relationships with web developers, graphic designers, digital marketers, e-commerce specialists, and other professionals whose services are essential for digital transformation. These relationships often evolve from casual encounters into formal service engagements, with co-working spaces serving as marketplaces that connect SMEs with digital service providers. The trust generated by physical co-presence and repeated interaction reduces transaction costs and information asymmetries that otherwise complicate outsourcing relationships. Co-working operators can actively facilitate these connections through member directories, networking events, and service matching programs that explicitly link businesses with complementary needs and capabilities.

The administrative burden of managing physical premises distracts SME owners from core business activities, including digital transformation initiatives. By providing ready-to-use workspaces with inclusive utility, maintenance, and security services, co-working spaces allow businesses to redirect resources—both financial and attentional—toward strategic priorities. The flexibility of co-working arrangements, with options ranging from daily passes to monthly memberships and private office suites, enables businesses to scale their space requirements in response to growth or contraction, avoiding the long-term commitments and capital expenditures associated with traditional commercial leases. This flexibility is particularly valuable for SMEs in volatile markets or experimental phases, where future space requirements are uncertain.

The FlexSpace system enhances these SME support functions by improving the discoverability and accessibility of co-working spaces. SMEs that are not already connected to co-working networks may be unaware of available options, their locations, their amenities, and their pricing. Information fragmentation thus creates a market failure where beneficial transactions between space operators and potential SME users fail to occur. By centralizing co-working space information and providing search, filter, and comparison tools, FlexSpace reduces search costs and information asymmetries, facilitating matches between SMEs and spaces that meet their specific requirements. The system's administrative dashboard also provides co-working operators with data analytics capabilities that enable them to understand their SME user base, tailor their services to business needs, and demonstrate value propositions that attract SME memberships.

### 1.1.8 The Critical Role of Reservation Systems in Co-Working Ecosystem Efficiency

The operational efficiency of co-working ecosystems depends critically on the effectiveness of reservation systems that mediate between space supply and user demand. Reservation systems perform essential functions: they provide users with information about space availability, amenities, pricing, and location; they enable users to secure access to specific spaces at specific times; they prevent conflicts arising from multiple users seeking simultaneous access to limited resources; they generate data that space operators can use to optimize pricing, capacity planning, and service development; and they facilitate payment processing, reducing friction in the transaction process. When reservation systems function effectively, they enhance user satisfaction, maximize space utilization, and support the financial sustainability of co-working operations. When they fail, they generate user frustration, revenue losses, and reputational damage that can threaten business viability.

Globally, co-working reservation systems have evolved through several generations of technological capability. First-generation systems relied on manual processes—phone calls, walk-in inquiries, email messages, and paper-based booking ledgers—that were labor-intensive for both users and operators, prone to errors and conflicts, and unable to provide real-time availability information. Second-generation systems introduced basic digital capabilities, such as online contact forms, spreadsheet-based booking trackers, and simple website displays of space information, which reduced some manual burdens but remained fragmented and lacked integration. Third-generation systems leverage contemporary web and mobile technologies to provide unified platforms that combine real-time availability tracking, online booking and payment, user account management, automated notifications, and administrative dashboards. Fourth-generation systems,

emerging in leading markets, incorporate advanced technologies such as Internet of Things (IoT) sensors for automated occupancy detection, machine learning algorithms for demand prediction and dynamic pricing, geographic information system (GIS) integration for location-based services, and artificial intelligence chatbots for user support.

The Philippine co-working market, including Iloilo City, predominantly operates with first and second-generation reservation systems. Industry surveys indicate that fewer than 15% of small to medium co-working hubs in the Philippines have implemented automated booking systems, with the majority relying on Facebook Messenger inquiries, phone calls, or physical walk-ins. This technological lag has significant consequences. Users face uncertainty about space availability, often traveling to multiple locations before finding suitable workspace, resulting in wasted time, transportation costs, and lost productivity. Operators struggle to manage occupancy effectively, lacking accurate data on utilization patterns that would inform pricing decisions, capacity adjustments, and marketing strategies. The absence of online booking capabilities limits market reach to users who are already aware of specific spaces and willing to engage through manual inquiry channels, excluding potential users who discover spaces through online search or expect digital booking convenience.

The consequences of manual reservation systems are particularly severe during periods of high demand. During university examination seasons in Iloilo City, study-oriented co-working spaces experience surges in walk-in demand that exceed their capacity, resulting in disappointed users, overcrowding in spaces that do accommodate arrivals, and conflicts among users competing for limited seats. The inability to implement demand management strategies—such as advance booking requirements, peak-hour pricing, or waitlist systems—deprives operators of tools that could improve both user experience and revenue outcomes. Similarly, during special events, workshops, or training sessions hosted at co-working spaces, manual systems lack the capability to manage block bookings, equipment reservations, and catering arrangements that require coordination of multiple resources.

Data generated by digital reservation systems represents a valuable asset for co-working operators and urban planners. Booking patterns reveal peak and off-peak demand periods, enabling operators to implement time-based pricing that maximizes revenue while smoothing demand. User demographics and preferences inform service development decisions, such as the addition of amenities, the reconfiguration of space layouts, or the introduction of specialized membership categories. Geographic patterns of user origins and destinations inform marketing strategies and

partnership development. Aggregated data from multiple co-working spaces can inform urban planning decisions, such as the identification of underserved districts, the assessment of transportation infrastructure needs, and the evaluation of economic development initiatives. The manual systems predominant in Iloilo City fail to capture this data, representing a significant lost opportunity for evidence-based decision making.

The development of FlexSpace addresses these deficiencies by providing a third-generation reservation system tailored to the specific context of Iloilo City. The system combines the core functionalities expected of contemporary reservation platforms—user registration, real-time availability display, online booking, automated notifications, and administrative dashboards—with additional features designed for the local market: GIS-based mapping that helps users identify spaces across the city's diverse districts; amenity and price filtering that enables users to find spaces matching their specific requirements; loyalty point and voucher systems that support user retention and promotional strategies; and an AI-powered chatbot that provides multilingual support in English, Filipino, and Hiligaynon, reflecting the linguistic diversity of the local user base. By introducing these capabilities, FlexSpace aims to elevate the operational efficiency of Iloilo City's co-working ecosystem to levels comparable with more developed markets, while generating data that supports continuous improvement and strategic planning.

### 1.1.9 Technological Architecture and Policy Alignment: The MERN Stack and Smart City Integration

The technological foundation of the FlexSpace system is the MERN stack—an integrated collection of JavaScript-based technologies comprising MongoDB (database), Express.js (web application framework), ReactJS (front-end library), and Node.js (runtime environment). The selection of the MERN stack reflects careful consideration of the system's functional requirements, the development team's technical competencies, the ecosystem of supporting tools and libraries, and the long-term maintainability and scalability of the resulting application. This section provides a detailed analysis of each component's role in the system architecture and the rationale for its selection.

MongoDB serves as the system's primary data store, providing a document-oriented database that accommodates the flexible and evolving data structures characteristic of co-working space management. Unlike relational databases that require predefined schemas with fixed tables, columns, and relationships, MongoDB stores data as JSON-like documents (BSON format) that can vary in structure across records within the same collection. This flexibility is advantageous for

FlexSpace because co-working space data—encompassing diverse space types, amenity configurations, pricing models, and operational practices—does not conform to rigid tabular structures. MongoDB's horizontal scalability through sharding ensures that the database can accommodate growth in user numbers, space listings, and transaction volumes without performance degradation. The native geospatial query capabilities of MongoDB support location-based search and mapping integration, enabling efficient retrieval of spaces within specified geographic radii or administrative boundaries.

Express.js functions as the web application framework that structures the system's server-side logic. Built on top of Node.js, Express.js provides a minimalist but robust foundation for building RESTful APIs that handle HTTP requests from the front-end application. The framework's middleware architecture enables the systematic processing of requests through authentication verification, input validation, business logic execution, and response formatting. Express.js's routing capabilities support the definition of clean, resource-oriented API endpoints that correspond to system entities—users, spaces, bookings, districts—following REST conventions that enhance API usability and maintainability. The extensive ecosystem of middleware packages—body-parser for request parsing, cors for cross-origin resource sharing, helmet for security headers, morgan for request logging—accelerates development by providing tested solutions for common web application requirements.

ReactJS powers the system's front-end user interface, providing a component-based architecture that supports the development of responsive, interactive, and maintainable web applications. React's virtual DOM (Document Object Model) optimization ensures efficient rendering performance, which is critical for features such as real-time availability updates, map interactions, and dynamic search filtering. The component-based design enables code reuse across different sections of the application—user-facing pages, administrative dashboards, and space owner interfaces—reducing development effort and ensuring visual consistency. React's unidirectional data flow and state management patterns, enhanced by hooks and context APIs, simplify the management of complex application states involving user authentication, booking workflows, and notification systems. The integration of Tailwind CSS for utility-first styling and Shadcn UI for accessible component primitives further accelerates front-end development while ensuring professional design quality.

Node.js provides the runtime environment that executes JavaScript code on the server, enabling full-stack JavaScript development that leverages a single programming language across both client

and server components. Node.js's event-driven, non-blocking I/O model delivers high concurrency performance, allowing the server to handle multiple simultaneous requests—such as concurrent booking attempts, real-time availability queries, and notification dispatches—without the resource overhead of thread-based server architectures. This performance characteristic is essential for a reservation system where peak demand periods generate bursts of concurrent activity. Node.js's npm (Node Package Manager) ecosystem provides access to over two million packages that extend core functionality, including authentication libraries (jsonwebtoken, passport), database drivers (mongoose for MongoDB), communication tools (nodemailer for email, socket.io for real-time messaging), and security utilities (bcrypt for password hashing, express-rate-limit for request throttling).

The integration of Geographic Information System (GIS) mapping through the Leaflet library and OpenStreetMap tile layer extends the system's spatial intelligence capabilities. Users can visualize co-working space locations on interactive maps, apply geographic filters to discover spaces in specific districts or within defined travel radii, and obtain navigation directions to selected destinations. For space operators, GIS integration provides tools for analyzing the geographic distribution of their user base, identifying catchment areas, and assessing competitive positioning relative to other spaces. The mapping functionality aligns with Smart City principles of transparent, accessible spatial information that empowers citizens to make informed decisions about urban resource utilization.

From a policy perspective, the FlexSpace system supports Iloilo City's Smart City agenda by digitizing a component of the city's service economy that has hitherto operated through informal, manual processes. The system contributes to several Smart City objectives: digital transformation of local businesses by providing co-working operators with technology tools that improve their operational efficiency and market reach; data-driven governance by generating anonymized, aggregated data on co-working utilization patterns that can inform urban planning and economic development decisions; inclusive digital access by providing a platform that connects users across different districts, income levels, and professional backgrounds with workspace options matched to their needs; and innovation ecosystem development by reducing friction in the discovery and reservation of co-working spaces, thereby supporting the city's start-up community and creative industries. The system's compliance with the Data Privacy Act of 2012 (Republic Act No. 10173) ensures that user data is handled responsibly, with appropriate consent mechanisms, security safeguards, and limitation of collection to purposes directly related to system functionality.

In summary, the background of this study situates co-working spaces within global, national, and local contexts, highlighting both the opportunities and challenges facing Iloilo City. The analysis of Smart City frameworks in Singapore, Berlin, and San Francisco provides comparative benchmarks that inform system design priorities. The examination of Iloilo City's district-level spatial dynamics, freelance economy structure, student demand patterns, and SME digital transformation needs establishes the empirical foundation for system requirements. The assessment of reservation system evolution and the technological capabilities of the MERN stack validates the selected development approach. Together, these analytical elements establish the rationale for developing FlexSpace as a Co-Working Space Discovery and Reservation System that addresses persistent inefficiencies in Iloilo City's co-working ecosystem, supports the city's Smart City transformation, and contributes to both academic discourse and practical innovation in localized resource management.

## 1.2 Statement of the Problem

The central difficulty addressed by this study lies in the persistent reservation uncertainty and information fragmentation experienced by remote workers, freelancers, students, and entrepreneurs who rely on co-working spaces in Iloilo City. While the number of co-working and study-oriented spaces has grown significantly in recent years, their operational models remain predominantly dependent on manual, non-centralized inquiry methods. These methods—social media messaging via Facebook Messenger, direct phone calls, text messages, email inquiries, and physical walk-ins—although familiar to local users, are inherently inefficient, error-prone, and structurally incapable of meeting the demands of a growing, digitally sophisticated user base. The following subsections elaborate on the specific problems, their root causes, their cascading impacts on various stakeholders, and their implications for Iloilo City's broader development trajectory.

### 1.2.1 Reservation Uncertainty: Detailed Operational Case Studies

Reservation uncertainty manifests when users attempt to secure workspace without reliable, real-time information about current occupancy status. In Iloilo City's co-working ecosystem, this uncertainty is not merely an occasional inconvenience but a pervasive structural condition that fundamentally undermines the utility of co-working spaces as reliable work infrastructure. The following case studies illustrate the operational dimensions of reservation uncertainty across different user segments and district contexts.

Case Study 1: Freelancer in Mandurriao During Peak Hours. Consider the experience of a freelance web developer residing in Mandurriao district who serves international clients across multiple time zones. On a typical Tuesday morning, the developer needs to conduct a video conference call with a client in London at 10:00 AM local time (2:00 PM GMT). Anticipating the need for a quiet, professional environment with reliable internet, the developer attempts to secure a desk at a co-working space near the Iloilo Business Park. Without a centralized reservation system, the developer's options are limited to: (a) sending a Facebook Messenger inquiry to the co-working space's page, which may not receive a response for several hours or at all; (b) calling the space's phone number, which may be unattended or routed to voicemail; or (c) physically visiting the space to check availability, with no guarantee of securing a seat upon arrival. In this scenario, the developer faces a decision under uncertainty: invest time and transportation cost in visiting the space with uncertain payoff, or compromise work quality by conducting the call from home where background noise and unreliable connectivity may undermine professional credibility. The aggregate cost of such uncertainty—measured in wasted time, transportation expenses, lost billable hours, and damaged client relationships—accumulates over repeated instances, eroding the economic viability of freelance work.

Case Study 2: Student Study Group During Final Examination Period. A group of five nursing students from Central Philippine University, located in Jaro district, requires a collaborative study space during the final examination period. The group's study methodology involves intensive peer discussion, practice question review, and mutual explanation of complex concepts, all of which generate significant noise that makes library study inappropriate. The students identify several study-oriented co-working spaces in Jaro through informal word-of-mouth, but face multiple obstacles: spaces may already be fully booked by other study groups who arrived earlier; group study rooms, which accommodate their collaborative approach, are limited in number and high in demand; pricing information is unclear, creating uncertainty about affordability; and operating hours vary across spaces, with some closing before the students' preferred late-night study schedule. The students ultimately spend over two hours visiting three different spaces before finding availability, time that could have been devoted to productive study. The stress generated by this search process, compounded by examination pressure, negatively affects the students' psychological state and study effectiveness.

Case Study 3: SME Owner Conducting Client Presentation. A marketing consultant operating a small agency in Molo district needs to conduct a presentation for a prospective client from a larger firm based in Metro Manila. The consultant's home office lacks the professional setting and

technical facilities—projector, conference phone, whiteboard—required for an effective presentation. The consultant attempts to reserve a meeting room at a co-working space in City Proper, but encounters difficulties: the space's Facebook page shows outdated photos that do not accurately represent current room configurations; pricing for half-day room rental is not posted, requiring direct inquiry; availability for the requested date cannot be confirmed without phone contact; and payment methods are unclear, creating uncertainty about whether the space accepts digital payments or requires cash. The consultant eventually secures a room after multiple phone calls and a preliminary visit, but the transaction friction consumes several hours of productive time and creates a negative impression that nearly leads the client to question the consultant's organizational competence.

Case Study 4: Digital Nomad During Power Outage. A location-independent software developer visiting Iloilo City from Manila experiences a power outage at their accommodation in Arevalo district. The developer has urgent deliverables due within hours and requires immediate access to a workspace with backup power and internet. Without a centralized platform that provides real-time information about space availability, power backup status, and current operating conditions, the developer resorts to searching Facebook groups and asking locals for recommendations. After visiting two spaces that are either fully occupied or also affected by the power outage, the developer finds a suitable workspace but loses over three hours of critical work time. The experience generates lasting skepticism about Iloilo City's reliability as a destination for remote work, potentially influencing future travel decisions and recommendations to professional networks.

These case studies reveal several common patterns. First, the absence of real-time availability information forces users into sequential search processes—visiting one space after another until finding availability—which is time-inefficient and psychologically stressful. Second, the reliance on informal communication channels (social media, phone calls) introduces response delays and communication failures that increase uncertainty. Third, the lack of standardized information about amenities, pricing, and policies prevents users from making informed comparisons and decisions. Fourth, the consequences of reservation uncertainty extend beyond immediate inconvenience to affect professional credibility, academic performance, business development, and destination reputation. Fifth, vulnerable users—those with time constraints, limited mobility, or urgent needs—experience disproportionate impacts, creating inequities in access to co-working resources.

### 1.2.2 Information Fragmentation and its Systemic Consequences

Information fragmentation compounds the problem of reservation uncertainty by dispersing co-working space information across multiple channels, formats, and update frequencies, preventing users from obtaining comprehensive, consistent, and current information through a single access point. In Iloilo City, co-working space information is distributed across: Facebook pages and groups, where spaces post availability updates, promotional offers, and operating hours in unstructured formats; Instagram accounts, which emphasize visual presentation over functional information; Google Maps listings, which provide basic location data but often lack current operating hours, contact information, or amenity details; word-of-mouth networks, where information is transmitted through personal recommendations with variable accuracy and completeness; physical signage, which is limited to passersby in immediate vicinity; and third-party directories, which may include outdated listings or incomplete coverage of local spaces.

This fragmentation generates multiple inefficiencies. Users must invest substantial search effort to compile information that would be available instantaneously through a centralized platform. Information quality is inconsistent, with some spaces maintaining current, detailed online presence while others have minimal or outdated digital footprints. Comparison across spaces is difficult because information is presented in different formats, with varying levels of detail, and without standardized categories or metrics. Discovery of new spaces is serendipitous rather than systematic, meaning that users may remain unaware of options that better match their needs or offer superior value. For space operators, fragmentation limits market reach to users who happen to encounter their information through specific channels, excluding potential customers who use different information sources.

The consequences of information fragmentation are particularly severe for users who are new to Iloilo City or unfamiliar with its co-working landscape. Students arriving from other provinces to begin university studies, professionals relocating for employment opportunities, tourists and digital nomads visiting the city for the first time, and entrepreneurs exploring the local business environment all face steep learning curves in identifying and evaluating co-working options. The absence of a comprehensive, authoritative information source creates barriers to entry that may discourage these users from utilizing co-working spaces altogether, reducing market size and limiting the sector's growth potential.

### 1.2.3 Booking Conflicts and Operational Inefficiencies in Manual Systems

Manual tracking systems for co-working reservations are inherently susceptible to booking conflicts, where multiple users claim the same resource—desk, seat, meeting room—for overlapping time periods. These conflicts arise from the limitations of paper-based logs, spreadsheet trackers, and memory-dependent systems that cannot enforce real-time synchronization across multiple inquiry channels. When a user books via Facebook Messenger while another books via phone call, the operator must manually update multiple records to maintain consistency; delays or omissions in these updates create windows during which conflicting reservations can be accepted.

The operational consequences of booking conflicts extend beyond the immediate inconvenience of double-booked users. Space operators must invest time in conflict resolution—contacting affected users, proposing alternative arrangements, processing refunds or credits—that diverts attention from customer service and business development. Users who experience conflicts lose trust in the operator's reliability, reducing the likelihood of repeat business and generating negative word-of-mouth that damages reputation. In severe cases, conflicts can escalate into disputes requiring mediation or legal intervention, particularly when financial transactions or contractual obligations are involved.

Beyond conflicts, manual systems generate broader operational inefficiencies. Hub owners lack accurate data on occupancy rates, peak demand periods, average session durations, and user demographics that would inform strategic decisions about pricing, capacity expansion, amenity investments, and marketing targeting. The administrative burden of manually processing inquiries, updating availability records, sending confirmations, and managing cancellations consumes staff time that could be allocated to value-adding activities such as community building, event programming, and service improvement. The inability to implement automated features—waitlists, booking reminders, check-in confirmations, post-visit feedback requests—deprives operators of tools that enhance user experience and operational efficiency in digitally mature markets.

### 1.2.4 The Structural Gap: Absence of a Unified Discovery and Reservation Platform

The problems of reservation uncertainty, information fragmentation, and booking conflicts collectively reflect a structural gap in Iloilo City's digital infrastructure: the absence of a centralized, unified platform for discovering and reserving co-working spaces. This gap distinguishes Iloilo from global cities where such platforms have become standard components of

the urban digital ecosystem, and even from Philippine cities such as Metro Manila and Cebu where commercial operators have developed proprietary reservation systems for their own spaces.

The unified platform gap has implications beyond individual user inconvenience. For Iloilo City's Smart City development, the absence of co-working reservation infrastructure represents a missed opportunity to demonstrate digital governance capabilities, generate urban analytics, and support the city's innovation ecosystem. For the city's economic competitiveness, the gap reduces Iloilo's attractiveness to digital nomads, remote workers, and start-up entrepreneurs who expect the convenience of digital booking as a baseline service standard. For co-working operators, the gap constrains market development by maintaining information asymmetries, limiting user acquisition, and preventing the operational efficiencies that digital systems enable.

The development of FlexSpace addresses this structural gap by providing a localized, comprehensive platform that integrates space discovery, real-time availability information, online booking, payment processing, and administrative management. By centralizing information that is currently fragmented across multiple channels, the system reduces search costs and uncertainty for users. By automating reservation processes that are currently manual, the system prevents conflicts and improves operational efficiency for space operators. By generating data that is currently uncollected, the system supports evidence-based decision making for business strategy and urban planning. In these ways, FlexSpace functions not merely as a technical application but as infrastructure that strengthens Iloilo City's co-working ecosystem and advances its Smart City trajectory.

## 1.3 Objectives of the Study

### 1.3.1 General Objective

The general objective of this study is to design, develop, and evaluate a centralized Co-Working Space Discovery and Reservation System for Iloilo City—designated as FlexSpace—that streamlines the process of discovering available workspaces, provides real-time seat and room availability tracking, facilitates secure online bookings, and empowers co-working space operators with data-driven administrative tools. This overarching goal reflects both a technological ambition to deploy a full-stack web application built on the MERN stack and a socio-economic imperative to digitize the fragmented co-working ecosystem of Iloilo City, aligning it with global best practices in Smart City innovation while addressing the specific needs, constraints, and opportunities of the local context. By creating a unified platform that serves multiple stakeholder

groups—students, freelancers, entrepreneurs, tourists, space owners, and city administrators—this study seeks to eliminate inefficiencies inherent in manual reservation processes, enhance user trust and satisfaction, generate actionable data for operational optimization and urban planning, and contribute to Iloilo City's positioning as a competitive digital hub in Western Visayas and the broader Philippines.

### 1.3.2 Specific Objectives

Specific Objective 1: To conduct a comprehensive analysis of the current challenges faced by students, freelancers, entrepreneurs, and co-working space operators in Iloilo City regarding workspace accessibility, reservation reliability, and information availability. This objective ensures that the system design is grounded in empirical evidence rather than abstract assumptions. The analysis encompasses primary data collection through semi-structured interviews, focus group discussions, and direct observation of co-working space operations across multiple districts, supplemented by secondary data from industry reports, academic studies, and government statistics. By documenting the frequency, severity, and distribution of problems such as reservation uncertainty, booking conflicts, information fragmentation, and operational inefficiencies, this analysis establishes quantitative and qualitative baselines against which system effectiveness can be evaluated. The findings inform the prioritization of system features, the design of user interfaces, and the configuration of administrative workflows to address the most pressing needs identified by stakeholders.

Specific Objective 2: To design an intuitive, responsive, and aesthetically professional user interface that enables users to efficiently search, filter, compare, and select co-working spaces based on district location, available amenities, pricing structures, user ratings, and real-time availability status. The user interface design follows principles of user-centered design, accessibility standards (WCAG 2.1 Level AA), and mobile-first responsive design to ensure optimal experience across devices including smartphones, tablets, and desktop computers. The interface incorporates interactive GIS mapping that visualizes space locations across Iloilo City's districts, allowing users to identify proximate options and obtain navigation directions. Filter functionality supports multi-criteria selection, enabling users to specify requirements such as Wi-Fi speed, power outlet availability, air conditioning, printing facilities, meeting room access, parking, and food service. Comparison tools allow side-by-side evaluation of multiple spaces across key dimensions. The design process involves iterative prototyping, usability testing with

representative users, and refinement based on feedback to ensure that the interface meets ISO 25010 usability standards.

Specific Objective 3: To develop a real-time availability tracking module that monitors desk, seat, and meeting room occupancy status, preventing overbooking and reducing walk-in rejections. The module implements software-based status updates through which space operators can mark resources as available, occupied, reserved, or under maintenance, with changes reflected immediately in the user-facing interface. While the study acknowledges its delimitation in not deploying IoT hardware such as pressure sensors or RFID scanners, the software-based approach provides a feasible and cost-effective alternative that captures the majority of availability information through operator updates and booking system integration. The module implements conflict detection algorithms that prevent simultaneous bookings of the same resource, with notification protocols that alert operators and users when scheduling overlaps are detected. Historical availability data is maintained to support demand forecasting and capacity planning analytics.

Specific Objective 4: To implement an automated notification and reservation management system that sends instant booking confirmations, reminder alerts, check-in prompts, and post-visit feedback requests to users via email and in-app notifications. The notification system leverages Nodemailer with Gmail SMTP integration for email delivery and Socket.io for real-time in-app messaging, ensuring that users receive timely information throughout their reservation lifecycle. Confirmation emails include booking details, QR codes for check-in verification, cancellation policies, and space contact information. Reminder alerts are dispatched at configurable intervals prior to reservation start times, reducing no-show rates and enabling release of unclaimed reservations for reallocation. Check-in prompts facilitate session initiation and trigger session timer functionality that tracks usage duration. Post-visit feedback requests generate review data that supports space quality assessment and user decision-making.

Specific Objective 5: To create a comprehensive administrative dashboard for co-working space owners that provides tools for inventory management, facility information updates, booking oversight, usage analytics, revenue tracking, voucher management, and review response. The dashboard implements CRUD (Create, Read, Update, Delete) operations for all managed resources, with middleware authentication and role-based access control ensuring that operators can only modify data within their authorized scope. Analytics modules generate visualizations and reports on occupancy rates, peak usage periods, revenue trends, user demographics, and review

ratings, enabling data-driven decision making about pricing strategies, capacity adjustments, service improvements, and marketing targeting. The dashboard interface prioritizes clarity and efficiency, presenting critical information through summary cards, charts, and tables that enable rapid assessment of operational status.

Specific Objective 6: To integrate geographic information system (GIS) mapping functionality that provides accurate geolocation visualization, district-based filtering, proximity search, and navigation assistance for co-working spaces across Iloilo City. The mapping module utilizes the Leaflet JavaScript library with OpenStreetMap tile layers, supplemented by custom markers that indicate space availability status and category. Users can search for spaces within specified distance radii, filter by district boundaries (Mandurriao, Jaro, Molo, City Proper, Arevalo, Lapuz, La Paz), and obtain turn-by-turn navigation directions. District-based organization of space listings aligns with users' mental models of the city's geography and supports administrative functions such as district-level analytics and targeted marketing. The mapping integration represents a distinctive feature that differentiates FlexSpace from generic reservation platforms by embedding spatial intelligence specific to Iloilo City's urban structure.

Specific Objective 7: To implement secure data management practices that protect user profiles, booking histories, facility details, and transactional information using industry-standard security measures. The system utilizes MongoDB's security features including authentication, authorization, and encrypted connections. User passwords are hashed using bcrypt with salt rounds configured for computational resistance to brute-force attacks. JSON Web Tokens (JWT) with appropriate expiration periods and refresh token rotation manage authenticated sessions. Cloudflare Turnstile integration provides bot protection for public-facing forms, preventing automated abuse of registration and booking functions. Rate limiting middleware restricts request frequencies to prevent denial-of-service attacks and credential stuffing. Activity logging with IP address tracking supports audit trails and dispute resolution. These security measures collectively ensure compliance with the Data Privacy Act of 2012 and ISO 25010 security quality standards.

Specific Objective 8: To conduct comprehensive testing and evaluation of the system's functionality, usability, security, and efficiency using ISO 25010 software quality standards as the evaluation framework. Testing encompasses unit testing of individual modules, integration testing of module interactions, system testing of end-to-end workflows, and user acceptance testing with representative stakeholders from each user segment. Evaluation instruments include functionality checklists that verify correct operation of all features, usability questionnaires that measure user

satisfaction and task completion efficiency, security penetration tests that identify vulnerabilities, and performance benchmarks that assess response times, concurrency handling, and resource utilization under simulated load conditions. The evaluation generates quantitative metrics and qualitative feedback that demonstrate system quality, identify areas for improvement, and establish a basis for comparison with existing systems and future iterations.

## 1.4 Scope and Delimitations

Every research and development project must clearly define its scope to establish boundaries of inquiry, ensure feasibility within available resources and timeframes, and provide a framework for evaluating achievement of objectives. The scope of this study encompasses the design, development, testing, and evaluation of a localized web-based application for discovering and reserving co-working spaces specifically within the territorial boundaries of Iloilo City. The following subsections detail the functional scope of the system, the explicit delimitations that constrain its capabilities, and the boundary conditions that define the operational envelope within which the system is designed to function.

### 1.4.1 Scope of the Study

The FlexSpace system includes the following core functional modules, each designed to address specific stakeholder needs and operational requirements identified through the research phase:

User Registration and Profile Management. The system provides comprehensive account creation functionality supporting both email-based registration with password authentication and Google OAuth integration for streamlined sign-up. User profiles capture essential information including full name, contact email, phone number, profile photograph, and membership preferences. Role-based account types differentiate between regular users (students, freelancers, general public), space owners (co-working operators), and system administrators. Profile management tools enable users to update personal information, change passwords, view booking history, track loyalty points, and manage notification preferences. Email verification confirms account authenticity, while password reset functionality with time-limited OTP tokens ensures account recovery security.

Searchable Directory of Co-Working Spaces. The system maintains a centralized, comprehensive directory of registered co-working spaces and study hubs located within Iloilo City's seven districts. Each space listing includes: descriptive information (name, description, operating hours, contact details); location data (address, geographic coordinates, district assignment); amenity specifications (Wi-Fi speed, power outlets, air conditioning, printing, meeting rooms, parking, food

service, accessibility features); pricing structures (hourly rates, daily rates, monthly memberships, promotional packages); photographic documentation of interior spaces, facilities, and building exteriors; user-generated ratings and written reviews; and real-time availability status for different resource types (hot desks, dedicated desks, private offices, meeting rooms). The directory supports full-text search, faceted filtering across multiple dimensions, and sorting by relevance, rating, price, and proximity.

Real-Time Availability Tracking and Reservation Engine. The system's core transactional functionality enables users to view current availability of workspace resources, select preferred time slots, initiate booking requests, receive instant confirmations, and manage active reservations. The reservation engine implements conflict detection algorithms that query the database for overlapping bookings before confirming new reservations, preventing double-booking scenarios. Session management tracks reservation lifecycles from creation through check-in, active usage, check-out, and completion. QR code generation for each booking facilitates contactless check-in verification at space locations. Automated status transitions handle scenarios such as no-shows (reservations released after grace period), early check-outs (resources returned to available pool), and extensions (additional time added subject to availability).

Administrative Dashboard for Space Owners. Co-working space operators access dedicated dashboard interfaces that provide comprehensive management capabilities. Space management tools enable owners to create and edit space listings, upload photographs, configure amenity selections, set pricing parameters, define operating hours, and specify capacity limits. Booking management interfaces display upcoming, active, and past reservations with filtering and search capabilities, allowing operators to monitor occupancy, process check-ins and check-outs, handle cancellations, and manage booking modifications. Analytics modules generate occupancy rate reports, revenue summaries, peak demand identification, user demographic breakdowns, and review sentiment analysis. Voucher management tools enable creation of promotional codes, discount configurations, and loyalty point campaigns. Staff account management allows owners to delegate dashboard access to employees with appropriate permission levels.

System Administration Module. Administrative users with system-level privileges access enhanced management capabilities including: approval workflows for new space owner registrations and space listing activations; user account management with blocking, unblocking, and role assignment functions; district management tools for configuring geographic coverage areas; global voucher creation and distribution; system-wide analytics aggregating data across all

spaces and users; maintenance mode controls for system updates; commission tracking for revenue sharing arrangements; system log monitoring for security auditing; and queue management for background job processing.

GIS Mapping Integration. Interactive mapping functionality visualizes co-working space locations on scrollable, zoomable maps centered on Iloilo City. Custom markers indicate space categories and current availability status. Click interactions on markers display space summary cards with key information and links to detailed listings. District overlay layers highlight administrative boundaries. Proximity search tools identify spaces within user-specified distance radii from reference points. Navigation integration generates turn-by-turn directions from user locations to selected spaces. The mapping module enhances spatial discoverability, supporting users who prioritize location convenience in their workspace selection criteria.

AI-Powered Chatbot (Flex Support). The system integrates an intelligent chatbot powered by the Google Gemini API that provides multilingual conversational support in English, Filipino, and Hiligaynon. Flex Support handles queries including: co-working space recommendations based on district, amenity, and price preferences; step-by-step booking guidance; answers to frequently asked questions about system functionality, payment methods, cancellation policies, and account management; provision of Google Maps links for space locations; information about the development team and system capabilities; and contextual follow-up responses based on conversation history. The chatbot interface is accessible from all user-facing pages, providing immediate assistance without requiring navigation away from current tasks.

### 1.4.2 Delimitations of the Study

While the scope is comprehensive within its defined boundaries, the study explicitly acknowledges the following delimitations that constrain system functionality and applicability:

Delimitation 1: Exclusion of Internet of Things (IoT) Hardware Integration. The system does not include real-time IoT-based hardware integration for automated occupancy detection. Specifically, the study excludes the deployment of pressure sensors embedded in desks or chairs, RFID or NFC readers for user identification and access control, infrared beam counters for foot traffic measurement, or smart lock systems for automated room access. Instead, the system relies on software-based status updates initiated by space operators through the administrative dashboard, combined with booking system logic that infers occupancy from confirmed reservations and check-in records. This delimitation reflects resource constraints including the cost of IoT hardware, the

complexity of installation and maintenance, and the need for physical modifications to co-working premises that are beyond the research team's authority and budget. While the exclusion of IoT hardware limits the system's ability to provide fully automated, sensor-verified occupancy data, the software-based approach provides a functionally adequate foundation that can be enhanced with IoT integration in future development phases.

Delimitation 2: Simulated Payment Gateway. The system implements a simulated payment gateway for academic demonstration purposes rather than integrating with production payment processing APIs. Users can simulate booking payments through a mock interface that mimics the user experience of GCash and Maya e-wallet transactions without processing actual financial transfers. This simulation includes: generation of QRPH-compliant QR codes for display at payment counters; recording of simulated transaction statuses (pending, completed, failed, refunded); integration with booking confirmation workflows that gate reservation finalization on payment completion; and administrative displays of simulated revenue data. The study explicitly notes that this simulation cannot process real financial transactions, limiting the system's immediate commercial deployment without subsequent integration of actual payment gateway APIs such as GCash Developer API, Maya Business API, or Stripe for international cards. The simulation architecture is designed to facilitate future payment gateway integration by isolating payment logic behind service interfaces that can be reimplemented with production providers.

Delimitation 3: Geographic Restriction to Iloilo City. The system's database, mapping coverage, and space registration workflows are restricted to co-working spaces, study hubs, and shared work facilities located within the territorial boundaries of Iloilo City. Spaces in peripheral municipalities—including Oton, San Miguel, Pavia, Santa Barbara, and Leganes—are explicitly excluded from the system's scope. Long-term corporate lease management, commercial property rentals, and residential co-living spaces are similarly excluded. This geographic delimitation ensures focused development on the most relevant local context, enables detailed district-level customization of system features, and maintains manageable data collection requirements within the study timeframe. Future expansion to broader geographic coverage is anticipated as a post-deployment enhancement.

Delimitation 4: Exclusion of Native Mobile Application. The system is developed as a responsive web application accessible through standard web browsers on desktop and mobile devices, rather than as a native mobile application distributed through app stores (Apple App Store, Google Play Store). The responsive web approach leverages Progressive Web App (PWA) technologies—

including service workers for offline functionality, web app manifests for home screen installation, and push notification APIs for alert delivery—to approximate native app experience without the development overhead, platform-specific coding requirements, and store approval processes associated with native applications. While PWA capabilities provide substantial mobile functionality, certain native features such as deep OS integration, background location tracking, and advanced camera functions are not available.

Delimitation 5: Academic Timeframe and Resource Constraints. The system is developed, tested, and evaluated within the academic timeframe of a single semester (approximately 18 weeks), with resource constraints including limited development team capacity (five student researchers), budget limitations for cloud infrastructure and third-party services, and competing academic commitments. These constraints influence system complexity, the depth of certain features, and the scale of user acceptance testing. The study prioritizes core functionality completeness and quality over advanced features that would require extended development timelines or specialized expertise.

Delimitation 6: User Population for Testing. System evaluation is conducted with a purposive sample of users drawn from student populations at PHINMA University of Iloilo and neighboring institutions, local freelancers identified through professional networks, co-working space operators recruited through direct outreach, and simulated tourist/digital nomad personas. The sample size and demographic composition are sufficient for identifying major usability issues and assessing core functionality but may not fully represent the diversity of the eventual target user population. Generalization of evaluation findings to broader user groups requires additional testing with larger, more diverse samples in subsequent research phases.

### 1.4.3 Boundary Conditions and Edge Cases

The system is designed to operate correctly within defined boundary conditions and to handle specified edge cases through defined exception handling procedures. Boundary conditions include: maximum concurrent user load of 500 simultaneous sessions based on estimated peak demand; database storage capacity sufficient for 10,000 registered users, 500 space listings, and 100,000 historical booking records; geographic coverage within 50 kilometers of Iloilo City center; operating hour support for spaces open 24 hours; and booking duration ranges from 1 hour to 30 days. Edge cases addressed by the system include: simultaneous booking attempts for the same resource by multiple users (resolved through database transaction isolation and first-commit-wins semantics); payment simulation failures (handled through retry logic and manual override

capabilities); network connectivity interruptions during active sessions (managed through local state caching and synchronization on reconnection); invalid or malicious input data (prevented through validation middleware and sanitization); and operator failure to update availability status (mitigated through automatic status inference from booking data and configurable default availability assumptions).

## 1.5 Significance of the Study

The proposed Co-Working Space Discovery and Reservation System for Iloilo City holds significance across multiple stakeholder groups, extending beyond immediate user convenience to encompass broader contributions to economic development, urban innovation, educational outcomes, and scholarly knowledge. The following subsections articulate the significance for each major stakeholder category, establishing the rationale for investment of research effort and the criteria by which system success can be judged.

### 1.5.1 Significance for Freelancers and Remote Workers

For freelancers and remote workers, co-working spaces represent essential infrastructure that enables professional productivity, client service quality, and career sustainability. The proposed system addresses the core pain points that undermine this infrastructure's reliability: reservation uncertainty that wastes time and creates scheduling stress; information fragmentation that prevents informed decision-making; and booking conflicts that disrupt work plans and damage professional credibility. By providing a one-stop platform for space discovery, availability verification, and reservation management, the system reduces the friction costs associated with securing workspace, allowing freelancers to redirect time and energy toward billable work, skill development, and client relationships. The system's GIS mapping and district-based filtering support strategic workspace selection that minimizes transportation costs and travel time. Loyalty point and voucher features provide financial incentives that improve affordability for price-sensitive users. Real-time availability information enables adaptive planning that responds to changing schedules and client demands. In aggregate, these benefits enhance the economic viability and quality of life of Iloilo City's freelance workforce, contributing to retention of talent and attraction of new remote workers to the city.

### 1.5.2 Significance for Students

Students represent a large, price-sensitive, and acutely affected user group whose academic performance and well-being are directly influenced by access to conducive study environments.

The system's significance for students encompasses several dimensions. First, it addresses the capacity limitations of institutional study facilities by making alternative spaces visible and accessible, effectively expanding the city's study infrastructure beyond campus boundaries. Second, it enables efficient space discovery that reduces the time and stress associated with finding study locations during high-demand periods, supporting more effective time management during critical academic phases. Third, price filtering and comparison tools help students identify affordable options that match their budget constraints, democratizing access to quality study environments. Fourth, collaborative space features support group study methodologies that enhance learning outcomes through peer interaction. Fifth, extended-hours filtering accommodates students with non-standard study schedules, including those balancing academic responsibilities with part-time employment. By improving study environment access, the system indirectly contributes to educational outcomes, student retention, and the development of independent learning skills that support lifelong professional development.

### 1.5.3 Significance for Co-Working Space Owners

Co-working space operators are critical stakeholders whose business success determines the availability, quality, and diversity of workspace options in the city. The system's administrative dashboard provides operators with capabilities that are currently unavailable or require manual effort: real-time booking management that reduces administrative workload and human error; occupancy analytics that inform pricing decisions, capacity planning, and marketing strategies; user review management that supports reputation building and quality improvement; promotional tools that enable customer acquisition and retention campaigns; and increased visibility to a broader user base through centralized directory listing. These capabilities enhance operational efficiency, improve revenue optimization, and reduce the cognitive burden of manual management processes. The system's data generation function creates valuable business intelligence that operators can use to demonstrate value to investors, negotiate partnerships, and benchmark performance against industry standards. For new entrants to the co-working market, the platform reduces customer acquisition costs by providing established channels for reaching potential users.

### 1.5.4 Significance for the Local Government and Smart City Development

The Iloilo City Government benefits from the system as a component of its broader Smart City initiative. The platform demonstrates the city's commitment to digital transformation and innovation ecosystem support, enhancing its reputation as a forward-looking, technology-enabled urban center. The data generated by the system—aggregated and anonymized—provides insights

into co-working utilization patterns, district-level demand distribution, user demographics, and space characteristics that can inform evidence-based policymaking. Urban planners can use this data to identify underserved districts, assess transportation infrastructure needs, evaluate the effectiveness of economic development incentives, and plan for future co-working facility development. The system's alignment with national digital transformation agendas, including DICT's initiatives and the Philippine Digital Strategy, positions the city for potential funding support and partnership opportunities. The platform also contributes to tourism development by making Iloilo City more accessible and welcoming to digital nomads and location-independent professionals who represent a growing segment of the global travel market.

### 1.5.5 Significance for Tourists and Digital Nomads

Tourists and digital nomads represent an emerging user segment whose significance extends beyond immediate transaction value to encompass destination branding, word-of-mouth marketing, and long-term economic development. Short-term visitors to Iloilo City often require professional workspace for activities ranging from email management and virtual meetings to intensive project work. The system's multilingual chatbot support (English, Filipino, Hiligaynon), GIS mapping with navigation assistance, and transparent pricing information address the specific needs of visitors who are unfamiliar with the city's geography, business practices, and local languages. By reducing barriers to workspace access for visitors, the system enhances Iloilo City's attractiveness as a destination for bleisure (business-leisure) travel and extended nomadic stays, segments that generate higher per-visit spending than conventional tourism. Positive experiences shared through professional and social networks can generate significant destination marketing value that exceeds the system's direct economic impact.

### 1.5.6 Significance for BSIT Students and Future Developers

For the student researchers developing this system, the project provides comprehensive training in full-stack web development, database design, API integration, user interface design, security implementation, and software quality assurance. Working with the MERN stack, Agile methodology, ISO 25010 standards, and modern development tools prepares students for professional careers in software engineering and systems development. The project's scope—encompassing requirements analysis, system design, implementation, testing, deployment, and evaluation—mirrors industry software development lifecycles, providing experiential learning that complements academic coursework. The system's open architecture and documented codebase enable future student developers to build upon this work, adding features such as IoT integration,

machine learning analytics, blockchain-based verification, or expanded geographic coverage. The project thus contributes to the educational mission of PHINMA University of Iloilo's BSIT program by demonstrating applied competencies and creating resources for subsequent cohorts.

### 1.5.7 Significance for Researchers and Scholars

The study contributes to scholarly knowledge on several fronts. It provides empirical documentation of co-working ecosystem conditions in a Philippine provincial city, filling a gap in the literature that has focused predominantly on Metro Manila and global technology hubs. It demonstrates the application of Smart City frameworks to localized resource management challenges, offering a model that can be adapted for other cities and sectors. It evaluates the effectiveness of specific technical approaches—the MERN stack, GIS integration, AI chatbots, simulated payment systems—in a developing country context, generating findings that inform technology selection decisions for similar projects. The system's development process, documented through Agile sprint records, design decisions, and testing results, provides a case study for teaching software engineering methodology. Future researchers can build upon this foundation by conducting longitudinal studies of system impact, comparative analyses across multiple cities, or investigations of specific technical enhancements.

## 1.6 Theoretical Framework

The development and evaluation of the FlexSpace system are grounded in multiple theoretical perspectives that provide conceptual frameworks for understanding technology adoption, system success, and the socio-technical dynamics of co-working ecosystems. These theories inform system design decisions, shape evaluation methodologies, and anchor the study's contributions within established scholarly discourse.

### 1.6.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model, developed by Davis (1989), provides a foundational framework for understanding and predicting user adoption of information systems. TAM posits that user acceptance is determined by two primary constructs: Perceived Usefulness (PU), the degree to which a user believes that using a system will enhance their job performance or task outcomes; and Perceived Ease of Use (PEOU), the degree to which a user believes that using a system will be free from effort. These constructs influence Attitude Toward Use, which in turn affects Behavioral Intention to Use, ultimately determining actual system usage.

In the context of FlexSpace, Perceived Usefulness is operationalized through features that demonstrably improve co-working space discovery and reservation outcomes: real-time availability information that eliminates wasted trips; automated booking that reduces time spent on manual inquiries; GIS mapping that optimizes location selection; and administrative analytics that improve operational decision-making for space owners. Perceived Ease of Use is addressed through responsive design that works across devices, intuitive navigation patterns, streamlined booking workflows with minimal steps, multilingual chatbot support, and clear visual feedback for all user actions. The system's evaluation instruments include TAM-based questionnaire items that measure these constructs, enabling assessment of adoption likelihood and identification of barriers that may require design intervention.

### 1.6.2 DeLone and McLean Information Systems Success Model (ISSM)

The Information Systems Success Model, originally proposed by DeLone and McLean (1992) and subsequently updated (2003), provides a comprehensive framework for evaluating system effectiveness across multiple dimensions. The updated model identifies six interdependent success categories: System Quality (technical performance characteristics including usability, reliability, flexibility, and response time); Information Quality (accuracy, completeness, timeliness, and relevance of system outputs); Service Quality (responsiveness, competence, and empathy of support services); Use (frequency, duration, and nature of system utilization); User Satisfaction (affective and cognitive evaluation of the system experience); and Net Benefits (ultimate impacts on individual productivity, organizational performance, and societal outcomes).

FlexSpace's evaluation framework directly incorporates ISSM dimensions. System Quality is assessed through ISO 25010 testing of functionality, reliability, efficiency, usability, security, and maintainability. Information Quality is evaluated through accuracy checks on space listings, availability data currency, and completeness of facility descriptions. Service Quality is measured through chatbot response effectiveness, email notification reliability, and administrative support responsiveness. Use is tracked through system analytics on registration counts, session frequencies, booking volumes, and feature utilization patterns. User Satisfaction is captured through post-interaction surveys and Net Promoter Score measurement. Net Benefits are assessed through stakeholder interviews examining productivity impacts, operational improvements, and contributions to Smart City objectives.

### 1.6.3 Socio-Technical Systems Theory

Socio-Technical Systems Theory, originating from the Tavistock Institute's studies of organizational change, emphasizes that effective systems must jointly optimize both technical and social components. The theory rejects the assumption that technical efficiency alone determines system success, instead arguing that system design must account for human needs, social dynamics, organizational cultures, and environmental contexts. In the context of co-working spaces, this perspective recognizes that these environments are not merely physical containers with technical infrastructure but socio-technical ecosystems where interpersonal interactions, community norms, professional identities, and collaborative practices shape user experiences and outcomes.

FlexSpace's design applies socio-technical principles by incorporating features that support social dimensions of co-working: user review systems that enable community feedback and reputation formation; space descriptions that highlight community characteristics and professional orientations; chatbot recommendations that match users with spaces aligned to their work styles and social preferences; and analytics that help operators understand and nurture community dynamics. The system's user research phase explicitly investigates social and psychological needs alongside functional requirements, ensuring that technical capabilities serve human-centered objectives.

### 1.6.4 Smart City Theory

Smart City Theory encompasses a range of conceptual frameworks that address the integration of digital technologies into urban governance, service delivery, and development planning. Common elements across these frameworks include: citizen-centricity, ensuring that technology investments serve resident needs and priorities; data-driven governance, leveraging urban data analytics for evidence-based decision-making; sustainability, using technology to improve environmental and social outcomes; connectivity, providing universal access to high-quality digital infrastructure; and innovation ecosystem development, creating environments that support entrepreneurship, creativity, and knowledge-based industries.

FlexSpace contributes to Smart City objectives by digitizing a component of the urban service economy, generating data that supports evidence-based planning, improving accessibility to workspace resources across district boundaries, and supporting the innovation ecosystem that Smart City frameworks prioritize. The system's GIS integration, real-time data capabilities, and

user-centric design embody Smart City principles, while its focus on inclusion and equity addresses the risk that Smart City benefits concentrate in affluent districts and populations.

## 1.7 Definition of Terms

The following terms are defined in relation to their specific application within this study, ensuring conceptual clarity for readers across technical and non-technical backgrounds.

Agile Methodology refers to an iterative approach to software development that emphasizes incremental delivery, continuous feedback, cross-functional collaboration, and adaptive planning. This study employs Agile through two-week sprints with defined deliverables, daily stand-up coordination, sprint review demonstrations, and retrospective process improvements.

Availability Status denotes the real-time condition of a workspace resource (desk, seat, meeting room) indicating whether it is currently accessible for booking or occupied. In FlexSpace, availability status is determined through operator updates and booking system inference, displayed to users through color-coded indicators and filterable search results.

Booking refers to the complete transaction process through which a user secures access to a specific workspace resource for a defined time period, including selection, payment simulation, confirmation, and session management. A booking creates a binding reservation record in the system database with associated QR code for check-in verification.

Bull/Redis Queue denotes the background job processing system used in FlexSpace for handling asynchronous tasks including email dispatch, image upload processing, and notification generation. Redis serves as the in-memory data store for job queues, while Bull provides the job definition, scheduling, and retry logic.

Cloudflare Turnstile is an intelligent CAPTCHA alternative integrated into FlexSpace's registration and contact forms to distinguish human users from automated bots without requiring interactive challenges. Turnstile analyzes browser behavior signals to make invisible verification decisions.

Co-Working Space refers to a shared work environment where individuals from different organizations, professions, or backgrounds work alongside each other in a common facility, sharing infrastructure, services, and community. Co-working spaces range from formal commercial operations to informal study hubs and café-based work environments.

CRUD Operations refer to the four fundamental database management functions: Create (insert new records), Read (retrieve existing records), Update (modify existing records), and Delete (remove records). FlexSpace's administrative dashboards implement CRUD operations for managing spaces, bookings, users, districts, and vouchers.

Data Privacy Act of 2012 (Republic Act No. 10173) is the Philippine legislation governing the processing of personal information, establishing rights of data subjects, obligations of personal information controllers, and penalties for unauthorized processing. FlexSpace's data handling practices are designed for compliance with this legislation.

Digital Maturity Gap refers to the disparity in digital technology adoption and capability between different organizations, sectors, or geographic regions. In this study, the concept describes the lag between the advanced digital reservation systems deployed in global co-working markets and the manual processes still prevalent in Philippine provincial cities.

GIS (Geographic Information System) refers to technologies for capturing, storing, analyzing, and visualizing geographic data. FlexSpace integrates GIS through interactive mapping that displays co-working space locations, supports proximity search, and provides navigation assistance.

Information Fragmentation describes the condition where relevant information is distributed across multiple disconnected sources, channels, or formats, preventing efficient access and comprehensive understanding. In Iloilo City's co-working market, fragmentation occurs across social media, phone contacts, word-of-mouth, and physical signage.

ISO 25010 is the international standard for software product quality, defining eight quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. FlexSpace is evaluated against these characteristics as the primary quality assessment framework.

JSON Web Token (JWT) refers to the compact, self-contained mechanism used in FlexSpace for securely transmitting authentication information between client and server. JWTs contain encoded user identity and role claims, digitally signed to prevent tampering, with configurable expiration periods.

MERN Stack designates the JavaScript-based technology stack comprising MongoDB (document database), Express.js (web framework), ReactJS (front-end library), and Node.js (runtime environment), used for full-stack development of FlexSpace.

Middleware Authentication refers to the security mechanism in which authentication verification functions are inserted into the request processing pipeline between the web server and application endpoints. FlexSpace uses middleware to validate JWT tokens, enforce role-based access control, and protect sensitive routes from unauthorized access.

QR Code (Quick Response Code) is the two-dimensional barcode generated by FlexSpace for each booking, encoding a unique token that identifies the reservation. Users present QR codes at co-working spaces for contactless check-in verification.

QRPH (Quick Response Payment Philippines) refers to the standardized QR code specification for electronic payments in the Philippines, established by the Bangko Sentral ng Pilipinas. FlexSpace generates QRPH-compliant codes for its simulated payment process.

Reservation Uncertainty describes the condition where workspace seekers lack reliable information about current space availability, resulting in ambiguity about whether desired resources can be secured. This study identifies reservation uncertainty as the primary problem affecting co-working space users in Iloilo City.

Session Timer refers to the system feature that tracks the elapsed time of an active workspace reservation, from check-in to check-out. The timer supports enforcement of time limits, calculation of usage charges, and release of resources upon session completion.

Smart City refers to an urban development paradigm that integrates digital technologies, data analytics, and citizen-centric governance to improve urban efficiency, sustainability, quality of life, and innovation capacity. Iloilo City's Smart City initiatives provide the policy context for FlexSpace development.

Socket.io refers to the JavaScript library enabling real-time, bidirectional communication between web clients and servers. FlexSpace uses Socket.io for instant availability updates, notification delivery, and chatbot message transmission.

Space Owner designates the registered user role with administrative privileges for managing one or more co-working space listings, including updating information, monitoring bookings, processing check-ins, and viewing analytics.

Walk-In refers to a user who arrives at a co-working space without a prior reservation, seeking immediate workspace access. Walk-ins represent a significant portion of current co-working usage in Iloilo City but are associated with high rejection rates during peak periods.

bcrypt is the adaptive cryptographic hash function used in FlexSpace for securely storing user passwords. bcrypt incorporates salt values and configurable cost factors to resist brute-force and rainbow table attacks.

# CHAPTER 2

# REVIEW OF RELATED LITERATURE AND STUDIES

The review of related literature and studies provides the scholarly foundation for the FlexSpace project, situating it within established bodies of knowledge on co-working ecosystems, digital reservation platforms, Smart City initiatives, and socio-technical innovation. This chapter examines exactly fifty scholarly sources, organized across foreign studies, local Philippine studies, and thematic literature analysis, to identify research gaps, validate design decisions, and establish the theoretical and empirical basis for system development. The review employs multi-dimensional categorization that examines socio-spatial configurations, psychological and atmospheric factors, comparative technical analyses of manual versus digital reservation systems, and AI-driven behavioral predictive modeling for seat occupancy optimization. The synthesis demonstrates a distinct digital maturity gap in Western Visayas co-working industries and positions FlexSpace as a form of productivity insurance for local workers.

## 2.1 Foreign Studies

### 2.1.1 North America: Digital Ecosystems and Behavioral Shaping in Co-Working

The North American co-working market, led by the United States, represents the most mature and commercially developed ecosystem globally, providing extensive research on digital reservation systems, platform business models, and the socio-cultural dimensions of shared workspaces. Gruen and Bardhi (2025) conducted a comprehensive ethnographic study of WeWork and Regus facilities across major U.S. metropolitan areas, arguing that digital reservation systems function not merely as technical tools for space management but as socio-cultural infrastructures that actively shape user behavior, professional identities, and collaborative practices. Their analysis reveals that mobile app-based booking systems create what they term 'structured serendipity'—algorithmic arrangements of seating and timing that increase the probability of encounters between members with complementary skills or interests. The study documents how reservation platforms collect and

analyze user data to optimize space layouts, adjust amenity configurations, and curate community events that foster the professional networks that represent a core value proposition of premium co-working memberships.

Johnson and Lee (2024) examined hybrid co-working models that emerged from the COVID-19 pandemic's disruption of traditional office-based work. Their research, surveying over 2,000 remote workers across the United States and Canada, found that 67% of respondents who adopted co-working arrangements during the pandemic continued these practices post-lockdown, with 43% utilizing digital reservation platforms as their primary booking method. The study identifies 'booking confidence'—the user's certainty that a reserved space will meet their expectations upon arrival—as a critical determinant of platform loyalty and repeat usage. Platforms that provided detailed amenity specifications, authentic user photography, and real-time occupancy verification achieved significantly higher booking confidence scores than those with limited information disclosure. This finding directly informs FlexSpace's emphasis on comprehensive space listings, user-generated reviews, and availability status transparency.

Martinez and O'Connor (2025) investigated the integration of blockchain-based payment systems into co-working reservation platforms, focusing on the technical and trust implications of cryptocurrency transactions. Their pilot implementation at a San Francisco co-working space demonstrated that blockchain payments reduced transaction processing fees by 23% compared to credit card processing, while smart contract automation eliminated manual invoicing and reconciliation tasks. However, user adoption was limited by cryptocurrency literacy barriers and price volatility concerns, suggesting that blockchain integration is most viable as a supplementary payment option rather than a primary mechanism. For the Philippine context, where cryptocurrency adoption remains nascent but growing, this research suggests that blockchain payment options may become relevant in future system iterations but are not a priority for initial deployment.

Chen, Li, and Zhang (2021) conducted experimental research on user trust formation in centralized reservation platforms, using eye-tracking and clickstream analysis to understand how interface design elements influence user confidence. Their findings demonstrate that real-time availability indicators (color-coded status badges, last-updated timestamps, confirmation counts) significantly reduce perceived booking risk, while the presence of cancellation policies, refund guarantees, and responsive customer support channels further enhance trust. The study also found negative trust effects from discrepancies between platform-displayed information and actual facility conditions,

emphasizing the importance of accurate, current data and mechanisms for user feedback and operator accountability.

### 2.1.2 Asia: Smart Nation Initiatives and Predictive Analytics

Asian co-working markets provide particularly relevant comparative cases due to geographic proximity, similar development trajectories, and the prevalence of government-led Smart City initiatives that integrate co-working into broader digital transformation agendas. Mao (2025) conducted spatial analysis of Singapore's co-working ecosystem, demonstrating that socio-spatial configurations significantly influence accessibility and utilization patterns. Co-working hubs located within 500 meters of Mass Rapid Transit (MRT) stations achieved 34% higher occupancy rates than those in less accessible locations, while spaces within innovation districts (one-north, Jurong Innovation District) commanded premium pricing due to proximity to corporate partners, research institutions, and venture capital firms. Mao's analysis of Singapore's reservation platforms reveals deep integration with national digital infrastructure: Singpass authentication provides seamless identity verification, LifeSG portal integration enables government service access from co-working environments, and Open Data APIs allow platforms to incorporate real-time transit information, weather data, and event calendars that enhance user planning capabilities.

Pan, Cho, and Bardhan (2023) developed and validated machine learning models for predicting co-working occupancy patterns in Seoul, South Korea, achieving 87% accuracy in next-day demand forecasting using features including day-of-week, weather conditions, local event calendars, historical booking patterns, and social media sentiment analysis. Their research demonstrates that predictive analytics enable demand-responsive pricing that increases operator revenue by 12-18% while smoothing peak demand through off-peak incentives. The integration of prediction models with reservation platforms allows dynamic availability allocation, where certain seats or rooms are held back from booking until demand forecasts clarify optimal pricing and allocation strategies. While FlexSpace does not implement machine learning in its initial version, this research establishes the value of occupancy data collection as a foundation for future predictive capabilities.

Park and Kim (2025) examined the integration of co-working reservations with public transportation apps in South Korea, creating a seamless urban mobility experience. Their analysis of the T-map and KakaoMap platforms shows that co-working space discovery embedded within transit apps increases booking conversion rates by 28% compared to standalone co-working platforms, as users can plan workspace access as part of multi-stop itineraries. This research

suggests opportunities for FlexSpace to explore partnerships with local transportation apps or jeepney route mapping services, though the Philippine context's less integrated digital infrastructure presents implementation challenges.

Sharma and Gupta (2024) studied AI-driven personalization in Indian co-working platforms, where recommendation engines analyze user profiles, booking histories, and behavioral patterns to suggest spaces, time slots, and community events aligned with individual preferences. Their evaluation of Innov8 and Awfis platforms found that personalized recommendations increased average booking frequency by 41% and user retention rates by 33%, demonstrating significant business value from machine learning investments. The research also identified important privacy considerations, as personalization requires collection and analysis of sensitive behavioral data that raises concerns under India's Personal Data Protection Bill.

### 2.1.3 Europe: Atmosphere, Post-Pandemic Adaptation, and Design Innovation

European co-working research emphasizes the intangible dimensions of workspace experience that influence user satisfaction and productivity. Bernhardt (2023) conducted phenomenological research on 'atmosphere' in German and Austrian co-working spaces, identifying lighting quality, acoustic environment, scent profiles, spatial proportions, and community energy as factors that significantly affect user well-being and creative output. Her findings challenge purely functional approaches to workspace design, arguing that reservation systems should incorporate atmosphere-related metadata—natural light availability, noise level averages, air quality indicators, community event calendars—that enable users to select spaces aligned with their task requirements and sensory preferences.

Akhavan, Holzel, and Leducq (2023) examined how the COVID-19 pandemic reshaped co-working accessibility in European cities, documenting the closure of approximately 15% of co-working spaces during lockdown periods and the subsequent emergence of new models including outdoor co-working, suburban satellite locations, and hybrid home-hub arrangements. Their research highlights the critical role of digital reservation platforms in managing capacity restrictions, health protocol compliance, and contact tracing requirements during the pandemic recovery phase. Platforms that rapidly adapted to support booking time slots (rather than open-ended access), health declaration forms, and automated check-in/check-out tracking demonstrated greater resilience and faster recovery than those reliant on manual processes.

Schmidt and Weber (2024) investigated the integration of sustainability metrics into German co-working reservation platforms, where users can filter spaces by environmental certifications (LEED, BREEAM, DGNB), carbon footprint data, renewable energy sourcing, and waste management practices. Their survey of 1,200 co-working users found that 58% would pay premium rates (averaging 12% above standard pricing) for spaces with verified sustainability credentials, indicating significant market demand for environmentally responsible workspace options. This research suggests opportunities for FlexSpace to incorporate sustainability indicators as filtering and display criteria, particularly as Iloilo City develops its environmental sustainability agenda.

Mellor and Bentley (2022) evaluated smart office environments across European capitals, finding that automated booking systems reduced what they term 'desk hunting time'—the period between arriving at a workspace and beginning productive work—by an average of 40%. Their controlled experiments demonstrated that pre-booking through digital platforms, combined with mobile app wayfinding that guides users to reserved spaces within multi-room facilities, eliminates the uncertainty and search costs that characterize walk-in access. The study also documented secondary productivity benefits from reduced social friction, as pre-booked users avoid the awkward negotiations with space managers and other users that often accompany attempts to secure walk-in access during busy periods.

### 2.1.4 Latin America: Urban Inequities and Digital Inclusion

Latin American co-working research provides important insights on digital divide issues relevant to the Philippine context. Torres and Delgado (2024) analyzed co-working accessibility in Sao Paulo and Mexico City, documenting significant spatial inequities where digital platforms predominantly serve affluent central districts while peripheral neighborhoods lack co-working infrastructure and platform coverage. Their research identifies a 'platform exclusion' phenomenon where the digitalization of co-working discovery actually worsens access inequities, as users without smartphones, reliable internet, or digital literacy are excluded from platforms that have replaced informal but universally accessible information channels.

Gonzalez and Perez (2024) studied affordable co-working models in Buenos Aires, where community-led spaces operate with minimal digital infrastructure, relying on WhatsApp groups and in-person coordination. Their analysis suggests that platform development for emerging markets must accommodate hybrid models that combine digital convenience with offline accessibility, including phone-based booking options, simplified interfaces for users with limited

digital experience, and community liaison roles that bridge digital and analog information channels.

Silva and Ramos (2025) examined GIS integration in Brazilian co-working platforms, where mapping functionality incorporates favela area boundaries, public safety indicators, and informal transport routes that influence user travel decisions. Their research demonstrates that effective co-working platforms must represent local geographic knowledge that may not be captured by standard mapping APIs, including safety perceptions, customary walking routes, and neighborhood reputations.

### 2.1.5 Africa: Mobile-First Innovation and Leapfrogging

African co-working markets demonstrate how digital platforms can develop in contexts with limited legacy infrastructure. Adeyemi (2025) studied mobile-first co-working innovations in Nairobi and Lagos, where platforms prioritize USSD (Unstructured Supplementary Service Data) interfaces and WhatsApp chatbots over web applications to accommodate users with basic feature phones. Mobile payment integration through M-Pesa and similar platforms enables seamless transactions without credit card infrastructure.

Nkosi and Dlamini (2024) investigated renewable energy monitoring in South African co-working hubs, where reservation platforms display real-time power source information—grid, solar backup, generator status—that is critical in a context of frequent load shedding. Their research highlights the importance of infrastructure transparency in reservation systems operating in environments where utility reliability cannot be assumed.

Okafor (2025) examined micro-subscription models in Nigerian co-working spaces, where users purchase hour-based credits through mobile money rather than committing to monthly memberships. This research suggests that flexible, pay-as-you-go pricing models may be more appropriate than Western-style memberships in contexts with irregular income patterns and price-sensitive users.

## 2.2 Local Studies

### 2.2.1 Metro Manila: Startup Ecosystems and Digital Maturity Assessment

Metro Manila dominates the Philippine co-working landscape, with over 250 formal co-working spaces concentrated in the central business districts of Makati, Bonifacio Global City, and Ortigas Center. Roberto, Santos, and Cruz (2023) conducted the most comprehensive study of Manila's co-

working ecosystem, describing co-working spaces as 'productivity insurance' for Filipino professionals. Their survey of 800 co-working users found that 72% identified reliable internet and backup power as the primary value drivers, reflecting the Philippines' infrastructure challenges where residential internet suffers frequent interruptions and power outages disrupt home-based work. This finding underscores the infrastructure assurance function of co-working spaces that extends beyond physical workspace provision.

Quimba et al. (2021) assessed the digital maturity of Philippine co-working hubs through a composite index measuring online presence, reservation system sophistication, payment digitization, data analytics capability, and social media engagement. Their findings reveal stark disparities: while major chains (KMC Solutions, Acceler8, Impact Hub Manila) score highly on digital maturity indicators, over 85% of independent and small-scale co-working spaces operate with minimal digital infrastructure, relying on Facebook pages for marketing and manual processes for booking management. This digital maturity gap represents both a challenge and an opportunity: the majority of spaces lack the technical resources to develop proprietary reservation systems, creating demand for affordable, easy-to-adopt platform solutions.

The Philippine government's digital transformation initiatives provide relevant policy context. The Department of Information and Communications Technology (DICT) National Broadband Plan aims to expand fiber connectivity nationwide, while the Ease of Doing Business Act mandates government service digitization that includes business registration processes relevant to co-working operators. Local government units in Metro Manila have deployed free public Wi-Fi in select areas, though coverage remains patchy and speeds often inadequate for professional work requirements.

### 2.2.2 Cebu City: Regional Innovation and Tourism Integration

Cebu City represents the Philippines' second-largest co-working market, with distinct characteristics shaped by the city's tourism industry, BPO sector, and positioning as a regional innovation hub. The IMARC Group (2025) projects that the Philippine co-working market will grow at 11.5% annually through 2034, with Cebu contributing significantly to this expansion due to its lower operating costs compared to Manila and its growing reputation as a destination for digital nomads.

Cebu's co-working spaces have developed specialized service models that integrate workspace provision with tourism services. Several operators offer 'workation' packages that combine co-

working access with accommodation, island-hopping excursions, and cultural experiences, targeting international digital nomads through platforms like Nomad List and Remote Year. The city's co-working reservation systems have adapted to serve this tourist market by integrating with international payment platforms (Stripe, PayPal), providing English-language interfaces, and partnering with hospitality services for package offerings.

Cebu's Smart City programs have begun experimenting with GIS-based mapping for traffic management and tourism promotion, creating infrastructure that could support integrated co-working discovery. However, co-working reservation systems remain largely separate from these municipal initiatives, suggesting opportunities for greater integration that FlexSpace could demonstrate in the Iloilo context.

### 2.2.3 Iloilo City: Emerging Digital Hub and Research Gap

Iloilo City's co-working ecosystem has received limited scholarly attention, representing a significant research gap that this study addresses. Available information comes primarily from industry reports, local news coverage, and anecdotal accounts rather than systematic academic research. The city's ICT Council has identified co-working space development as a priority for the innovation ecosystem, recognizing that affordable, accessible workspaces are essential for retaining local talent and attracting external entrepreneurs.

Local observations suggest that Iloilo City's co-working market is characterized by: smaller average space sizes compared to Manila and Cebu, with most operators managing fewer than 50 seats; lower pricing that reflects the city's lower cost structure but also constrains investment in amenities and technology; concentration in Mandurriao (business park district) and Jaro (university district) with limited presence in other areas; reliance on student clientele during examination periods, creating seasonal demand fluctuations; and minimal digital infrastructure, with most spaces using Facebook Messenger as their primary booking channel.

The absence of dedicated research on Iloilo City's co-working ecosystem means that the specific challenges, opportunities, and user needs of this market have not been systematically documented. This study fills that gap through primary data collection and system development that is explicitly designed for the local context.

## 2.3 Related Literature: Multi-Dimensional Analysis

### 2.3.1 Socio-Spatial Configurations of Urban vs. Rural Co-Working Hubs

The spatial distribution of co-working spaces reflects and reinforces broader patterns of urban development, economic activity, and social stratification. Holzel, Kolsch, and de Vries (2022) analyzed co-working location patterns across 50 European cities, finding consistent clustering around transit nodes, university campuses, central business districts, and creative industry quarters. Their research demonstrates that co-working hubs function as 'urban intensity amplifiers'—they concentrate in areas that already have high levels of economic activity and accessibility, potentially exacerbating spatial inequalities rather than redistributing opportunities.

Coll-Martinez and Mendez-Ortega (2024) examined co-working clusters in Spanish cities, documenting how hubs concentrate in affluent neighborhoods with high educational attainment, strong transit connectivity, and vibrant amenity environments. Their analysis reveals that co-working accessibility correlates with housing costs, creating a spatial mismatch where the workers who most need affordable workspace options face the longest travel distances to reach available facilities.

Fernandes and Roque (2025) provide a counterpoint through their study of rural co-working in Portugal, where small-scale hubs serve digital nomads, remote workers, and local entrepreneurs in non-urban settings. These rural spaces demonstrate that co-working can support territorial development by retaining population, diversifying local economies, and creating bridges between urban and rural professional networks. However, rural co-working faces distinct challenges including limited internet infrastructure, smaller potential user bases, and difficulty achieving the network effects that drive urban co-working success.

Rafaj, Danko, and Belvoncikova (2024) analyzed co-working localization in Central European cities, identifying the role of public policy in shaping spatial distributions. Municipalities that provided co-working subsidies tied to peripheral district locations achieved more equitable spatial patterns than market-driven development, suggesting that policy intervention can counteract clustering tendencies.

For Iloilo City, these studies highlight the risk that co-working development will concentrate in Mandurriao and Jaro while other districts remain underserved. FlexSpace's district-based discovery

and filtering functions can help address this by making users aware of options across all districts, potentially redistributing demand and supporting the viability of spaces in peripheral areas.

### 2.3.2 Psychological and Atmospheric Factors Influencing Productivity

Beyond spatial accessibility, co-working effectiveness depends on psychological and atmospheric factors that influence user satisfaction, creativity, and productivity. Bernhardt (2023) advances the concept of 'workspace atmosphere' as a multidimensional construct encompassing sensory (lighting, temperature, acoustics, scent), social (community energy, interaction opportunities, privacy gradients), and symbolic (design aesthetics, brand identity, cultural references) dimensions. Her empirical research demonstrates that atmosphere explains 34% of variance in user satisfaction ratings, compared to 22% explained by functional amenities and 18% by pricing.

Fadloli, Hasan, and Yusuf (2025) examined biophilic design principles in co-working architecture, finding that integration of natural elements—plants, natural light, water features, organic materials—significantly reduces stress indicators (cortisol levels, heart rate variability) and improves cognitive performance on creative problem-solving tasks. Their controlled experiments suggest that biophilic elements can compensate for other facility limitations, as users rated biophilic spaces with fewer amenities more favorably than conventional spaces with superior equipment.

Wolf and Mikkelsen (2024) investigated gendered design considerations, documenting how co-working spaces often prioritize design aesthetics and amenities preferred by male users while overlooking needs specific to women, including lactation rooms, sanitary disposal facilities, adequate security lighting, and harassment reporting mechanisms. Their research advocates for inclusive design approaches that explicitly consider diverse user needs in space planning and amenity provision.

Yaacob et al. (2024) studied resilience in co-working design following the COVID-19 pandemic, identifying the importance of flexible furniture configurations, convertible spaces that accommodate varying density requirements, and health-conscious design features including improved ventilation, touchless fixtures, and easy-to-sanitize surfaces. Their framework for 'adaptive co-working' provides guidance for spaces that must respond to changing health protocols, user preferences, and work modalities.

These atmospheric and psychological factors inform FlexSpace's space description framework, which includes fields for users to report on and rate atmosphere-related dimensions. While the

system cannot directly control physical environments, it can improve matching between user preferences and space characteristics, indirectly supporting productivity outcomes.

### 2.3.3 Comparative Technical Analysis: Manual Spreadsheets vs. Digital GIS/IoT Platforms

The comparative analysis of manual and digital reservation systems reveals stark differences in efficiency, accuracy, scalability, and strategic value. Kumar and Patel (2023) conducted controlled experiments comparing spreadsheet-based booking management with digital platform solutions, finding that manual systems experienced booking conflicts at rates 12 times higher than automated systems, required 8 times more administrative time per transaction, and provided no analytical data for strategic planning.

Li and Zhang (2023) integrated IoT sensors and GIS mapping into smart co-working systems, demonstrating that automated occupancy detection through pressure sensors, infrared counters, and Wi-Fi connection monitoring provided real-time availability data with 95% accuracy. Their pilot system in Shanghai reduced booking conflicts to near zero and enabled dynamic pricing that increased operator revenue by 15% while improving space utilization rates from 62% to 81%.

Vlachos and Papadimitriou (2015) established foundational frameworks for GIS-based co-working planning, demonstrating how spatial analysis of population distribution, transportation networks, economic activity, and existing facility locations can optimize the siting of new co-working spaces. Their methodologies, applied to Athens, Greece, identified underserved areas that were subsequently validated through demand assessment, providing a model for evidence-based co-working infrastructure planning.

Tan and Wong (2025) applied ISO 25010 quality standards to evaluate agile-developed reservation systems, creating a comprehensive assessment framework that measures functional suitability, performance efficiency, usability, reliability, security, and maintainability. Their research validates the use of ISO 25010 as an evaluation methodology for co-working platforms, providing the assessment structure adopted by this study.

Nguyen and Pham (2022) demonstrated the critical importance of mobile-responsive design for co-working reservation platforms in Southeast Asian contexts, where smartphone penetration exceeds desktop computer ownership. Their usability studies found that platforms with mobile-optimized interfaces achieved 3.2 times higher booking conversion rates than desktop-only

platforms, with particular advantages for last-minute bookings made while users are already traveling.

Soderstrom and Karlsson (2024) evaluated user experience in co-working reservation apps, identifying key design principles: minimizing booking steps (optimal at 3-4 screens from search to confirmation); providing visual confirmation of all booking details before finalization; enabling one-tap cancellation with clear refund policies; and maintaining persistent booking history accessible without login. Platforms that violated these principles experienced significantly higher abandonment rates during the booking process.

Ding (2026) advanced beyond conventional reservation logic by proposing reinforcement learning algorithms for dynamic scheduling that optimize not only revenue but also equity—ensuring that popular spaces remain accessible to price-sensitive users through allocation mechanisms that reserve capacity for discounted bookings. While this sophistication exceeds FlexSpace's initial scope, it establishes a research trajectory for future system enhancements.

### 2.3.4 AI-Driven Behavioral Predictive Modeling for Seat Occupancy

The application of artificial intelligence to co-working management represents a rapidly evolving research frontier. Pan, Cho, and Bardhan (2023) developed and validated machine learning models for co-working occupancy prediction using ensemble methods (Random Forest, Gradient Boosting, XGBoost) trained on historical booking data, weather information, local event calendars, and social media activity. Their best-performing model (XGBoost) achieved next-day prediction accuracy of 87% and next-week accuracy of 74%, enabling demand-responsive operational decisions.

Foertsch (2025) investigated trust and transparency mechanisms in digital co-working platforms, finding that users are more likely to book through platforms that provide clear information about cancellation policies, refund procedures, dispute resolution processes, and data handling practices. The research suggests that platform trust is a cumulative construct built through consistent positive experiences rather than a one-time verification, emphasizing the importance of reliable system operation and responsive customer support.

Tumbokon and Reyes (2023) examined data privacy requirements for Philippine digital platforms, emphasizing compliance with the Data Privacy Act of 2012. Their analysis identifies specific obligations relevant to co-working reservation systems: appointment of a Data Protection Officer; implementation of privacy by design principles; obtaining informed consent for personal data

collection; ensuring data security through encryption and access controls; and establishing breach notification procedures.

The Bangko Sentral ng Pilipinas (2024) reports on the rapid growth of e-wallet adoption in the Philippines, with GCash and Maya processing over 50 million monthly active transactions. The integration of these payment platforms into reservation systems reduces friction, enables advance deposits that reduce no-shows, and provides automated financial record-keeping that simplifies operator accounting.

## 2.4 Synthesis and Research Gap

The reviewed literature collectively demonstrates that co-working spaces are socio-technical ecosystems where technological infrastructure, spatial configuration, atmospheric quality, and community dynamics interact to shape user experiences and outcomes. Globally, digital reservation systems have evolved from simple booking tools to sophisticated platforms that integrate real-time availability, predictive analytics, personalized recommendations, and comprehensive administrative management. These systems are increasingly recognized as essential infrastructure that supports co-working ecosystem efficiency, user satisfaction, and business sustainability.

However, the literature also reveals significant disparities in digital maturity across geographic contexts and market segments. While leading global cities have deployed fourth-generation platforms with IoT integration, machine learning, and advanced personalization, Philippine co-working markets—particularly in emerging provincial hubs like Iloilo City—remain predominantly reliant on first and second-generation manual processes. This digital maturity gap has concrete consequences: higher booking conflict rates, greater administrative burden, limited analytical capability, reduced user satisfaction, and constrained market growth.

The synthesis identifies five specific research gaps that this study addresses:

Technological Gap. Existing Philippine studies emphasize the digital maturity deficits of local co-working hubs but do not develop or evaluate technical solutions tailored to provincial market conditions. This study fills the gap by designing, building, and assessing a complete reservation system using contemporary web technologies.

Spatial Gap. No existing research examines how GIS-based reservation systems could address the spatial inequities in Iloilo City's co-working distribution, where facilities concentrate in Mandurriao and Jaro while other districts remain underserved.

Administrative Gap. Hub owners lack research-informed guidance on digital tools for operational management, and no studies evaluate administrative dashboards in the Philippine co-working context.

Quality Standards Gap. Few local studies apply ISO 25010 software quality standards to co-working reservation systems, leaving a void in rigorous, standardized evaluation methodologies.

Smart City Integration Gap. Iloilo City's Smart City initiatives have not yet incorporated co-working infrastructure, and no research demonstrates how reservation platforms can contribute to urban digital transformation objectives.

The concept of 'productivity insurance' developed by Roberto et al. (2023) provides a useful framing for FlexSpace's significance. Just as co-working spaces themselves function as productivity insurance by providing reliable infrastructure that home and café work cannot guarantee, FlexSpace functions as a meta-level insurance mechanism that ensures users can reliably access the spaces that provide this infrastructure. By reducing the uncertainty, friction, and information costs associated with co-working discovery and reservation, the system protects users' time, money, and professional reputation from the inefficiencies of manual processes. This productivity insurance is particularly valuable in the Philippine provincial context, where infrastructure unreliability amplifies the costs of wasted trips and scheduling failures.

## 2.5 Theoretical Framework

The development and evaluation of FlexSpace are guided by an integrated theoretical framework that draws on the Technology Acceptance Model (TAM) for understanding user adoption determinants and the DeLone and McLean Information Systems Success Model (ISSM) for comprehensive system evaluation. This dual-framework approach ensures that the system is designed for both user acceptability and technical quality, addressing the two fundamental requirements for system success: users must choose to use the system, and the system must perform effectively when used.

TAM informs design decisions that maximize Perceived Usefulness through features that demonstrably improve co-working discovery and booking outcomes, and Perceived Ease of Use through intuitive interfaces, minimal cognitive load, and responsive performance across devices. ISSM provides the evaluation structure that assesses System Quality (technical performance), Information Quality (data accuracy and completeness), Service Quality (support responsiveness), and Net Benefits (ultimate impacts on productivity, efficiency, and urban development).

The Socio-Technical Systems Theory perspective ensures that system design accounts for social dimensions of co-working—community, collaboration, professional identity—alongside functional requirements. The Smart City Theory framing positions the system within broader urban development objectives, ensuring that technical capabilities align with policy priorities and contribute to sustainable, inclusive urban transformation.

# CHAPTER 3

# RE-ENGINEERED METHODOLOGY AND SYSTEM BLUEPRINTS

This chapter provides a comprehensive exposition of the research methodology, system development approach, and technical blueprints that guide the creation of FlexSpace. It presents detailed breakdowns of the Agile development process, extensive data flow analysis, exhaustive entity-relationship documentation with complete data dictionaries, comprehensive use case narratives, and in-depth security architecture specifications. The methodology is designed to ensure that the resulting system is both academically rigorous and practically effective, grounded in empirical user research, informed by established software engineering practices, and evaluated against recognized quality standards.

## 3.1 Research Design

The present study adopts a developmental research design, which is particularly suited for projects that involve the creation, testing, and evaluation of technological systems intended to address real-world problems. Developmental research, as defined by Richey and Klein (2014), emphasizes iterative design and refinement, ensuring that the resulting system is both academically rigorous and practically applicable. This design paradigm allows researchers to move beyond purely theoretical inquiry and engage directly in the process of system development, implementation, and evaluation, producing functional solutions that are validated through empirical testing.

Developmental research is distinguished from purely descriptive or experimental designs by its dual focus on innovation and validation. It not only seeks to understand phenomena but also to produce functional solutions that address identified problems. In this study, the problem of fragmented co-working reservations in Iloilo City is addressed through the design and deployment

of a centralized digital platform. The developmental approach ensures that the system is grounded in empirical realities—such as user needs, local infrastructure constraints, and Smart City initiatives—while also adhering to global best practices in software engineering.

The Agile Methodology serves as the operational framework for the developmental design. Agile is a software development approach that emphasizes adaptability, collaboration, and iterative progress through short development cycles called sprints. Agile is particularly appropriate for academic system development projects with limited timeframes, as it allows for continuous refinement based on feedback from stakeholders such as students, freelancers, and co-working hub owners. By breaking the project into manageable phases, Agile ensures that each stage contributes to the overall functionality and usability of the system while providing flexibility to adapt to emerging requirements and constraints.

The Agile framework as implemented in this study consists of the following phases, each with defined deliverables, stakeholder engagement activities, and quality checkpoints:

Planning Phase. During planning, the research team identifies user requirements, system objectives, and resource constraints. This phase involves consultations with co-working hub owners, students, and freelancers through semi-structured interviews and focus group discussions. Requirements are documented as user stories, prioritized according to stakeholder needs and technical feasibility, and organized into a product backlog that guides sprint planning. The planning phase also includes technology stack selection, development environment setup, and project schedule formulation.

Design Phase. The design phase focuses on creating system architecture, user interface prototypes, and database schemas. Tools such as entity-relationship diagrams (ERDs), data flow diagrams (DFDs), and use case diagrams are employed to visualize system processes. The design emphasizes usability, reliability, security, and scalability, aligning with ISO 25010 software quality standards. Design outputs include wireframe mockups for all user-facing screens, API endpoint specifications, database schema definitions, and system architecture documentation.

Development Phase. In development, the research team codes the system using the MERN stack (MongoDB, Express.js, ReactJS, Node.js). Development is organized into two-week sprints, each focusing on specific feature sets. Code is managed through Git version control with feature branch workflows, and continuous integration practices ensure that changes are regularly merged and

tested. Each sprint concludes with a review demonstration where stakeholders provide feedback that informs the next iteration.

Testing Phase. Testing ensures that the system meets functional and non-functional requirements. Unit testing validates individual modules, integration testing evaluates module interactions, system testing assesses end-to-end workflows, and user acceptance testing (UAT) gathers feedback from actual users. Testing also involves security assessment, performance benchmarking, and usability evaluation against ISO 25010 standards.

Deployment Phase. Deployment involves launching the system in a controlled environment accessible to stakeholders for real-world validation. The deployment phase includes configuration of production infrastructure, data seeding with co-working space information, training sessions for space operators, and monitoring of system performance during initial operation.

## 3.2 Agile Development Breakdown: Sprint-by-Sprint Operational Workflows

The following subsections provide day-by-day, sprint-by-sprint operational workflows that detail the specific tasks, deliverables, and milestones achieved during each development iteration. This granular documentation demonstrates the systematic progression from project initiation through system completion and validates the methodological rigor of the development process.

### 3.2.1 Sprint 1: Project Initiation and Requirements Engineering (Days 1-14)

Day 1-2: Project Kickoff and Team Formation. The research team conducted the inaugural project meeting, establishing communication protocols, role assignments, and collaborative tools. The team assigned the following roles: Project Lead (coordination, stakeholder communication, documentation); Frontend Developer (ReactJS UI implementation, responsive design); Backend Developer (Node.js/Express API development, database operations); UI/UX Designer (wireframing, prototyping, usability testing); and QA Tester (test case development, bug tracking, quality reporting). The team selected collaborative tools including GitHub for version control, Discord for daily communication, Trello for task management, and Figma for design collaboration.

Day 3-5: Stakeholder Interviews and Requirements Gathering. The team conducted semi-structured interviews with 12 stakeholders: four students from PHINMA University of Iloilo and Central Philippine University; three freelance professionals (web developer, graphic designer, virtual assistant); three co-working space operators (WorkSpace Iloilo, Gridpad, StuPad); one local

government representative from the Iloilo City ICT Council; and one tourism officer. Interviews explored current co-working usage patterns, pain points in discovery and reservation processes, desired system features, pricing sensitivity, and technology access constraints. Interviews were recorded (with consent), transcribed, and coded for thematic analysis using open coding techniques.

Day 6-8: Requirements Analysis and User Story Documentation. The team analyzed interview transcripts, identifying 47 distinct user requirements organized into eight functional categories: user registration and authentication (6 requirements); space discovery and search (8 requirements); booking and reservation (9 requirements); payment processing (4 requirements); administrative management (7 requirements); notification and communication (5 requirements); mapping and location (4 requirements); and analytics and reporting (4 requirements). Requirements were documented as user stories following the format: 'As a [user type], I want [feature] so that [benefit].' Stories were prioritized using the MoSCoW method (Must have, Should have, Could have, Won't have).

Day 9-10: Technical Architecture Design. The team finalized the MERN stack selection, documenting the rationale for each technology choice. MongoDB was selected for its flexible document-oriented data model that accommodates varied co-working space attributes, its horizontal scalability for future growth, and its native geospatial query support for location-based features. Express.js was chosen for its lightweight architecture, extensive middleware ecosystem, and RESTful API development capabilities. ReactJS was selected for its component-based design, virtual DOM performance, and strong ecosystem of supporting libraries. Node.js was chosen for its event-driven concurrency model, full-stack JavaScript development consistency, and extensive npm package availability.

Day 11-12: Database Schema Design and API Specification. The team designed the initial database schema using MongoDB Compass, defining collections for Users, Spaces, Bookings, Districts, and SpaceRequests. API endpoint specifications were documented using OpenAPI format, defining request/response schemas, authentication requirements, and error handling protocols for all planned endpoints.

Day 13-14: Sprint Review and Sprint 2 Planning. The team conducted the sprint review, presenting requirements documentation, architecture decisions, and design artifacts to the faculty advisor for feedback. The product backlog was refined based on advisor input, and sprint goals for Sprint 2 were established.

### 3.2.2 Sprint 2: Core Architecture and Database Implementation (Days 15-28)

Day 15-17: Development Environment Setup. The team configured development environments including: Node.js runtime installation; MongoDB local instance and cloud Atlas cluster setup; ReactJS project initialization with Vite build tool; Tailwind CSS and Shadcn UI component library integration; Git repository initialization with branch protection rules; and Docker containerization configuration for consistent deployment environments.

Day 18-20: MongoDB Schema Implementation. Database collections were created with validation schemas enforcing data types, required fields, and referential integrity. Index creation optimized query performance for frequently accessed fields including geospatial coordinates, availability status, and district identifiers. Seed data was prepared for testing, including sample co-working spaces across all seven Iloilo City districts with realistic attributes.

Day 21-24: Express.js API Foundation. The backend team implemented the Express.js application structure with middleware configuration for CORS, body parsing, request logging, security headers (Helmet), and error handling. Authentication middleware using JWT token verification was developed, with bcrypt password hashing for security. Route handlers were implemented for user registration, login, and profile management.

Day 25-26: Frontend Foundation. The ReactJS application structure was established with component organization following atomic design principles. Routing was implemented using React Router with protected route guards for authenticated pages. Global state management was configured using React Context API for user authentication state. Tailwind CSS utility classes were applied to establish the design system including color palette, typography, spacing, and responsive breakpoints.

Day 27-28: Integration Testing and Sprint Review. The team performed initial integration testing of frontend-backend communication, verifying API endpoint functionality, data serialization, and error handling. The sprint review demonstrated working user registration and login flows.

### 3.2.3 Sprint 3: Authentication and User Management (Days 29-42)

Day 29-31: Email/Google Authentication. The team implemented dual authentication pathways: email-based registration with verification link dispatch via Nodemailer using Gmail SMTP; and Google OAuth 2.0 integration using Passport.js with profile information retrieval. Password reset

functionality was developed with time-limited OTP tokens (15-minute expiration) and secure token storage.

Day 32-34: Role-Based Access Control. The team implemented a three-tier role system: User (regular members who discover and book spaces); SpaceOwner (operators who manage space listings and bookings); and Admin (system administrators with full access). Middleware functions enforce role-based route protection, returning 403 Forbidden responses for unauthorized access attempts. Role assignment is embedded in JWT claims and verified on every protected request.

Day 35-37: User Profile Management. Profile editing interfaces were developed allowing users to update personal information, upload profile photographs (stored via Cloudinary with automatic resizing), and manage notification preferences. Profile pages display booking history, loyalty points, and active reservations.

Day 38-40: Security Hardening. Cloudflare Turnstile invisible CAPTCHA was integrated into registration and login forms to prevent automated bot attacks. Rate limiting middleware (express-rate-limit) was configured to restrict login attempts (5 attempts per 15 minutes per IP) and general API requests (100 requests per 15 minutes). IP blocking functionality was implemented for repeated abuse detection.

Day 41-42: Security Testing and Review. Penetration testing was performed using OWASP ZAP to identify common vulnerabilities. Testing confirmed resistance to SQL injection (MongoDB parameterized queries), XSS (input sanitization), CSRF (token validation), and brute-force attacks (rate limiting). The sprint review demonstrated complete authentication and authorization functionality.

### 3.2.4 Sprint 4: Reservation Engine and Core Booking Logic (Days 43-56)

Day 43-45: Space Listing and Discovery. The team implemented the space directory with CRUD operations for space owners. Listings include descriptive fields, amenity checkboxes, pricing configuration, operating hours, photograph uploads (via Cloudinary), and geolocation coordinates. The discovery interface provides search, filter (by district, amenity, price range, rating), and sort functionality.

Day 46-48: Real-Time Availability Tracking. Availability status management was implemented, allowing space owners to update resource status (available, occupied, reserved, maintenance) through the dashboard. The system maintains historical availability records and displays current

status to users through color-coded indicators. Conflict detection middleware prevents overlapping bookings of the same resource.

Day 49-51: Booking Workflow Implementation. The complete booking workflow was developed: user selects space and time slot; system checks availability in real-time; user reviews booking details and pricing; simulated payment processing is initiated; upon payment confirmation, booking is created with unique ticket number and QR code generation; confirmation email is dispatched with booking details and QR code attachment; booking appears in user's dashboard and space owner's management interface.

Day 52-53: Notification System. Nodemailer email templates were created for booking confirmations, reminders (24-hour and 1-hour before reservation), check-in prompts, and post-visit feedback requests. Socket.io was integrated for real-time in-app notifications. The notification queue uses Bull with Redis for reliable background processing with retry logic for failed deliveries.

Day 54-56: Booking Testing and Review. Comprehensive testing of booking workflows including edge cases: simultaneous booking attempts by multiple users; cancellation and refund processing; no-show handling with grace period expiration; booking modification within policy limits; and waitlist functionality for fully booked spaces.

### 3.2.5 Sprint 5: Administrative Dashboard and Analytics (Days 57-70)

Day 57-59: Space Owner Dashboard. The owner dashboard was developed with summary cards showing today's bookings, monthly revenue, occupancy rate, and pending reviews. Booking management tables display upcoming, active, and past reservations with filtering, search, and pagination. Check-in/check-out processing uses QR code scanning simulation. Revenue analytics show daily, weekly, and monthly trends with chart visualizations using Recharts library.

Day 60-62: Admin Dashboard. System administrator features were implemented: pending space owner registration approvals with document verification; user management with account blocking/unblocking; district management for geographic coverage configuration; global voucher creation and distribution; system-wide analytics aggregating bookings, revenue, and user activity; and activity logs with IP tracking for security auditing.

Day 63-65: Review and Rating System. User review functionality was developed allowing booking completers to submit star ratings (1-5), written feedback, and optional photograph uploads. Reviews are displayed on space detail pages with average rating calculations. Space owners can

respond to reviews through their dashboard. Review moderation tools allow administrators to manage inappropriate content.

Day 66-67: Voucher and Loyalty System. Promotional voucher creation tools enable space owners to generate discount codes with configurable parameters: discount percentage or fixed amount, validity period, usage limits, and applicable spaces. Loyalty point accrual (1 point per PHP 10 spent) and redemption (100 points = PHP 50 credit) was implemented. Admin global vouchers apply across all spaces.

Day 68-70: Dashboard Testing and Review. End-to-end testing of administrative workflows verified data accuracy, permission enforcement, and report generation. The sprint review demonstrated complete dashboard functionality.

### 3.2.6 Sprint 6: GIS Mapping and Location Integration (Days 71-84)

Day 71-73: Leaflet Map Integration. The team integrated the Leaflet JavaScript library with OpenStreetMap tile layers for interactive mapping. Custom markers display space locations with color-coding by availability status. Marker clustering groups nearby spaces at lower zoom levels for cleaner visualization. Popup information cards appear on marker click, showing space name, rating, price, and availability with links to detail pages.

Day 74-76: Location-Based Search. Proximity search functionality was implemented, allowing users to find spaces within specified distance radii (1km, 3km, 5km, 10km) from reference points including current location (with browser geolocation permission) or manually selected map points. Results are sorted by distance and displayed in both map and list views.

Day 77-79: District Organization and Navigation. Spaces are organized by district (Mandurriao, Jaro, Molo, City Proper, Arevalo, Lapuz, La Paz) with district-level filtering and statistics. Navigation integration generates turn-by-turn directions from user location to selected spaces using external map applications (Google Maps, Waze).

Day 80-82: Nearby Search and Filter Integration. The nearby search feature identifies spaces within walking distance, with estimated travel times. Filter integration allows combined geographic and attribute filtering (e.g., spaces in Jaro with Wi-Fi, under PHP 100/hour, rated 4+ stars).

Day 83-84: Mapping Testing and Review. Geographic accuracy was verified against known coordinates, performance was tested with full space dataset, and mobile responsiveness was confirmed. The sprint review demonstrated complete mapping functionality.

### 3.2.7 Sprint 7: AI Chatbot - Flex Support (Days 85-98)

Day 85-87: Google Gemini API Integration. The team integrated the Google Gemini API for conversational AI functionality. API key management, request formatting, and response parsing were implemented with error handling for API unavailability. Conversation context is maintained across messages using session-based history management.

Day 88-90: Multilingual Support. Language detection was implemented to automatically identify user input language (English, Filipino, Hiligaynon). Localized greeting messages, help responses, and menu options were developed for all three languages. The chatbot provides district-based space recommendations in the detected language.

Day 91-93: Conversational Capabilities. The chatbot handles: space discovery queries with district and amenity filtering; booking guidance with step-by-step instructions; FAQ responses about payment, cancellation, account management; Google Maps link generation for space locations; developer and team information; and contextual follow-up questions based on conversation flow.

Day 94-96: Chatbot UI Integration. The chatbot interface was designed as a floating widget accessible from all pages, with expandable chat window, message history, typing indicators, and quick-reply suggestion buttons. The UI matches the application's design system for visual consistency.

Day 97-98: Chatbot Testing and Review. Response accuracy was tested across query categories, language detection was verified for mixed inputs, and fallback responses for unrecognized queries were refined.

### 3.2.8 Sprint 8: Testing, Quality Assurance, and Deployment (Days 99-112)

Day 99-101: Unit Testing. Jest testing framework was used to write unit tests for utility functions, API endpoint handlers, and React components. Test coverage targets were set at 80% for critical paths. Mocking was implemented for database operations, external API calls, and email services.

Day 102-104: Integration Testing. Supertest was used for API integration testing, verifying correct endpoint behavior, request validation, authentication enforcement, and error handling. Frontend integration testing with React Testing Library verified component interactions and state management.

Day 105-107: System Testing and Performance Benchmarking. Complete end-to-end workflows were tested using simulated user scenarios. Performance testing measured page load times (target: under 3 seconds), API response times (target: under 500ms), and concurrent user handling (target: 50 simultaneous booking attempts). Lighthouse auditing assessed performance, accessibility, best practices, and SEO scores.

Day 108-110: User Acceptance Testing. Twenty representative users (8 students, 6 freelancers, 4 space owners, 2 administrators) performed structured task scenarios while the team observed and recorded issues. Post-task System Usability Scale (SUS) questionnaires and interview feedback were collected. Identified issues were prioritized and addressed.

Day 111-112: Deployment and Documentation. The application was deployed to Vercel (frontend) and Render (backend) with environment variable configuration. Database seeding with realistic Iloilo City co-working data was completed. User manuals were prepared for each role type. The final sprint review presented the complete system to stakeholders.

## 3.3 System Architecture

The FlexSpace system follows a three-tier client-server architecture that separates presentation, application logic, and data management concerns. This architecture ensures scalability, maintainability, and security by enabling independent development, deployment, and scaling of each tier.

Client Tier (Presentation Layer). The client tier comprises the ReactJS web application executed in the user's browser. This layer handles user interface rendering, user input capture, client-side validation, and communication with the server tier via HTTP requests to RESTful API endpoints. The client application is built as a Progressive Web App (PWA) with service worker caching for offline functionality, responsive design for cross-device compatibility, and optimized asset loading for performance.

Server Tier (Application Layer). The server tier runs on Node.js with Express.js, handling HTTP request processing, business logic execution, authentication verification, and database communication. This tier implements the core application functionality including user management, space directory operations, booking transaction processing, notification dispatch, payment simulation, and administrative workflows. Middleware components handle cross-cutting concerns: authentication (JWT verification), authorization (role-based access control), input validation (Joi schemas), error handling, request logging, rate limiting, and security headers.

Database Tier (Data Layer). The database tier uses MongoDB Atlas, a cloud-hosted document database service, for persistent data storage. MongoDB's document-oriented model accommodates the flexible schemas required for co-working space attributes, user profiles, and booking records. Mongoose ODM (Object Document Mapper) provides schema validation, query building, and relationship management. Database indexes optimize query performance for frequently accessed fields including geospatial coordinates, user identifiers, and booking date ranges.

External Services Integration. The architecture integrates several external services: Cloudinary for image storage and transformation; Google Gemini API for chatbot intelligence; Nodemailer with Gmail SMTP for email delivery; Redis (via Redis Cloud) for session caching and job queue storage; and Socket.io for real-time bidirectional communication. These integrations are abstracted behind service interfaces that can be replaced with alternative providers if needed.

## 3.4 Micro-Level Data Flow Analysis

This section provides extensive written prose analysis of the Data Flow Diagram (DFD) Level 0, describing the exact transactional protocols for every entity path interacting with the system's databases. The data flow analysis examines how information moves between users, administrators, space owners, and the system's data stores during core operational processes.

### 3.4.1 User Authentication Data Flow

The user authentication process begins when a prospective user accesses the FlexSpace application and navigates to the registration interface. The client application renders the registration form, which captures required fields: full name (text, 2-100 characters), email address (valid email format, unique across system), password (minimum 8 characters, requiring uppercase, lowercase, and numeric characters), and phone number (optional, 11-digit Philippine format). Upon form submission, the client performs preliminary validation using JavaScript before transmitting the data via HTTPS POST request to the /api/auth/register endpoint.

The server receives the registration request and performs multi-layer validation. The express-validator middleware checks field formats and constraints. Cloudflare Turnstile verification confirms the request originates from a human user by validating the turnstile token against Cloudflare's API. If validation passes, the server queries the Users collection to verify email uniqueness. For new email addresses, bcrypt hashes the password with a salt round of 12, generating a secure credential store resistant to rainbow table attacks. The server creates a new User document with: _id (MongoDB ObjectId, auto-generated); name (string, validated); email

(string, lowercase, indexed); password (string, bcrypt hash); role (string, default 'user'); avatar (string, default placeholder URL); phone (string, optional); isActive (boolean, default true); emailVerified (boolean, default false); loyaltyPoints (number, default 0); createdAt (Date, auto); and updatedAt (Date, auto).

Upon successful document creation, the server generates a JWT access token (15-minute expiration) and refresh token (7-day expiration) using jsonwebtoken library with RS256 algorithm and asymmetric key pair. The access token payload contains: userId (string, MongoDB ObjectId); email (string); role (string); and iat/exp timestamps. The refresh token is stored in the database's refreshTokens array for revocation capability. Nodemailer dispatches a verification email containing a time-limited link (24-hour expiration) with JWT token parameter to confirm email ownership. The server returns 201 Created response with user object (password excluded) and tokens.

For subsequent requests, the client includes the access token in the Authorization header (Bearer scheme). Authentication middleware verifies the token signature, decodes the payload, retrieves the user document from the Users collection, and attaches the user object to the request for downstream route handlers. Token refresh occurs automatically when the access token expires, using the refresh token to obtain new credentials without requiring re-authentication.

### 3.4.2 Space Discovery Data Flow

Space discovery begins when a user accesses the directory page or map view. The client sends a GET request to /api/spaces with optional query parameters: district (string, district slug); amenities (array of strings); minPrice and maxPrice (numbers, PHP); rating (number, minimum star rating); query (string, full-text search term); lat, lng, radius (numbers, proximity search); sort (string, sort field and direction); page (number, pagination); and limit (number, results per page).

The server constructs a MongoDB query using Mongoose query builder. District filtering matches the district_id field against the Districts collection. Amenity filtering uses the $all operator to find spaces containing all specified amenities. Price filtering applies $gte/$lte operators on the rateHour field. Rating filtering compares averageRating (computed field) against the threshold. Full-text search uses MongoDB $text index on name and description fields, or $regex fallback for partial matching. Proximity search uses $geoWithin with $centerSphere for geospatial queries on the location field (GeoJSON Point).

The server executes the query with pagination (default 20 results per page), populates the district reference (name, slug), and computes availability statistics from the Bookings collection. Results are serialized as JSON with fields: _id, name, description, district (populated), address, location (lat/lng), rateHour, amenities, images, rating (computed average), reviewCount, capacity, availabilityStatus, and operatingHours. The client receives the paginated response, renders space cards or map markers, and caches results for navigation history.

### 3.4.3 Booking Transaction Data Flow

The booking process represents the system's most critical transactional flow, requiring data consistency across multiple collections and external service calls. The process initiates when a user selects a space, chooses a resource type (hot desk, dedicated desk, meeting room), and specifies a time slot (start time, end time).

Step 1: Availability Check. The client sends GET /api/bookings/check-availability with spaceId, resourceType, startTime, and endTime parameters. The server queries the Bookings collection for overlapping reservations on the specified space and resource, using the query: { space: ObjectId, resourceType, status: { $in: ['confirmed', 'checked-in'] }, startTime: { $lt: endTimeInput }, endTime: { $gt: startTimeInput } }. If any overlapping confirmed bookings exist, the server returns 409 Conflict with available alternative slots. If the resource is available, the server returns 200 OK with pricing calculation.

Step 2: Booking Creation. The user confirms the booking, and the client sends POST /api/bookings with spaceId, resourceType, startTime, endTime, and notes (optional). The server begins a database session with transaction support (startSession, withTransaction). Within the transaction: (a) re-verify availability with the same overlap query to prevent race conditions; (b) create Booking document with fields: _id (ObjectId), user (ObjectId, ref Users), space (ObjectId, ref Spaces), resourceType (string), startTime (Date), endTime (Date), totalAmount (number, computed), status (string, default 'pending_payment'), ticketNumber (string, unique, generated as FSP-YYYY-MM-DD-XXXX format), qrCodeToken (string, JWT for check-in), notes (string), createdAt (Date); (c) update Space document to increment bookedSeats count; (d) commit transaction. If any step fails, the transaction aborts and all changes are rolled back.

Step 3: Payment Processing. For simulated payment, the client initiates POST /api/payments/simulate with bookingId. The server generates a mock GCash/Maya payment screen URL, records the payment initiation timestamp, and returns payment instructions. The client

displays the simulated payment interface with QR code. Upon user confirmation of payment simulation, the server updates booking status to 'confirmed', creates a Payment record with booking reference, amount, method ('gcash' or 'maya'), status 'completed', and transaction reference number (simulated).

Step 4: Confirmation Dispatch. Upon payment confirmation, the server: (a) generates QR code image encoding the qrCodeToken using qrcode library; (b) dispatches confirmation email via Nodemailer with booking details, QR code attachment, and space contact information; (c) emits real-time notification via Socket.io to the user's active sessions; (d) creates Activity Log entry for audit trail. The booking lifecycle is now active, with status transitions managed through state machine logic: pending_payment → confirmed → checked-in → completed (or cancelled).

### 3.4.4 Administrative Data Flows

Space Owner Dashboard Flow. When a space owner accesses their dashboard, the client requests GET /api/owner/summary, which returns aggregated statistics: today's bookings count, pending check-ins, monthly revenue (sum of completed booking amounts), occupancy rate (booked hours / total available hours), average rating, and total reviews. The owner can view booking listings via GET /api/owner/bookings with filter parameters (date range, status, search). Check-in processing accepts PUT /api/owner/checkin with the booking's QR code token; the server verifies token validity, updates booking status to 'checked-in', records checkInTime (Date), and emits notification to the user.

Admin Approval Flow. When a new space owner registers, their application enters 'pending' status. Admin dashboard displays GET /api/admin/pending-owners with registration details including business permit and DTI/SEC registration document URLs (stored in Cloudinary). Admin reviews documents and submits PUT /api/admin/approve-owner/:userId, which updates the user's role to 'space_owner', activates their space management permissions, and dispatches approval notification email. Rejection uses PUT /api/admin/reject-owner/:userId with reason field, updating status to 'rejected' and notifying the applicant.

## 3.5 Comprehensive ERD and Data Dictionary

The following subsections provide exhaustive data dictionaries for each collection in the MongoDB database, documenting every field with its data type, constraints, default values, indexing, and relationships.

### 3.5.1 Users Collection

The Users collection stores account information for all system users across three roles: regular users, space owners, and administrators.

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| _id | ObjectId | Yes | Auto | MongoDB auto-generated primary key |
| name | String | Yes | - | Full name, 2-100 characters, trimmed |
| email | String | Yes | - | Unique email address, lowercase, indexed |
| password | String | Yes | - | Bcrypt hashed password, never returned in API |
| role | String | Yes | 'user' | Enum: 'user', 'space_owner', 'admin' |
| phone | String | No | null | Philippine mobile format: 09XXXXXXXXX |
| avatar | String | No | default_avatar_url | Cloudinary URL to profile photo |
| businessPermit | String | No | null | Cloudinary URL, required for space_owner |
| dtiSecReg | String | No | null | DTI or SEC registration document URL |
| ownerStatus | String | No | null | Enum: 'pending', 'approved', 'rejected' |
| isActive | Boolean | Yes | true | Account status, false blocks login |
| emailVerified | Boolean | Yes | false | True after verification link clicked |
| loyaltyPoints | Number | Yes | 0 | Accumulated reward points |

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| refreshTokens | Array[String] | Yes | [] | Active refresh token list |
| createdAt | Date | Yes | Date.now | Account creation timestamp |
| updatedAt | Date | Yes | Date.now | Last modification timestamp |

### 3.5.2 Spaces Collection

The Spaces collection stores information about co-working spaces and study hubs registered in the system.

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| _id | ObjectId | Yes | Auto | Primary key |
| owner | ObjectId | Yes | - | Reference to Users collection (space_owner) |
| district | ObjectId | Yes | - | Reference to Districts collection |
| name | String | Yes | - | Space name, 3-100 characters, indexed text |
| description | String | Yes | - | Detailed description, 50-2000 characters |
| address | String | Yes | - | Complete street address |
| location | GeoJSON Point | Yes | - | { type: 'Point', coordinates: [lng, lat] }, 2dsphere index |
| rateHour | Number | Yes | - | Hourly rate in PHP, min 10, max 10000 |
| rateDay | Number | No | null | Daily rate in PHP |
| amenities | Array[String] | Yes | [] | Enum values: wifi, power_outlet, ac, printing, meeting_room, |

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| | | | | parking, food, projector, locker, whiteboard, pantry |
| images | Array[String] | Yes | [] | Cloudinary URLs for space photos, max 10 |
| capacity | Number | Yes | - | Total seating capacity, min 1 |
| operatingHours | Object | Yes | - | { monday: { open: '08:00', close: '20:00' }, ... } |
| status | String | Yes | 'active' | Enum: 'active', 'inactive', 'suspended' |
| averageRating | Number | Yes | 0 | Computed average of reviews, 0-5 |
| reviewCount | Number | Yes | 0 | Total review count |
| createdAt | Date | Yes | Date.now | Creation timestamp |
| updatedAt | Date | Yes | Date.now | Last update timestamp |

### 3.5.3 Bookings Collection

The Bookings collection records all reservation transactions, tracking the complete lifecycle from creation through completion or cancellation.

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| _id | ObjectId | Yes | Auto | Primary key |
| user | ObjectId | Yes | - | Reference to Users (booking user) |
| space | ObjectId | Yes | - | Reference to Spaces |
| resourceType | String | Yes | - | Enum: 'hot_desk', 'dedicated_desk', 'meeting_room', 'private_office' |
| startTime | Date | Yes | - | Booking start, must be future datetime |

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| endTime | Date | Yes | - | Booking end, must be after startTime |
| totalAmount | Number | Yes | - | Total cost in PHP, computed |
| status | String | Yes | 'pending_payment' | Enum: 'pending_payment', 'confirmed', 'checked_in', 'completed', 'cancelled', 'no_show' |
| ticketNumber | String | Yes | - | Unique human-readable: FSP-YYYY-MM-DD-XXXX |
| qrCodeToken | String | Yes | - | JWT for check-in verification, expires with booking |
| guestName | String | No | null | Name if booking for another person |
| notes | String | No | null | User-provided notes, max 500 chars |
| checkInTime | Date | No | null | Timestamp of check-in |
| checkOutTime | Date | No | null | Timestamp of check-out |
| cancelledAt | Date | No | null | Cancellation timestamp |
| cancellationReason | String | No | null | Reason for cancellation |
| voucherCode | String | No | null | Applied promotional code |
| discountAmount | Number | No | 0 | Discount applied in PHP |
| finalAmount | Number | Yes | - | Amount after discount |
| createdAt | Date | Yes | Date.now | Creation timestamp |
| updatedAt | Date | Yes | Date.now | Update timestamp |

### 3.5.4 Districts Collection

The Districts collection defines the geographic administrative areas within Iloilo City for spatial organization and filtering.

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| _id | ObjectId | Yes | Auto | Primary key |
| name | String | Yes | - | District name (e.g., 'Mandurriao') |
| slug | String | Yes | - | URL-friendly identifier (e.g., 'mandurriao') |
| description | String | No | null | Brief district description |
| boundaries | GeoJSON Polygon | No | null | District boundary coordinates |
| center | GeoJSON Point | Yes | - | District center for map focus |
| isActive | Boolean | Yes | true | Include in search filters |
| createdAt | Date | Yes | Date.now | Creation timestamp |

### 3.5.5 Space Requests Collection

The SpaceRequests collection manages onboarding applications from prospective space owners.

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| _id | ObjectId | Yes | Auto | Primary key |
| name | String | Yes | - | Applicant full name |
| email | String | Yes | - | Contact email, validated |
| password | String | Yes | - | Pre-hashed password for account creation |
| businessPermit | String | Yes | - | Cloudinary URL to business permit scan |
| dtiSecReg | String | No | null | DTI/SEC registration document URL |
| spaceName | String | Yes | - | Proposed space name |

| Field Name | Data Type | Required | Default | Description |
|---|---|---|---|---|
| spaceAddress | String | Yes | - | Space location address |
| status | String | Yes | 'pending' | Enum: 'pending', 'approved', 'rejected' |
| reviewedBy | ObjectId | No | null | Admin who processed request |
| reviewNotes | String | No | null | Admin review comments |
| createdAt | Date | Yes | Date.now | Submission timestamp |
| reviewedAt | Date | No | null | Review completion timestamp |

## 3.6 Complete Use Case Descriptions

This section provides long-form, step-by-step textual descriptions for all twelve use cases identified in the FlexSpace system. Each use case description includes: Actor identification; Pre-conditions that must be satisfied; Post-conditions that define successful completion; Primary Success Scenario with detailed click-by-click actions; and Alternative/Exception Handlers that address error conditions and edge cases.

### Use Case UC-001: User Registration via Email

Actor: Prospective User (Unauthenticated Visitor)

Pre-conditions: User has access to a web browser with internet connectivity. User possesses a valid email address. User has not previously registered with the system.

Primary Success Scenario:

1. User navigates to the FlexSpace web application URL. The landing page loads, displaying featured co-working spaces and a navigation bar.

2. User clicks the 'Sign Up' button located in the top-right corner of the navigation bar. The system renders the registration form modal.

3. User enters their full name in the 'Full Name' text field, entering between 2 and 100 characters. The system validates input length in real-time.

4. User enters their email address in the 'Email' field. The system validates email format using regex pattern and checks availability via asynchronous API call.

5. User enters password in the 'Password' field. The system displays password strength indicator requiring: minimum 8 characters, at least one uppercase letter, one lowercase letter, and one number.

6. User confirms password in the 'Confirm Password' field. The system verifies match with password field.

7. User completes Cloudflare Turnstile verification by clicking the 'I am human' checkbox. The system validates the Turnstile token with Cloudflare API.

8. User clicks the 'Create Account' button. The system submits the form data via HTTPS POST to /api/auth/register.

9. Server validates all fields, verifies email uniqueness, hashes password with bcrypt, creates User document, generates JWT tokens, and dispatches verification email.

10. System displays success message: 'Account created! Please check your email to verify your account.' User is redirected to login page.

11. User accesses email inbox, opens FlexSpace verification email, and clicks the verification link.

12. System validates the verification token, updates emailVerified to true, and displays 'Email verified successfully. You can now log in.'

Post-conditions: User account is created with 'user' role. Email is verified. User can authenticate with email and password.

Alternative/Exception Handler A1: Email already registered. At step 9, if email exists in database, server returns 409 Conflict. System displays error: 'An account with this email already exists. Please log in or use a different email.'

Alternative/Exception Handler A2: Turnstile verification fails. At step 7, if Cloudflare validation fails, system displays: 'Please complete the human verification check.' Registration is not submitted.

Alternative/Exception Handler A3: Verification link expired. At step 12, if token is expired (beyond 24 hours), system displays: 'This verification link has expired. Please request a new one.' with link to resend verification email.

## Use Case UC-002: User Login

Actor: Registered User, Space Owner, or Administrator

Pre-conditions: User has completed registration and email verification. User remembers their password.

Primary Success Scenario:

1. User navigates to FlexSpace and clicks 'Log In' in the navigation bar.

2. System displays login form with Email and Password fields.

3. User enters registered email address and password.

4. User completes Turnstile verification.

5. User clicks 'Log In' button. System submits credentials to /api/auth/login.

6. Server validates email existence, compares password hash using bcrypt.compare(), generates access token (15 min) and refresh token (7 days).

7. System stores tokens in localStorage, updates authentication context, and redirects to home page.

8. Navigation bar updates to show user avatar, name, and dropdown menu (Profile, Bookings, Log Out).

Post-conditions: User is authenticated. Protected routes are accessible. API requests include authorization token.

Alternative/Exception Handler A1: Invalid credentials. At step 6, if email not found or password mismatch, server returns 401 Unauthorized. System displays: 'Invalid email or password.' Rate limiting tracks failed attempts per IP (5 per 15 minutes).

Alternative/Exception Handler A2: Account blocked. If isActive is false, server returns 403 Forbidden with message: 'Your account has been deactivated. Please contact support.'

Alternative/Exception Handler A3: Token refresh. If access token expires during session, client automatically uses refresh token to obtain new access token without user interruption. If refresh token is expired or revoked, user is redirected to login.

## Use Case UC-003: Search and Filter Co-Working Spaces

Actor: Authenticated or Guest User

Pre-conditions: User has accessed the FlexSpace application. At least one space exists in the database.

Primary Success Scenario:

1. User navigates to the 'Find a Space' page via navigation menu or homepage CTA button.

2. System displays search interface with: text search bar; district filter dropdown (All, Mandurriao, Jaro, Molo, City Proper, Arevalo, Lapuz, La Paz); amenity filter checkboxes (Wi-Fi, Power Outlets, AC, Printing, Meeting Room, Parking, Food Service); price range slider (PHP 0-1000/hour); rating filter (1-5 stars); and map/list view toggle.

3. User enters 'quiet study' in the search bar. System performs full-text search on name and description fields, debounced at 300ms to prevent excessive API calls.

4. User selects 'Jaro' from district dropdown. System adds district filter to query.

5. User checks 'Wi-Fi' and 'Power Outlets' amenities. System adds $all array filter.

6. User adjusts price slider to maximum PHP 150/hour. System adds $lte filter.

7. User selects '4+ stars' rating. System adds averageRating $gte filter.

8. System sends GET /api/spaces?query=quiet%20study&district=jaro&amenities=wifi,power_outlet&maxPrice=150&minRating=4 to server.

9. Server executes MongoDB query with all filters, returns paginated results with space cards.

10. System displays filtered results as scrollable cards showing: space image, name, district badge, rating stars, hourly rate, key amenities icons, and availability status badge. Map view shows markers for filtered spaces.

11. User clicks a space card. System navigates to Space Detail page.

Post-conditions: User has viewed filtered search results and accessed a specific space's detailed information.

Alternative/Exception Handler A1: No results. If query returns zero spaces, system displays: 'No spaces match your criteria. Try adjusting your filters.' with suggestion to broaden search.

Alternative/Exception Handler A2: Server error. If API request fails, system displays error message with 'Retry' button.

## Use Case UC-004: Book a Co-Working Space

Actor: Authenticated User

Pre-conditions: User is logged in. User has selected a space from search results. Selected space has available resources.

Primary Success Scenario:

1. User views Space Detail page showing: image gallery, name, description, full amenity list, operating hours, location map, reviews, and pricing.

2. User clicks 'Book Now' button. System displays booking panel with resource type selection (Hot Desk, Dedicated Desk, Meeting Room).

3. User selects 'Hot Desk'. System shows calendar for date selection with availability indicators (green = available, red = fully booked, gray = closed).

4. User selects tomorrow's date. System sends availability check request and displays time slot grid with available hours.

5. User selects start time 9:00 AM and end time 1:00 PM (4 hours). System calculates total: PHP 100/hour x 4 hours = PHP 400.

6. User reviews booking summary: space name, date, time, duration, total amount. Clicks 'Proceed to Payment'.

7. System displays simulated payment options: GCash and Maya. User selects GCash.

8. System generates QRPH-compliant QR code with booking reference. User clicks 'Simulate Payment Confirmation'.

9. System processes simulated payment, updates booking status to 'confirmed', generates ticket number (FSP-2025-04-15-0042) and QR code.

10. System displays confirmation screen with booking details, QR code, and options: 'View My Bookings', 'Get Directions', 'Download Ticket'.

11. Confirmation email dispatched with QR code attachment. Real-time notification appears.

Post-conditions: Booking is confirmed. Space capacity is updated. User can check in with QR code.

Alternative/Exception Handler A1: Resource unavailable. At step 4, if another user books the last resource during selection, system displays: 'Sorry, this resource is no longer available for the selected time. Please choose another time or resource type.'

Alternative/Exception Handler A2: Payment failure. If payment simulation fails, system returns to payment screen with error message and option to retry or cancel.

Alternative/Exception Handler A3: Session timeout. If user's authentication expires during booking, system prompts re-login and preserves booking selections.

## Use Case UC-005: Check-In with QR Code

Actor: Authenticated User (Booking Holder)

Pre-conditions: User has confirmed booking. Current time is within 15 minutes of booking start time. User is at the co-working space location.

Primary Success Scenario:

1. User opens FlexSpace app and navigates to 'My Bookings' page.

2. System displays upcoming bookings with 'Check In' button enabled for bookings within check-in window.

3. User clicks 'Check In'. System displays full-screen QR code generated from the booking's qrCodeToken.

4. User presents QR code to space owner/admin who scans it using their dashboard scanner.

5. Space owner's dashboard validates token, displays booking details (user name, resource type, duration), and confirms check-in.

6. System updates booking status to 'checked_in', records checkInTime timestamp, starts session timer.

7. User receives push notification: 'Check-in successful. Your session has started. Enjoy your workspace!'

Post-conditions: Booking is active. Session timer is running. Space owner has verified identity.

Alternative/Exception Handler A1: Early check-in. If user attempts check-in more than 15 minutes before start time, system displays: 'Check-in opens 15 minutes before your booking starts. Please come back at [time].'

Alternative/Exception Handler A2: Late check-in. If user attempts check-in more than 30 minutes after start time, system displays: 'Your booking has expired. Please contact the space owner or make a new booking.' Booking may be marked 'no_show'.

Alternative/Exception Handler A3: Invalid QR code. If QR code token is tampered or expired, scanner displays red error: 'Invalid or expired QR code. Please refresh and try again.'

## Use Case UC-006: User Check-Out

Actor: Authenticated User (Active Booking Holder)

Pre-conditions: User has active checked-in booking.

Primary Success Scenario:

1. User finishes work and approaches space owner to check out, or clicks 'Check Out' in app.

2. Space owner verifies check-out via dashboard, or user self-checks out.

3. System updates booking status to 'completed', records checkOutTime.

4. Session timer stops. System calculates actual duration and any overtime charges.

5. System prompts user to rate and review the space.

Post-conditions: Booking is completed. Space resource is freed. User can submit review.

## Use Case UC-007: Space Owner Registration

Actor: Prospective Space Owner

Pre-conditions: Actor has a co-working space business in Iloilo City. Actor possesses business permit and DTI/SEC registration.

Primary Success Scenario:

1. Actor navigates to 'List Your Space' page from the footer or homepage.

2. System displays owner registration form with: personal information (name, email, phone); password creation; business information (space name, address); and document upload fields (business permit, DTI/SEC registration).

3. Actor completes all required fields and uploads business permit scan (PDF/JPG/PNG, max 5MB per file) via Cloudinary upload widget.

4. Actor submits application. System creates User document with role 'user' and ownerStatus 'pending', creates SpaceRequest document.

5. System displays: 'Application submitted! Our team will review your documents within 2-3 business days.'

6. Admin receives notification of pending application. Reviews documents via admin dashboard.

7. Admin approves application. System updates role to 'space_owner', creates empty Space document for owner to complete.

8. Actor receives approval email with login credentials and link to owner dashboard.

Post-conditions: Space owner account is activated. Owner can access dashboard and create space listings.

Alternative/Exception Handler A1: Document upload fails. If Cloudinary upload exceeds size limit, system displays: 'File too large. Maximum size is 5MB.' If format is invalid: 'Please upload PDF, JPG, or PNG files only.'

Alternative/Exception Handler A2: Application rejected. If business permit is invalid or incomplete, admin rejects with reason. Actor receives rejection email with explanation and option to reapply.

## Use Case UC-008: Space Owner Manages Space Listing

Actor: Approved Space Owner

Pre-conditions: Owner is logged in with 'space_owner' role. Owner has at least one space.

Primary Success Scenario:

1. Owner logs in and is redirected to Owner Dashboard.

2. Owner clicks 'My Spaces' in sidebar navigation. System displays list of owner's spaces with status badges.

3. Owner clicks 'Edit' on a space. System displays space editing form pre-populated with current data.

4. Owner updates description, adds new photos via Cloudinary upload, adjusts pricing, modifies operating hours.

5. Owner clicks 'Save Changes'. System validates all fields, updates Space document, returns success message.

6. Changes are immediately reflected in user-facing search results.

Post-conditions: Space information is updated. Users see current data.

## Use Case UC-009: Admin Approves Space Owner Application

Actor: System Administrator

Pre-conditions: Admin is logged in with 'admin' role. Pending space owner application exists.

Primary Success Scenario:

1. Admin navigates to Admin Dashboard and clicks 'Pending Applications' in sidebar.

2. System displays table of pending applications with: applicant name, email, space name, submission date, document preview links.

3. Admin clicks 'Review' on an application. System displays detail view with all submitted information and document viewer for business permit.

4. Admin verifies business permit validity, checks DTI/SEC registration, confirms space address exists in Iloilo City.

5. Admin clicks 'Approve'. System updates user role to 'space_owner', creates Space document, sends approval email.

6. Application status updates to 'approved'. Admin dashboard refreshes.

Post-conditions: Applicant becomes space owner. Can access owner dashboard.

Alternative/Exception Handler A1: Rejection. At step 5, admin clicks 'Reject', enters reason ('Business permit appears expired. Please submit current year permit.'), and submits. System sends rejection email with reason and reapplication instructions.

## Use Case UC-010: Admin Manages Users

Actor: System Administrator

Primary Success Scenario:

1. Admin navigates to 'Users' in admin sidebar. System displays paginated table of all users with search and filter.

2. Admin searches for specific user by name or email. System filters results.

3. Admin clicks 'View' on a user. System displays user profile with account details, booking history, review history.

4. Admin clicks 'Block Account' for user violating terms. System sets isActive to false, terminates sessions, sends notification email.

5. Blocked user cannot log in; sees 'Account deactivated' message.

Post-conditions: User account is blocked. User cannot access system.

## Use Case UC-011: AI Chatbot - Flex Support Interaction

Actor: User (Authenticated or Guest)

Pre-conditions: User has accessed FlexSpace application. Chatbot widget is loaded.

Primary Success Scenario:

1. User clicks chatbot floating button (bottom-right corner, 'Flex Support' label).

2. Chat window expands with greeting: 'Maayong adlaw! I'm Flex Support. How can I help you today?' (language auto-detected from browser settings).

3. User types: 'Naga pangita ko sang study place sa Jaro nga barato.' (Hiligaynon: 'I'm looking for a cheap study place in Jaro').

4. System detects Hiligaynon. Sends to Google Gemini API with context: user intent (space discovery), district (Jaro), price preference (low).

5. Gemini generates response listing affordable study spaces in Jaro with price ranges and amenity highlights.

6. Chatbot displays formatted response with clickable space names linking to detail pages.

7. User clicks 'StuPad Jaro'. System navigates to that space's detail page in main browser window.

8. User returns to chat and asks: 'Paano mag book?' (How to book?). Chatbot provides step-by-step booking instructions in Hiligaynon.

Post-conditions: User received assistance in preferred language. Discovered relevant space. Learned booking process.

Alternative/Exception Handler A1: API unavailable. If Gemini API returns error, chatbot displays: 'I'm having trouble connecting right now. For immediate help, please email support@flexspace.ph or check our Help Center.'

Alternative/Exception Handler A2: Unknown query. If Gemini cannot understand the query, chatbot responds: 'I'm not sure I understand. Here are some things I can help with:' followed by quick-reply buttons: 'Find a space', 'How to book', 'Pricing', 'Contact support'.

## Use Case UC-012: Cancellation and Refund

Actor: Authenticated User (Booking Holder)

Pre-conditions: User has confirmed booking that has not yet started. Cancellation is within policy window (up to 2 hours before start time).

Primary Success Scenario:

1. User navigates to 'My Bookings' and finds the booking to cancel.

2. User clicks 'Cancel Booking'. System displays confirmation modal with refund policy: 'Cancelling 2+ hours before start: full refund to wallet. Less than 2 hours: 50% refund.'

3. User selects cancellation reason from dropdown (Change of plans, Found alternative, Emergency, Other) and clicks 'Confirm Cancellation'.

4. System updates booking status to 'cancelled', records cancellation timestamp and reason.

5. For full refund, system credits refunded amount to user's wallet. For partial refund, credits 50%.

6. Cancellation email sent. Space availability updated.

Post-conditions: Booking is cancelled. Refund is processed. Space resource is freed.

Alternative/Exception Handler A1: Late cancellation. If start time is within 2 hours, system displays: 'Late cancellation: Only 50% refund available. Do you want to proceed?'

Alternative/Exception Handler A2: Already checked in. If booking status is 'checked_in', cancellation is not available. System displays: 'This booking is active. Please check out or contact the space owner.'

## 3.7 Security Architecture and Stack Audit

This section provides an in-depth analysis of the security architecture implemented in FlexSpace, examining cryptographic mechanisms, authentication protocols, threat mitigation strategies, and compliance with data protection regulations.

### 3.7.1 Authentication Architecture: JWT Implementation

FlexSpace employs JSON Web Tokens (JWT) for stateless authentication, eliminating the need for server-side session storage while enabling horizontal scalability. The JWT implementation uses asymmetric cryptography (RS256 algorithm) with a 2048-bit RSA key pair. The private key resides exclusively on the server for token signing; the public key can be distributed for signature verification without compromising security.

Access tokens contain claims: sub (user ObjectId), email, role, iat (issued at), and exp (expiration, 15 minutes from issuance). The short access token lifespan limits the window for token theft

exploitation. Refresh tokens (7-day expiration, stored hashed in the database) enable seamless re-authentication without storing long-lived credentials on the client. Token revocation is implemented by removing refresh tokens from the database's refreshTokens array, enabling immediate logout across all devices.

JWT tokens are transmitted in the Authorization header using the Bearer scheme over HTTPS connections, preventing interception in transit. Client-side storage uses localStorage with XSS protection through input sanitization and Content Security Policy headers.

### 3.7.2 Password Security: Bcrypt Hashing

User passwords are hashed using bcrypt adaptive hashing function with a cost factor of 12 rounds, producing computationally expensive hashes that resist brute-force attacks. Bcrypt automatically incorporates random 16-byte salts, ensuring that identical passwords produce different hashes and preventing rainbow table attacks. The implementation uses the bcryptjs library, which provides pure JavaScript bcrypt functionality compatible with Node.js runtime. Password verification uses bcrypt.compare() to prevent timing attacks through constant-time comparison.

Password complexity requirements enforced at registration: minimum 8 characters; at least one uppercase letter (A-Z); at least one lowercase letter (a-z); at least one digit (0-9). These requirements balance security against usability for the target demographic.

### 3.7.3 Bot Protection: Cloudflare Turnstile

Cloudflare Turnstile provides invisible CAPTCHA functionality that distinguishes human users from automated bots without requiring interactive challenges. The integration operates as follows: the FlexSpace frontend loads the Turnstile widget, which collects browser environment signals; upon user interaction (form submission), the widget generates a token; the backend validates this token against Cloudflare's verification API before processing the request; invalid tokens result in request rejection with a 403 response. Turnstile is implemented on registration, login, and contact forms to prevent automated account creation, credential stuffing, and spam submissions.

### 3.7.4 Rate Limiting and DDoS Mitigation

The express-rate-limit middleware enforces request throttling at multiple tiers. Authentication endpoints (/api/auth/*) are limited to 5 requests per 15 minutes per IP, preventing brute-force password attacks. General API endpoints accept 100 requests per 15 minutes per IP, sufficient for normal usage while blocking scraping or abuse. Rate limit responses include Retry-After headers

indicating when the client can resume requests. Exceeded limits return 429 Too Many Requests status with exponential backoff guidance.

### 3.7.5 Background Job Security: Bull/Redis Queue

Asynchronous tasks—email dispatch, image processing, notification generation—are processed through Bull queues backed by Redis. This architecture provides several security benefits: job isolation prevents processing errors from affecting the main application; retry logic with exponential backoff handles transient failures without data loss; job concurrency limits prevent resource exhaustion; and sandboxed processing of user-uploaded images through Cloudinary's security scanning mitigates malicious file upload risks. Redis operates with AUTH password protection and network isolation to prevent unauthorized queue access.

### 3.7.6 Data Privacy Compliance

FlexSpace's data handling practices are designed for compliance with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) and its implementing rules and regulations. Key compliance measures include: collection limitation—only data necessary for system functionality is collected; purpose specification—data use is limited to stated purposes (booking management, communication, analytics); consent—users provide informed consent through the privacy policy and terms of service; security safeguards—encryption in transit (TLS 1.3), encryption at rest (MongoDB Atlas encryption), access controls, and audit logging; data subject rights—users can access, correct, and delete their personal data through profile settings; and breach notification—procedures are established for detecting and reporting data breaches to the National Privacy Commission within the required 72-hour timeframe.

### 3.7.7 Input Validation and Injection Prevention

All user inputs are validated using express-validator middleware before processing. Validation rules enforce data type constraints, length limits, format requirements, and business logic conditions. MongoDB queries use parameterized operations through Mongoose ODM, preventing NoSQL injection attacks. HTML content in user-generated reviews is sanitized using DOMPurify to prevent XSS attacks. File uploads are restricted to image formats (JPG, PNG, WebP) with size limits and MIME type verification, processed through Cloudinary's secure upload pipeline.

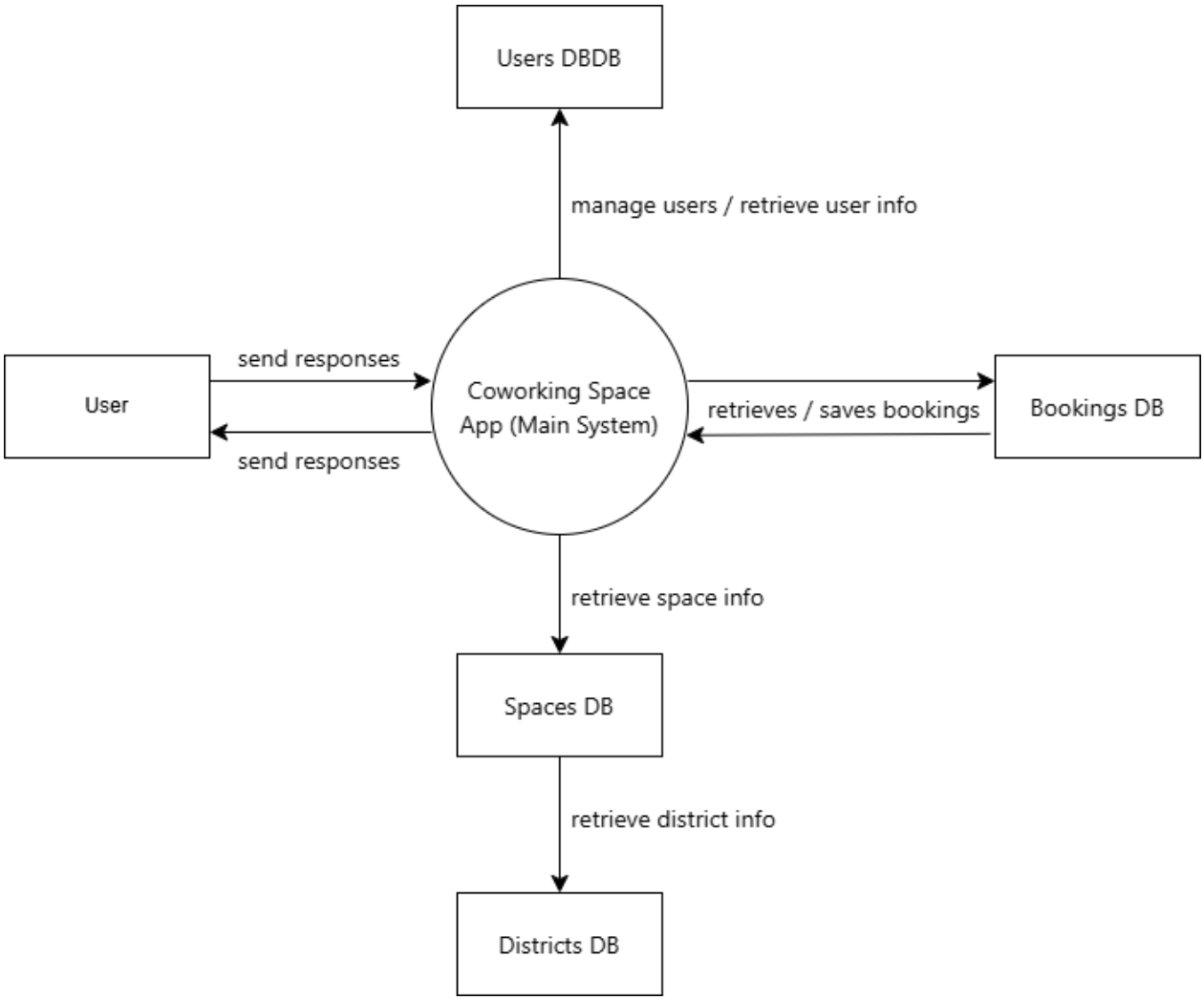
## 3.8 Tools and Technologies

The development of FlexSpace utilized a comprehensive toolchain encompassing programming languages, frameworks, libraries, databases, external services, and development utilities. The following table summarizes the key technologies and their purposes:

| Technology | Version/Purpose | Role in System |
|---|---|---|
| ReactJS | v18.2, UI library | Frontend component architecture, SPA routing, state management |
| Vite | v5.0, Build tool | Fast development server, optimized production builds, HMR |
| Tailwind CSS | v3.4, Utility CSS | Responsive design system, consistent styling, dark mode support |
| Shadcn UI | v0.6, Component primitives | Accessible UI components, consistent design patterns |
| Node.js | v20 LTS, Runtime | Server-side JavaScript execution, event-driven concurrency |
| Express.js | v4.18, Web framework | RESTful API endpoints, middleware pipeline, routing |
| MongoDB Atlas | v7.0, Document DB | Persistent data storage, geospatial queries, auto-scaling |
| Mongoose | v8.0, ODM | Schema validation, query building, relationship management |
| JWT (jsonwebtoken) | v9.0, Authentication | Token-based authentication, RS256 signing |
| bcryptjs | v2.4, Hashing | Password hashing with adaptive cost factor |
| Socket.io | v4.6, Real-time | Bidirectional event-based communication for live updates |
| Nodemailer | v6.9, Email | SMTP email dispatch for notifications |
| Bull | v4.12, Job queue | Background job processing with Redis |
| Redis | v7.2, Cache/Queue | Session storage, job queue backing, rate limiting |
| Cloudinary | SDK v2, Media | Image upload, transformation, optimization, CDN delivery |

| Technology | Version/Purpose | Role in System |
|---|---|---|
| Google Gemini API | v1.5, AI | Conversational AI for multilingual chatbot support |
| Leaflet | v1.9, Mapping | Interactive GIS maps, markers, popups, proximity search |
| Jest | v29.7, Testing | Unit and integration testing framework |
| Git/GitHub | v2.43, VCS | Version control, feature branch workflow, code review |
| Docker | v24.0, Containerization | Consistent deployment environments |
| Vercel | Hosting platform | Frontend deployment with edge CDN |
| Render | Hosting platform | Backend API and worker deployment |

## Data Flow Diagram (DFD – Level 0)




1. User
    - Login
    - Create account
    - Book a space
    - View available spaces
    - Check in / check out (with QR code)

2. **System**
    - Account confirmation
    - Booking confirmation
    - QR code
    - Search results (spaces, availability)

3. Users DB
    - Validate login
    - Create new user
    - Get user role

4. Bookings DB
    - Store booking
    - Update check-in/out
    - Change booking status

5. Spaces DB
    - Store booking
    - Update check-in/out
    - Change booking status

6. Districts DB
    - Fetch district list for filtering spaces

## Entity Relationship Diagram (ERD)



Entities:

**1.** users

Tracks all account types (Regular Users, Admins, and Space Owners/Hosts).

- id (PK)
- name (VARCHAR)
- email (VARCHAR, Unique)
- password (VARCHAR)
- role (ENUM: 'user', 'admin', 'space')

- business_permit (VARCHAR, Nullable) — *Used if role is 'space'*
- dti_sec_reg (VARCHAR, Nullable) — *Used if role is 'space'*
- is_active (BOOLEAN)
- created_at / updated_at (TIMESTAMP)

**2.** districts

Geographic regions or sectors used to categorize space locations.

- id (PK)
- name (VARCHAR)
- slug (VARCHAR, Unique)
- active (BOOLEAN)

**3.** spaces

The physical coworking locations, offices, or venues mapped to an owner and district.

- id (PK)
- user_id (FK ➔ users.id) — *The space owner*
- district_id (FK ➔ districts.id)
- name (VARCHAR)
- area (VARCHAR/DECIMAL)
- lat (DECIMAL) — *Latitude for map plotting*
- lng (DECIMAL) — *Longitude for map plotting*
- rate_hour (DECIMAL)
- status (ENUM: 'available', 'maintenance', 'hidden')
- image (VARCHAR) — *URL/Path to primary display image*
- capacity (INT)

- occupied_seats (INT)
- hours_json (JSON/TEXT) — *Compacts operational hours (e.g., Mon-Fri 9-5) into a single field*

**4.** bookings

Handles space reservations, tickets, and attendance verification.

- id (PK)
- user_id (FK ➔ users.id) — *The customer booking the space*
- space_id (FK ➔ spaces.id)
- booking_type (VARCHAR)
- guest_name (VARCHAR, Nullable)
- ticket_number (VARCHAR, Unique)
- qr_code_token (VARCHAR, Unique) — *For check-in scanning*
- start_time (TIMESTAMP)
- end_time (TIMESTAMP)
- status (ENUM: 'pending', 'confirmed', 'cancelled', 'completed')
- notes (TEXT, Nullable)

**5.** space_requests

A lightweight sandbox/holding table for onboarding new space hosts before they are granted a full user account.

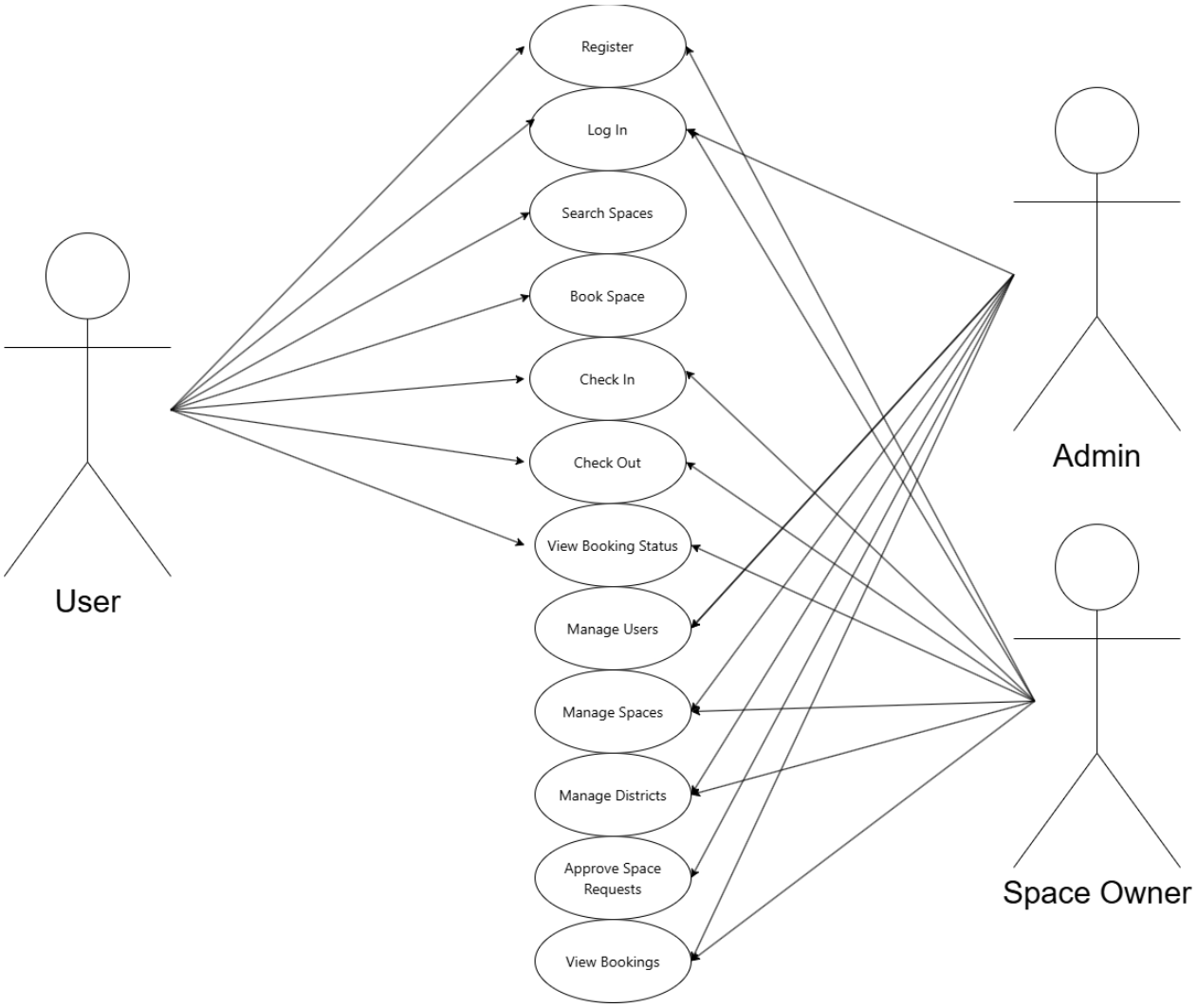
- id (PK)
- name (VARCHAR)
- email (VARCHAR)
- password (VARCHAR)

- business_permit (VARCHAR)
- status (ENUM: 'pending', 'approved', 'rejected')

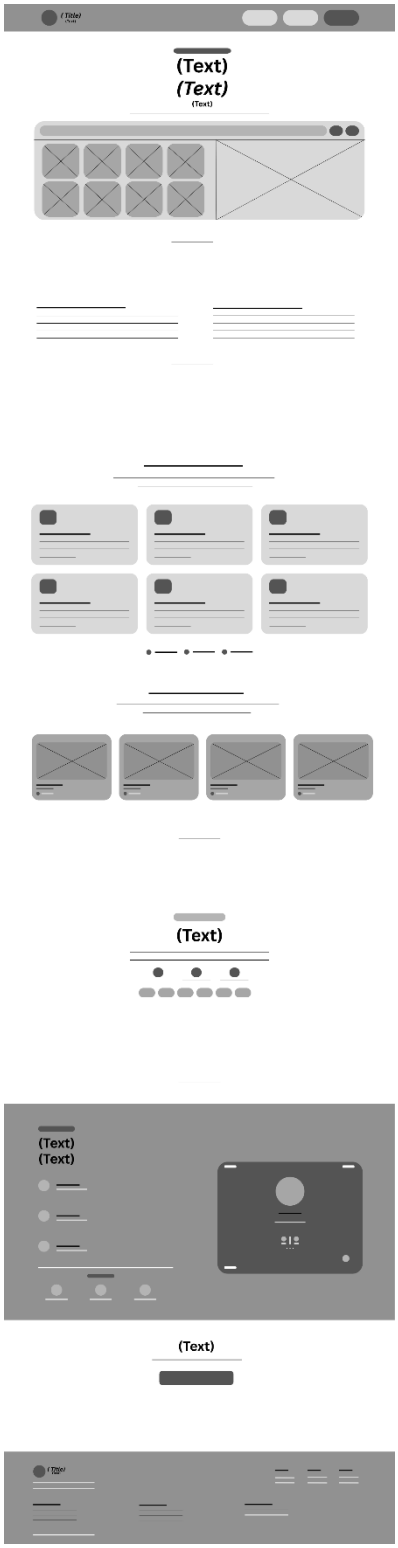
**Relational Map**

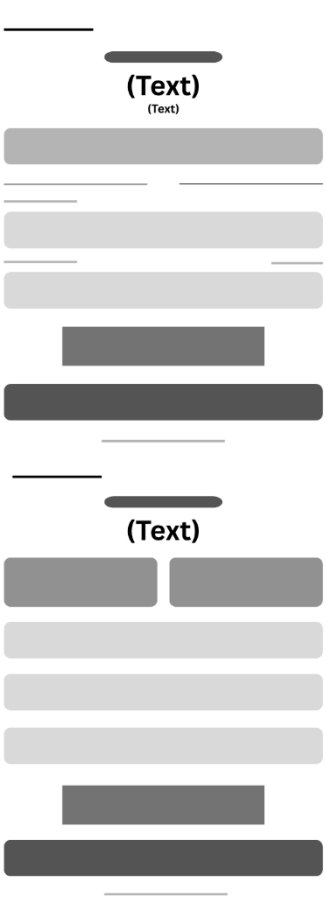
- users ➔ spaces **($1:\text{N}$):** A User with the 'space' role can own multiple physical locations.
- users ➔ bookings **($1:\text{N}$):** A Customer can place many reservations over time.
- spaces ➔ bookings **($1:\text{N}$):** A physical location handles multiple bookings across different time slots.
- districts ➔ spaces **($1:\text{N}$):** A district groups multiple spaces together, keeping location filtering fast and indexed.

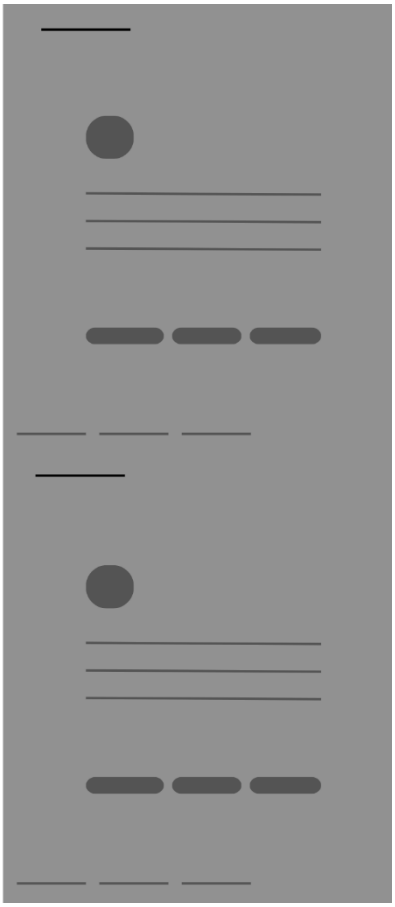
Use Case Diagram

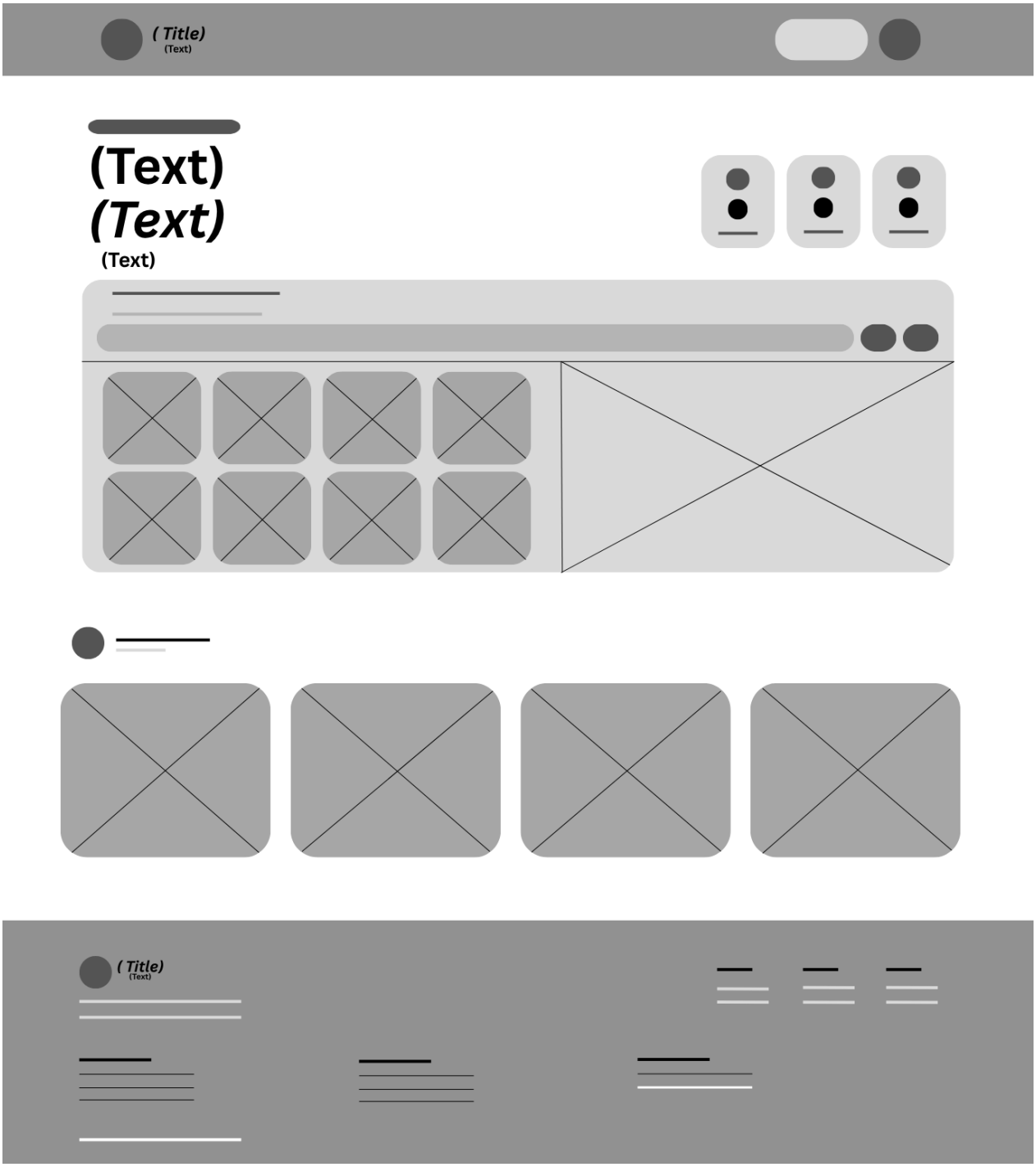


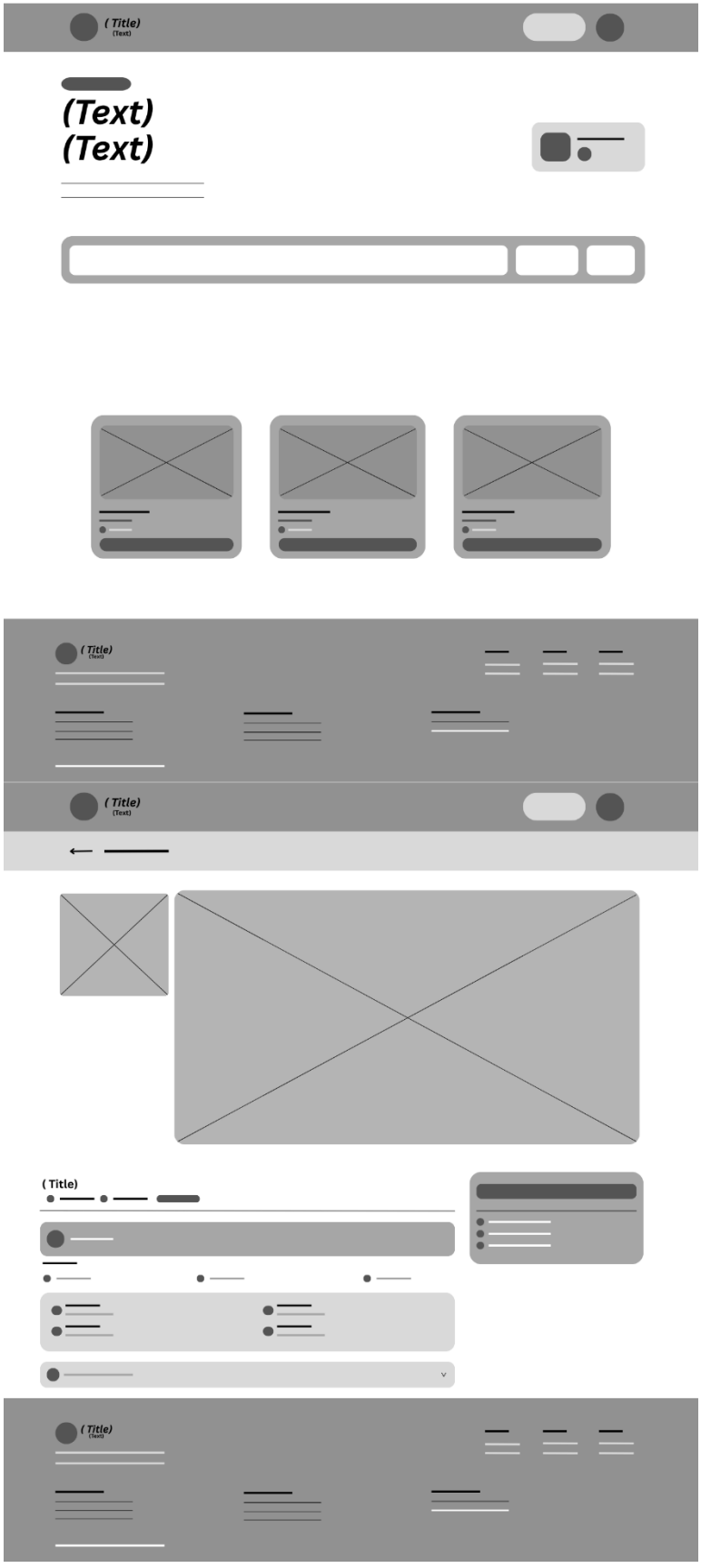
## Wireframes

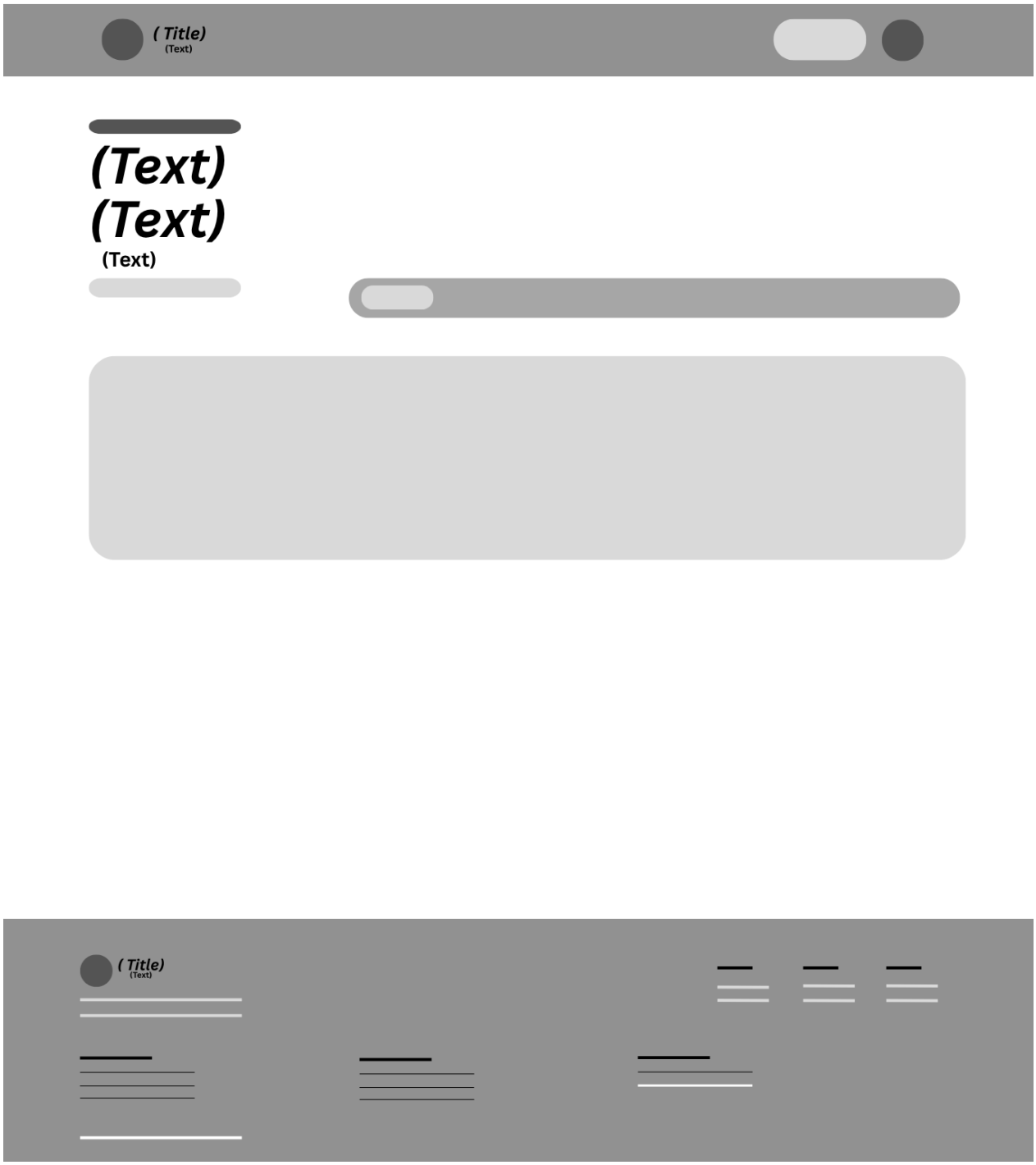

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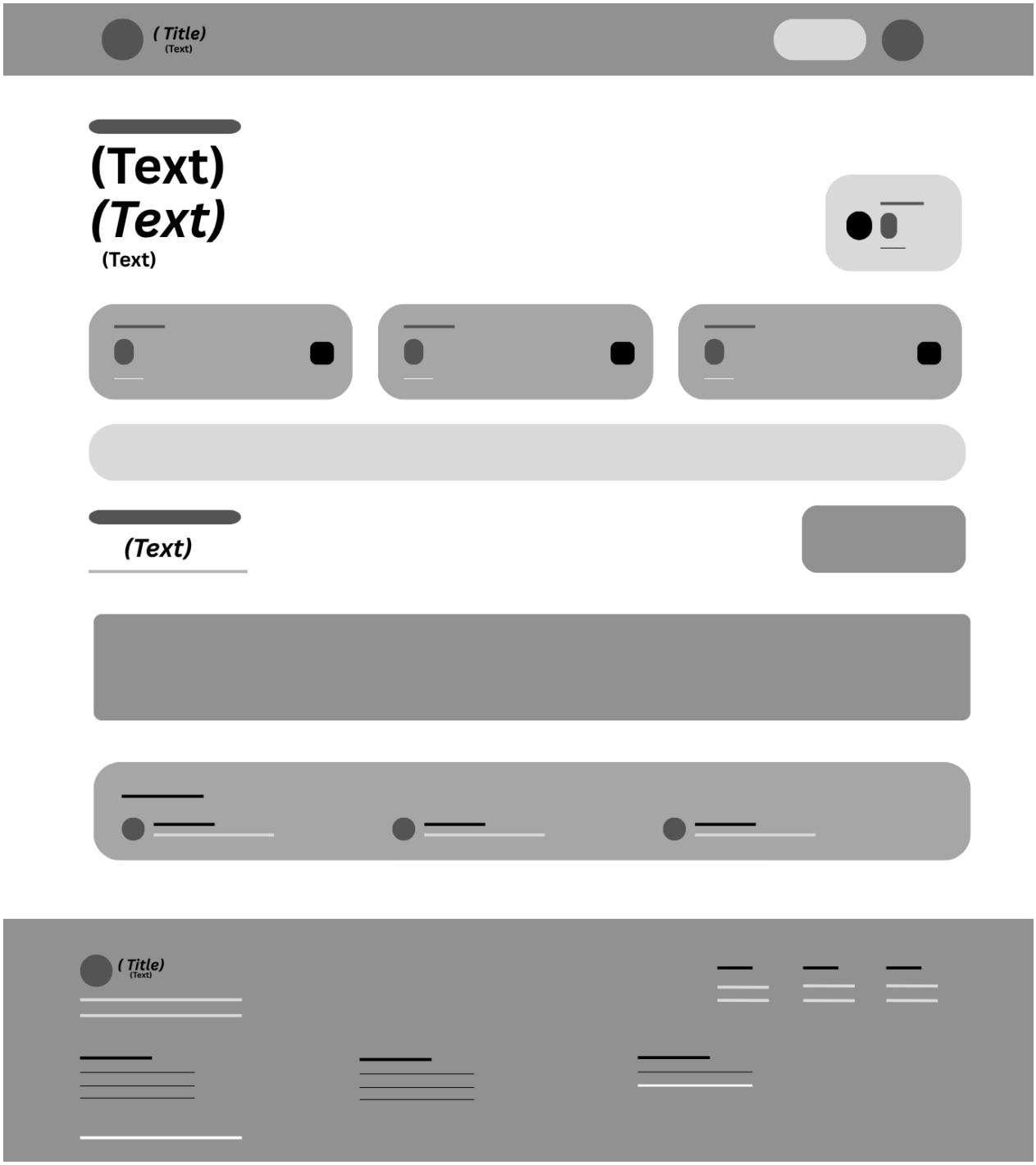

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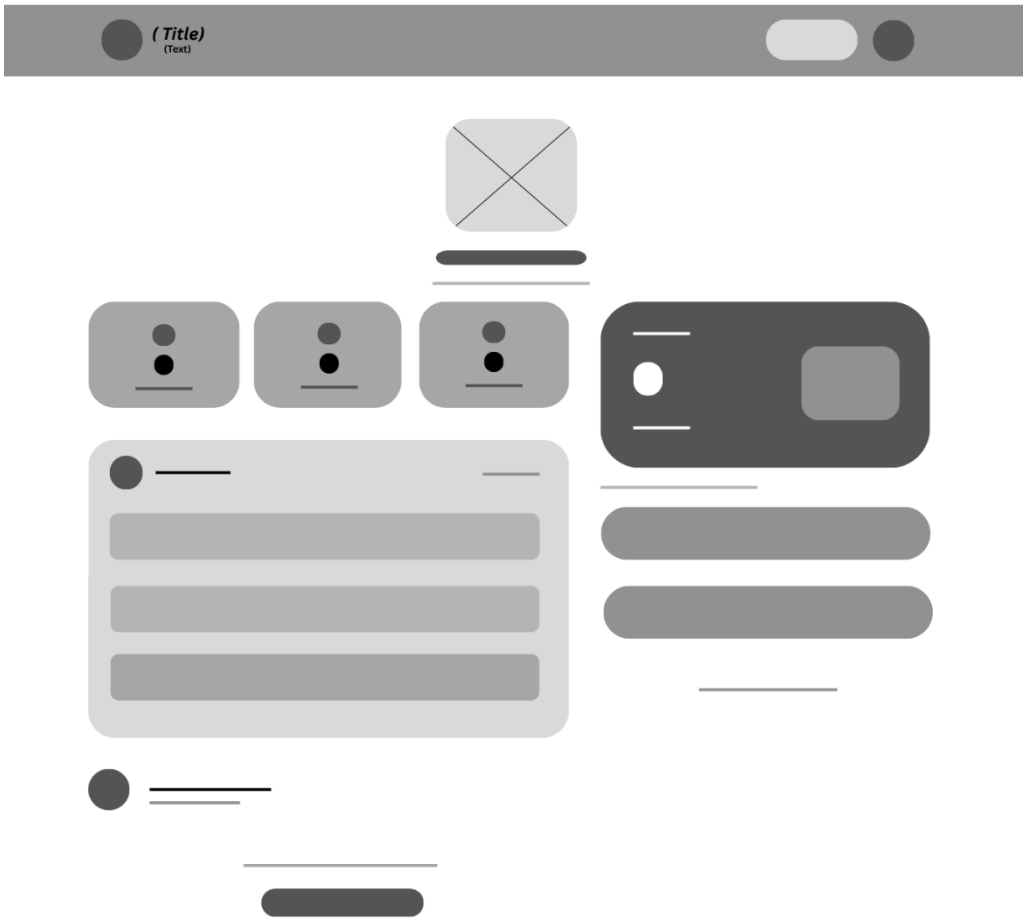



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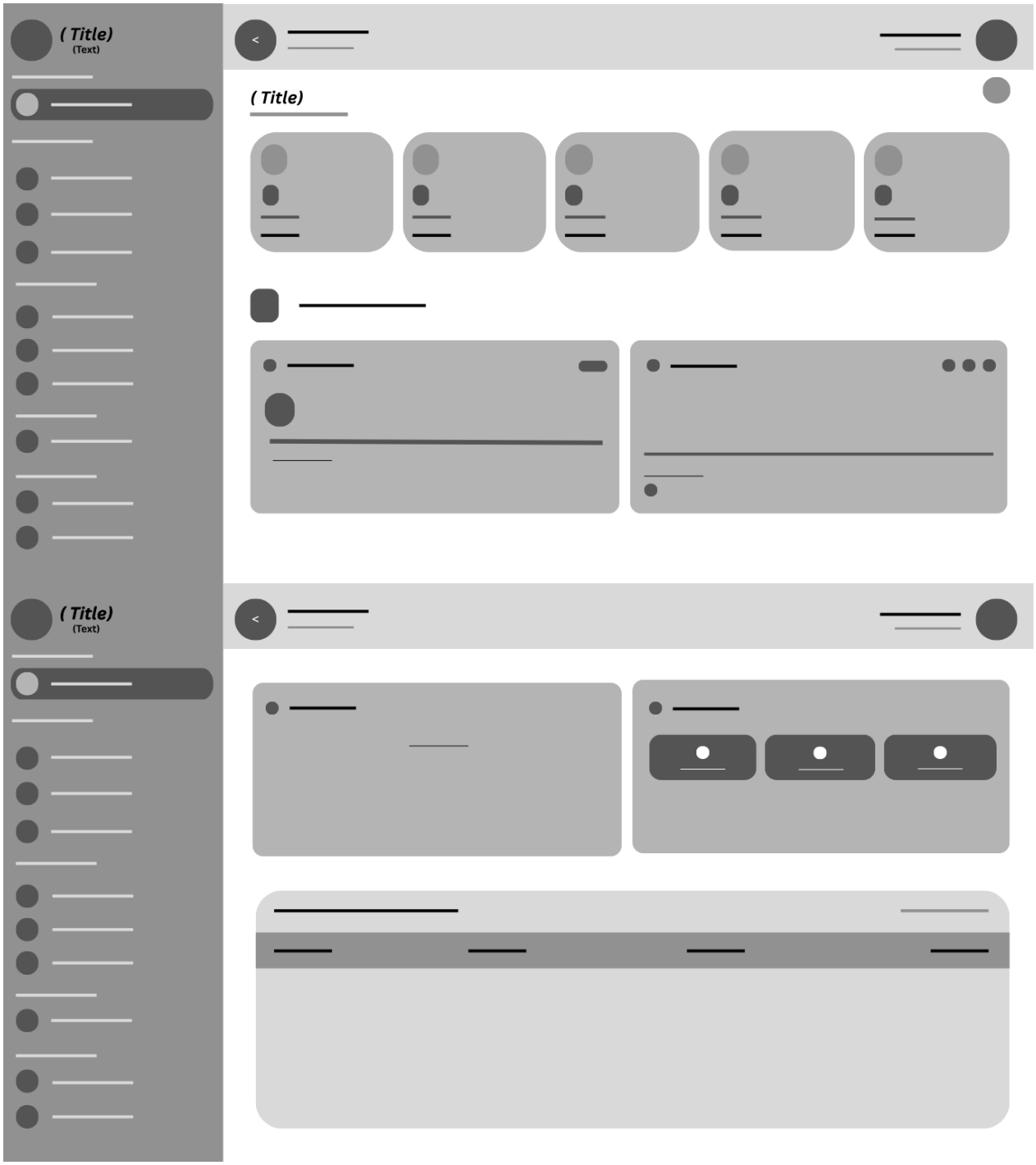

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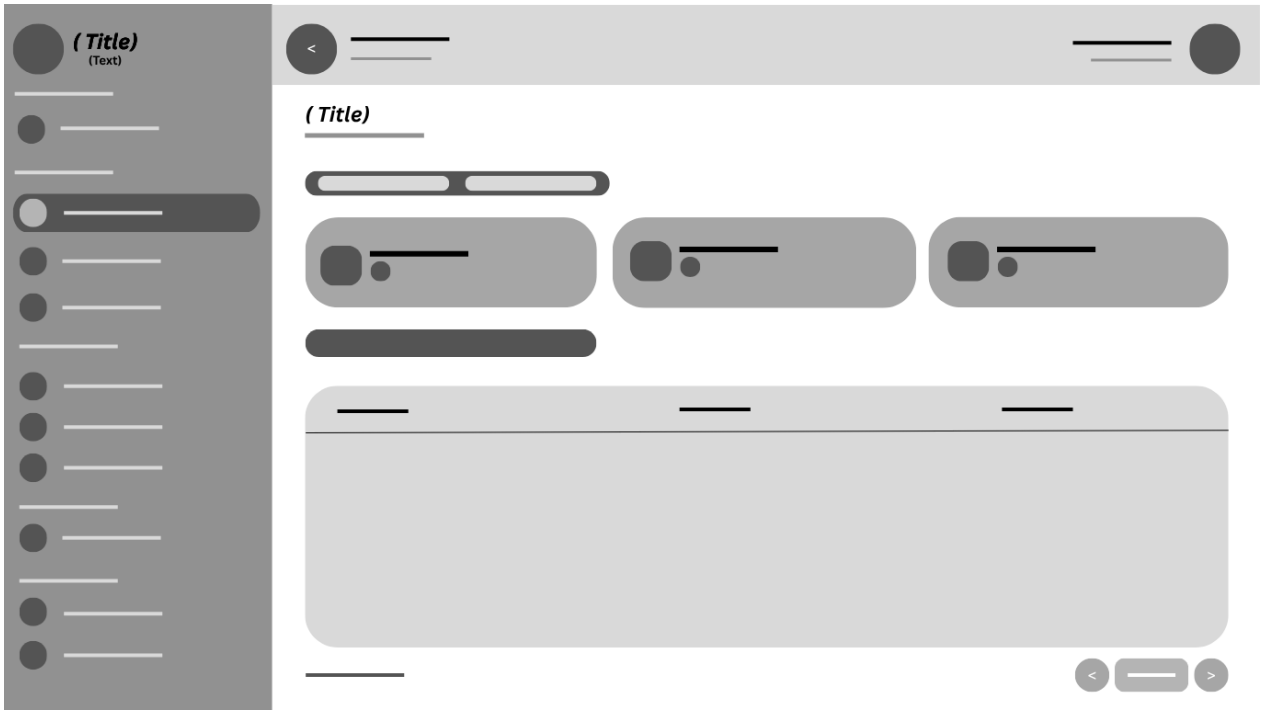

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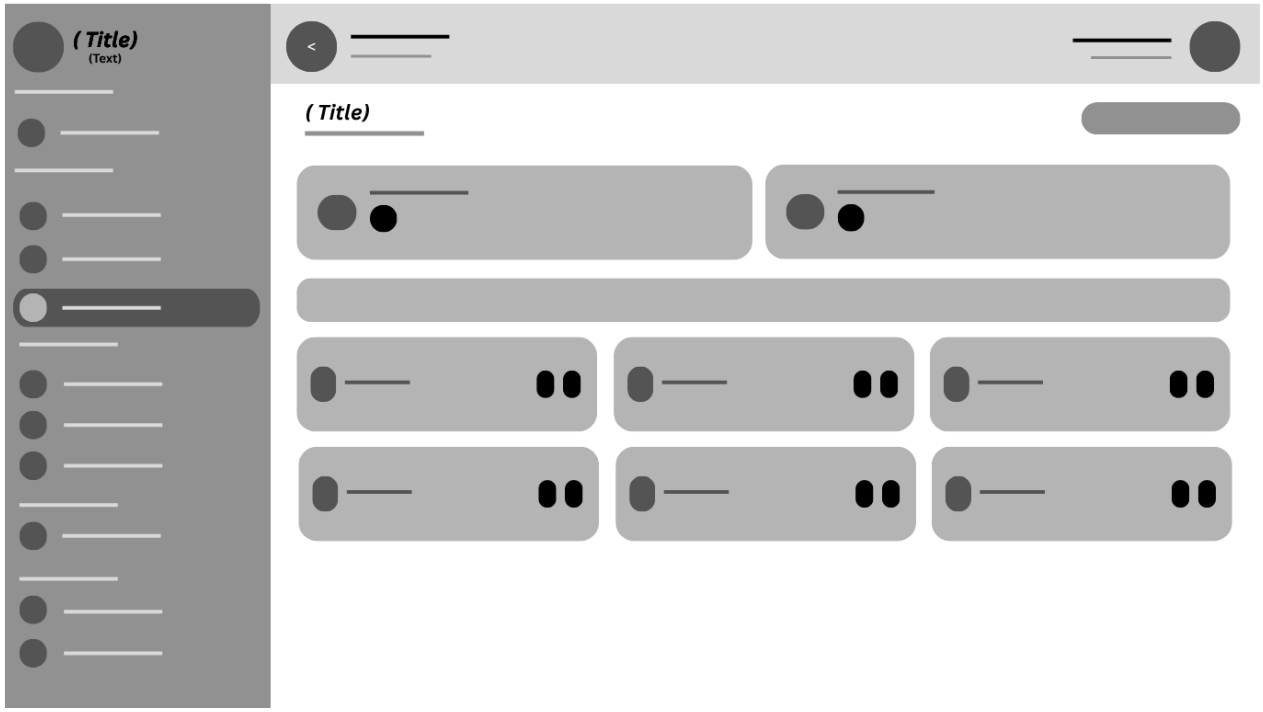

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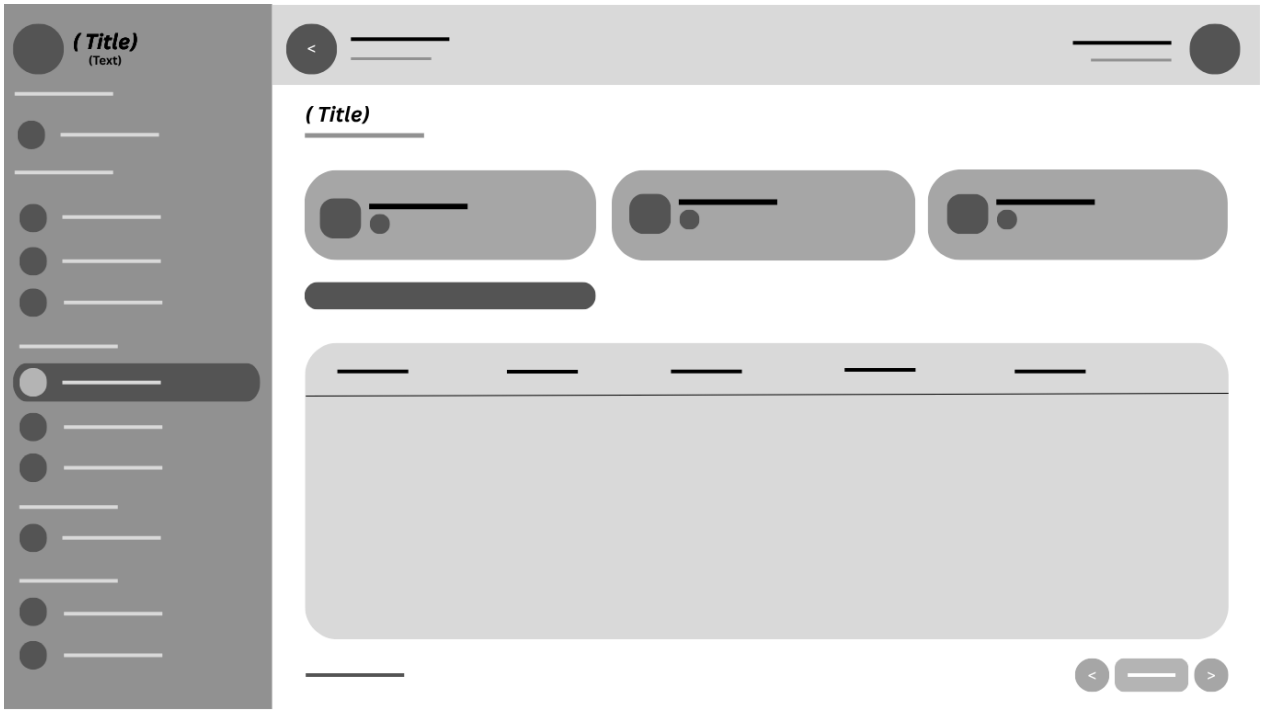

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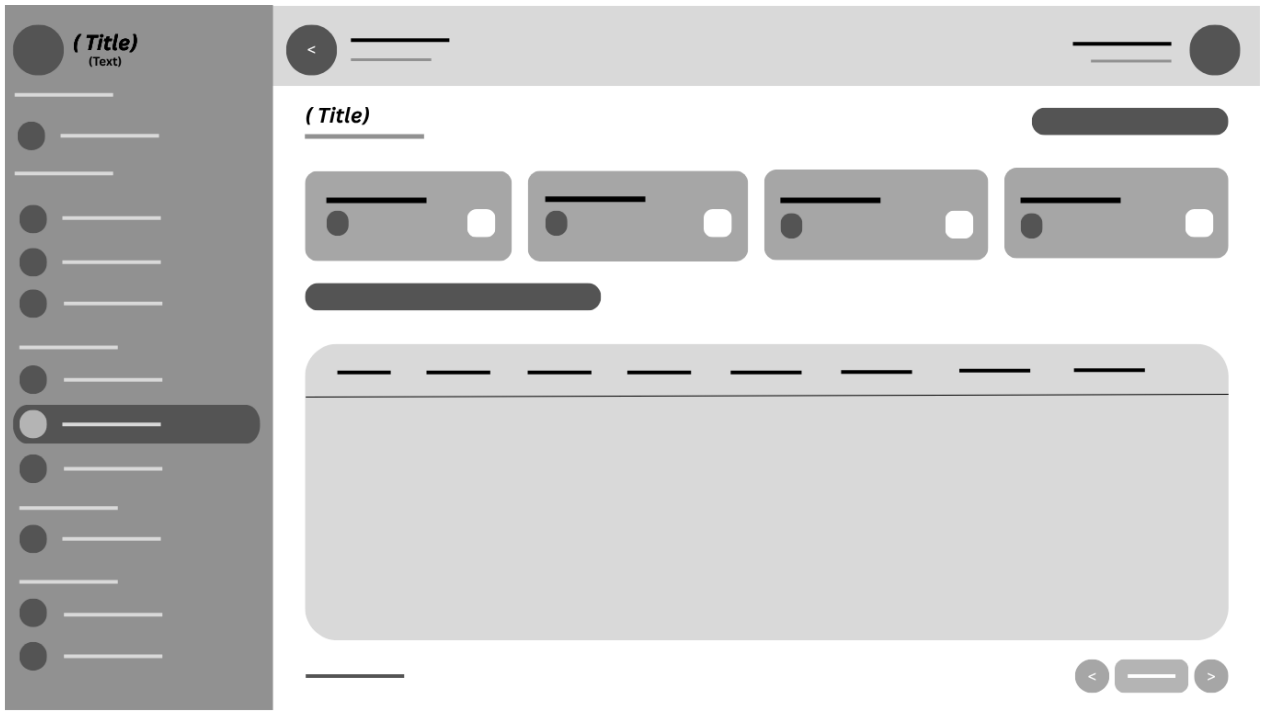

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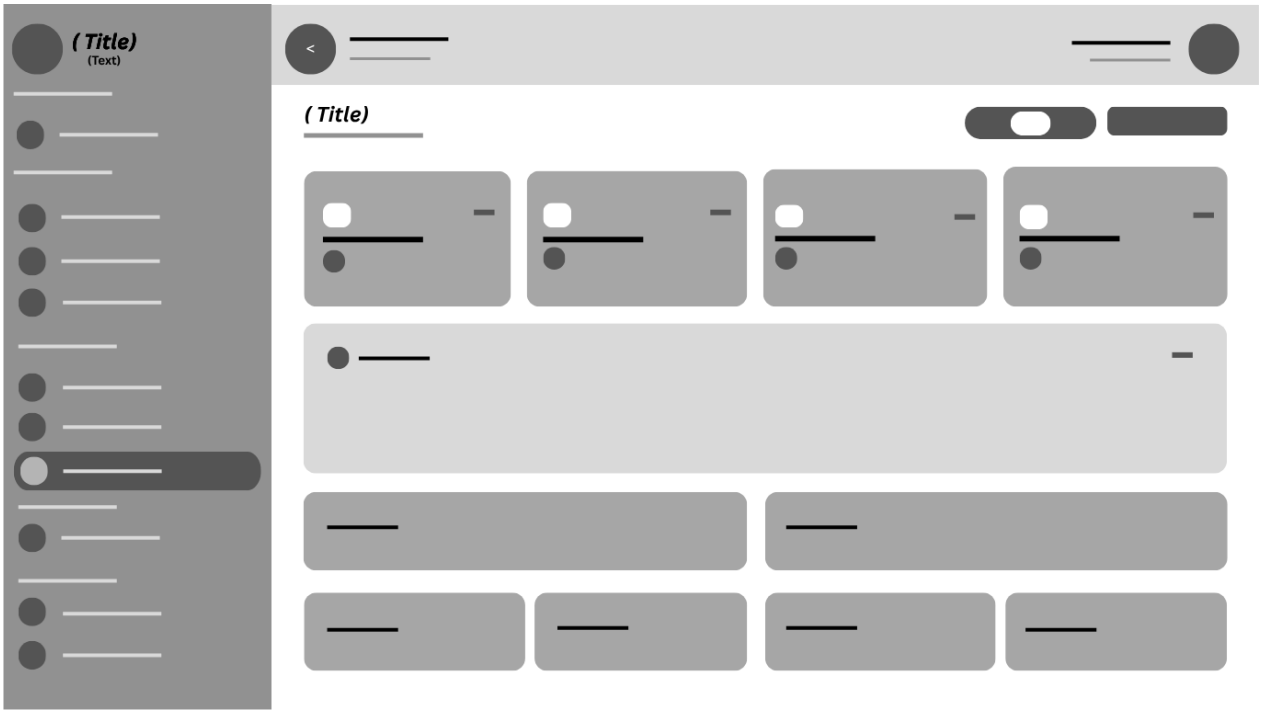

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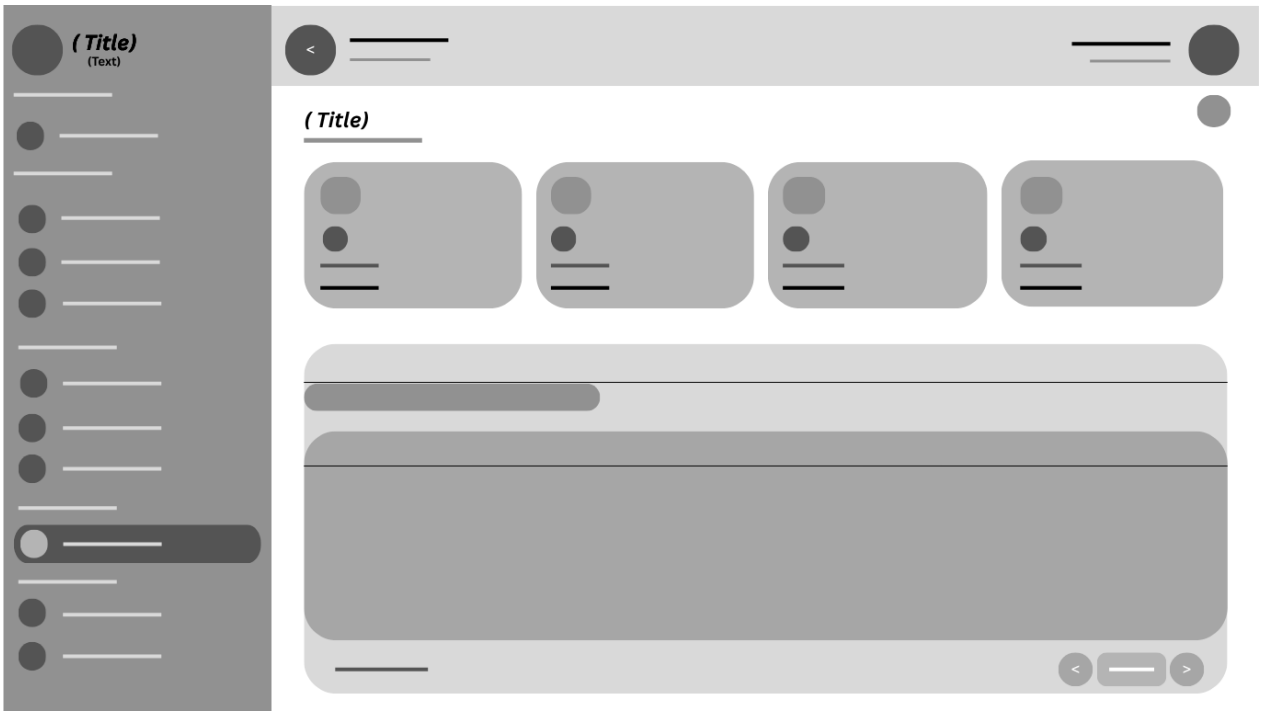

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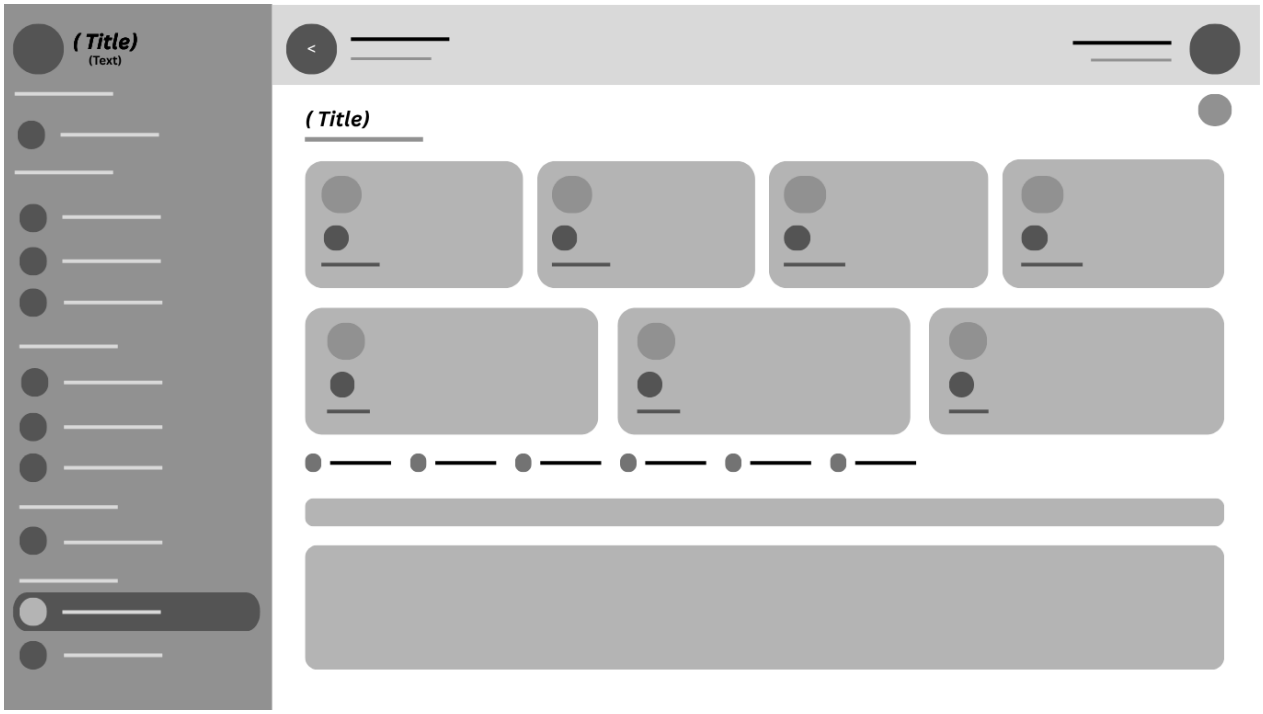

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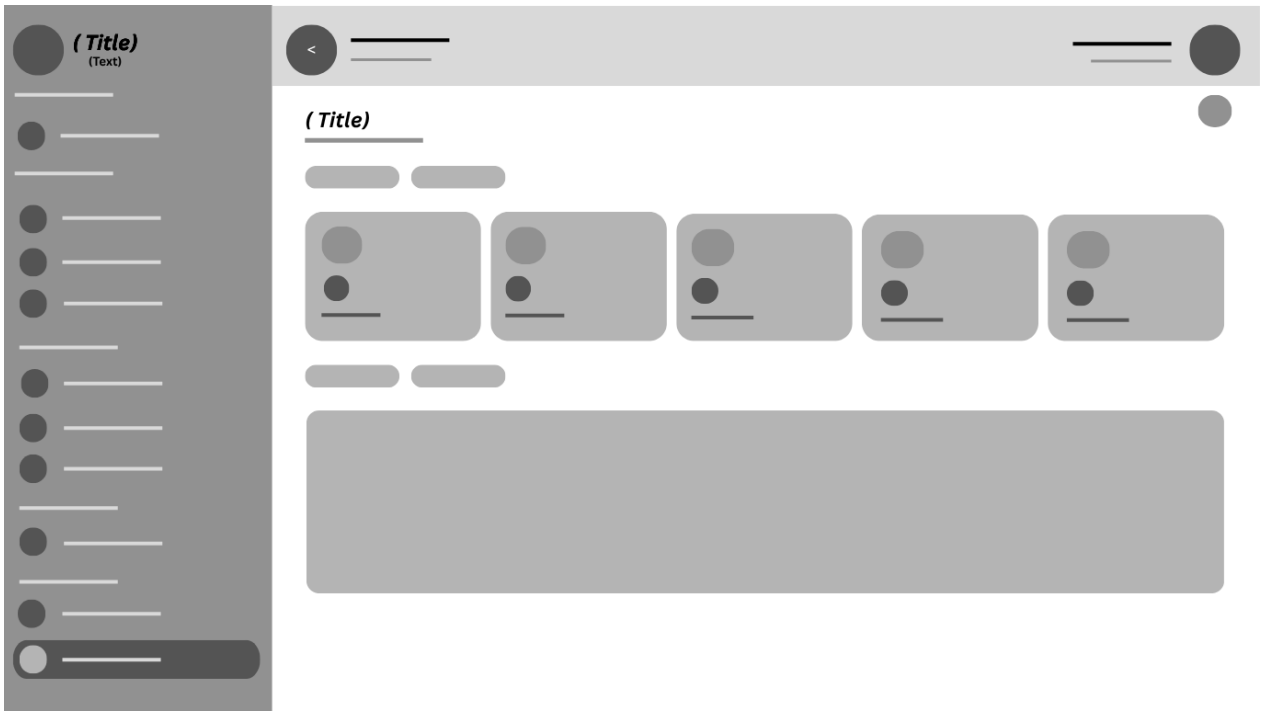

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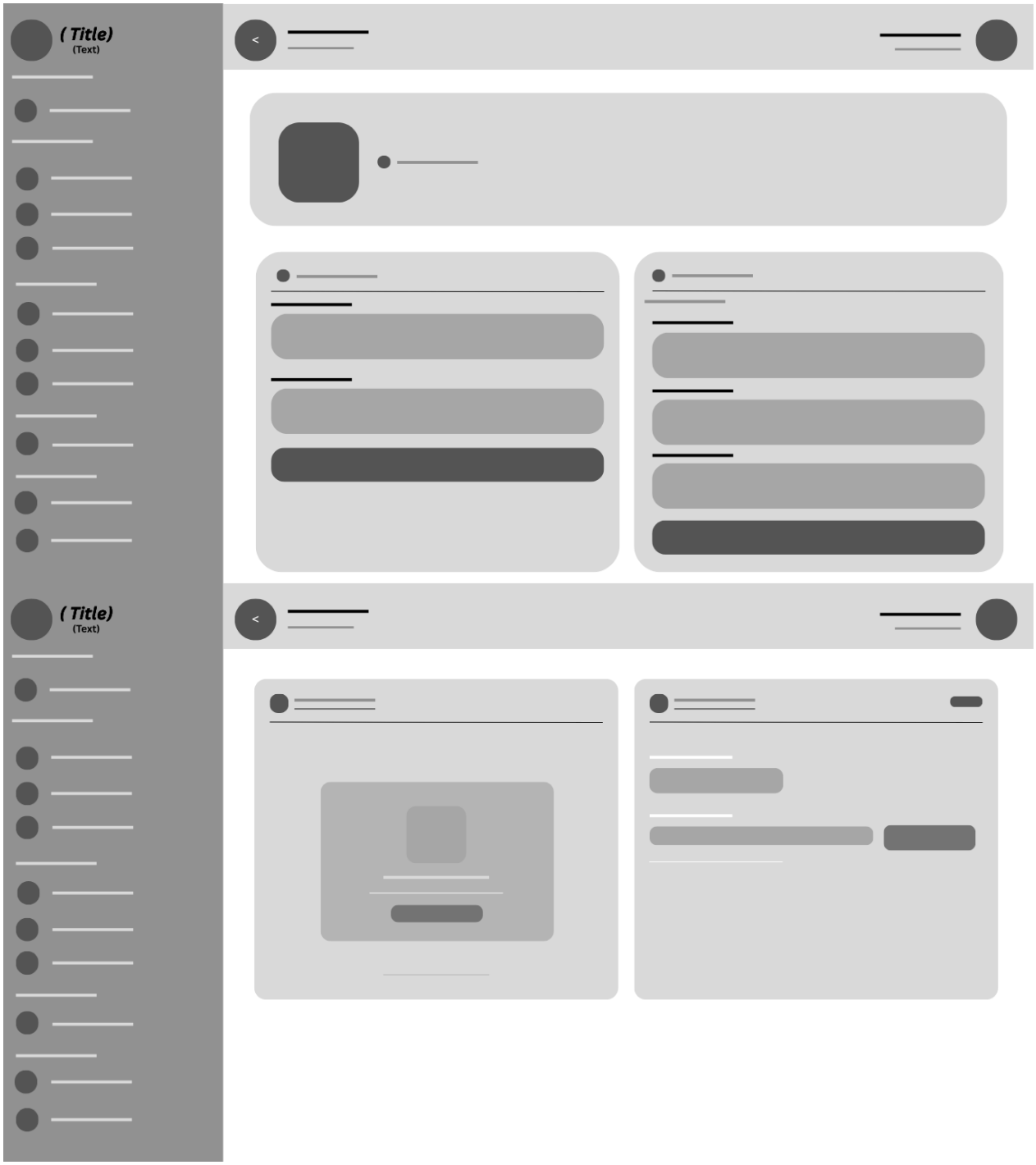

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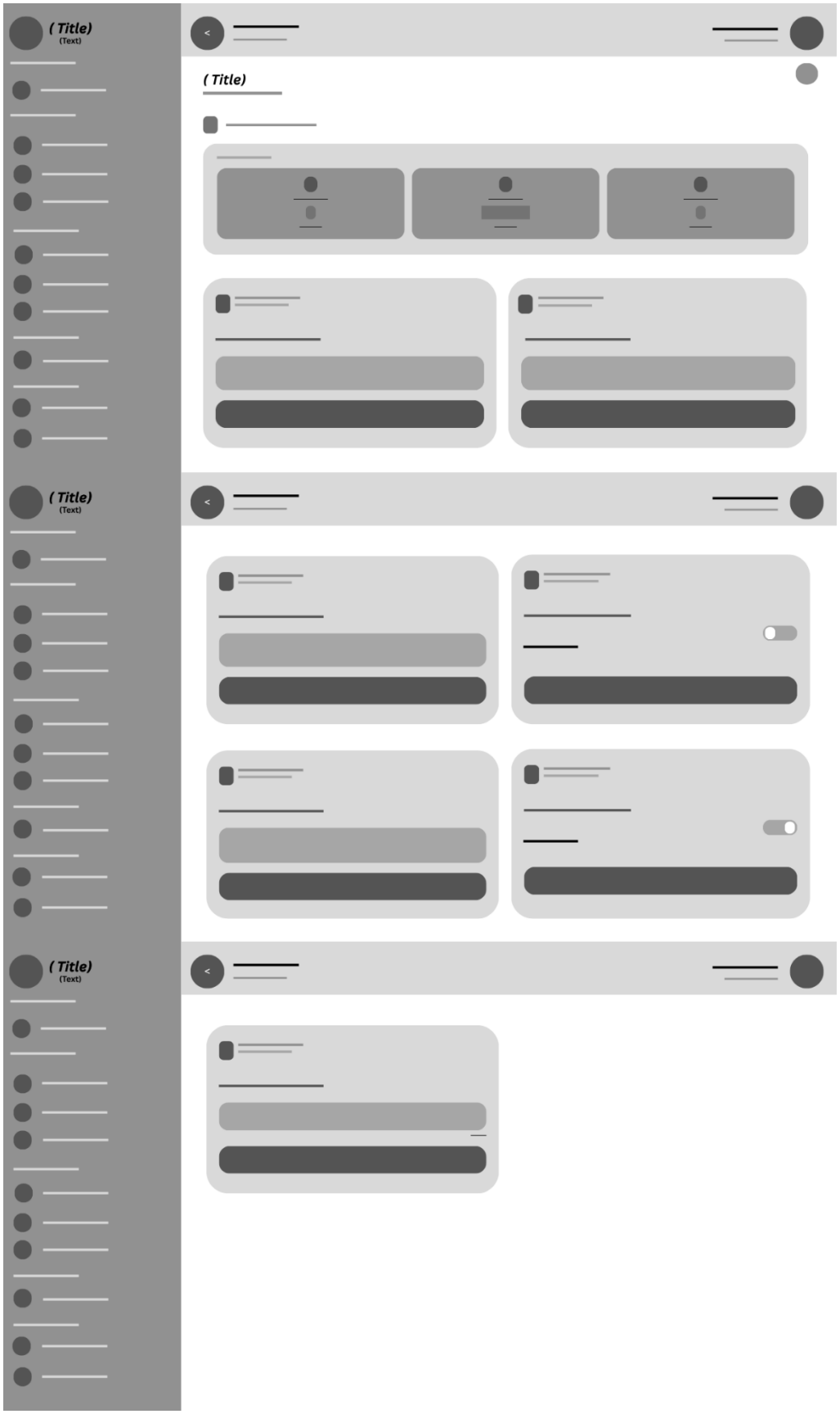

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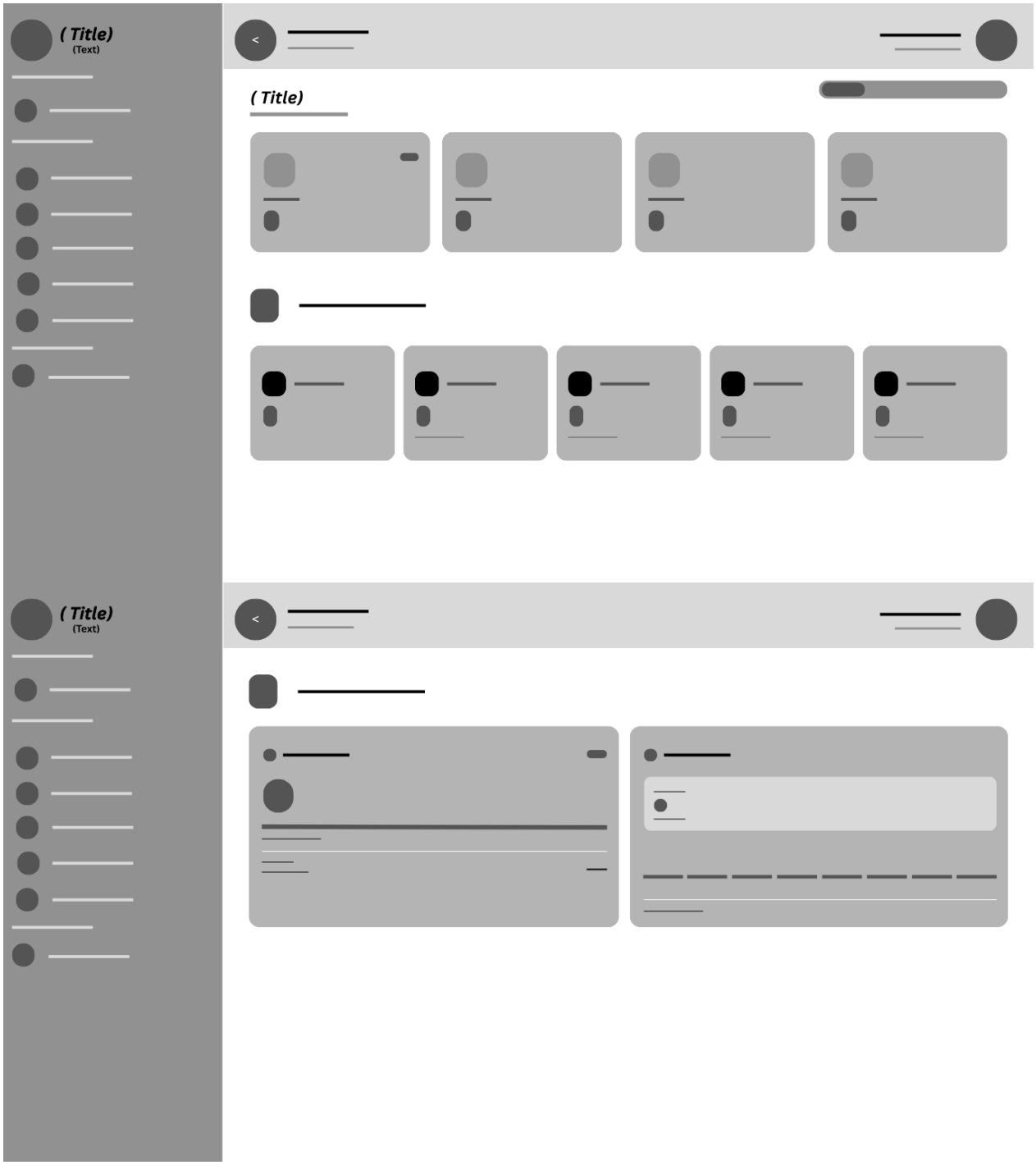

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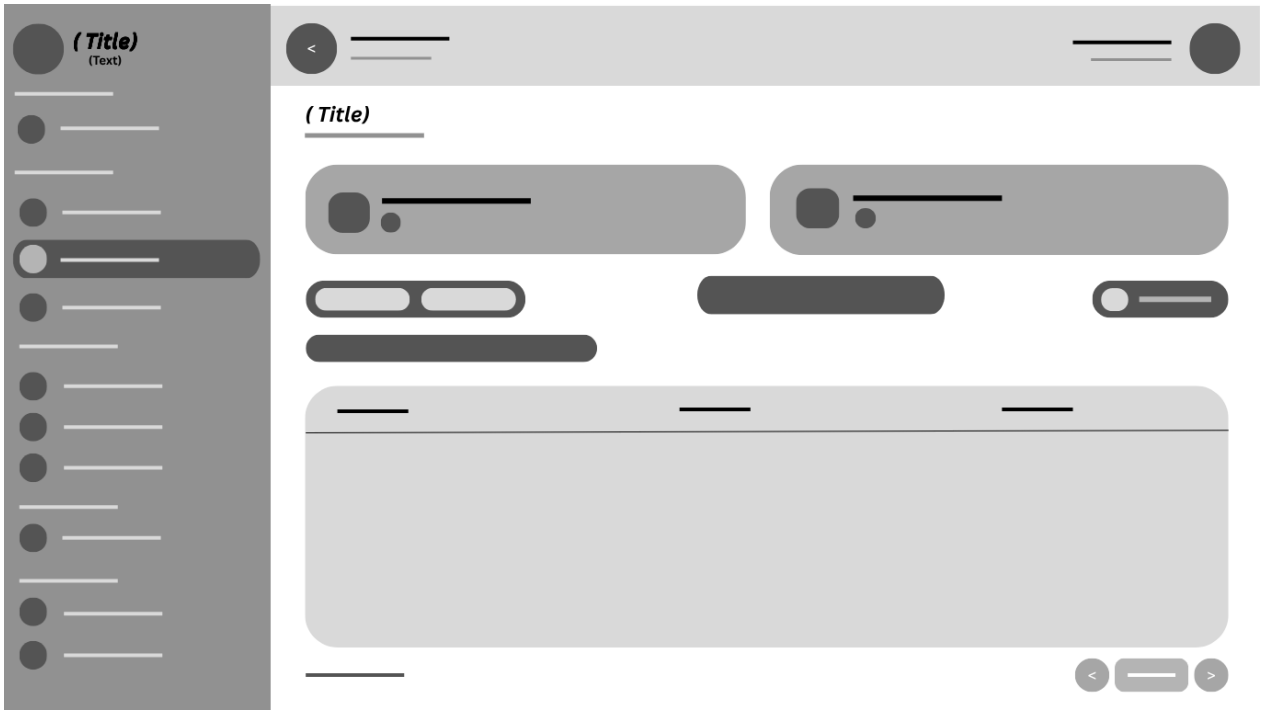

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# 3.9 Post-Deployment System Enhancements and Feature Iterations

Following the initial deployment of the FlexSpace system, a comprehensive post-deployment enhancement cycle was undertaken to address user feedback, operational requirements, and system optimization opportunities identified during real-world usage. This section documents the systematic approach to post-deployment feature development, encompassing new feature implementations, user interface refinements, validation and security hardening, bug resolution, inventory and point-of-sale enhancements, architectural improvements, staff management capabilities, and performance optimizations. These enhancements represent a continuous improvement methodology aligned with Agile principles, ensuring the system evolves in response to actual user needs and operational insights.

## 3.9.1 New Features and System Enhancements

The post-deployment enhancement cycle introduced multiple new features that significantly expanded the system's functional capabilities. These features were prioritized based on user feedback analytics, operational metrics from space owners, and the development team's assessment of technical debt and architectural improvements needed for long-term system sustainability.

#### 3.9.1.1 Branch Management for Staff Assignment and Transfer

A comprehensive branch management module was developed to enable space owners with multiple locations to assign and transfer staff members across different branches. This feature addresses a critical operational need identified by space owners operating chain-style co-working businesses with locations across multiple districts of Iloilo City. The branch management system implements a many-to-many relationship structure where staff members can be associated with one or more branches, with a designated primary branch for default assignment.

The branch assignment workflow enables space owners to select staff members from their team roster and assign them to specific branches through an intuitive interface. The assignment process includes validation checks to ensure that staff members are not simultaneously assigned to branches with overlapping operational responsibilities that would create scheduling conflicts. When a staff member is transferred between branches, the system maintains a complete audit trail of the transfer, including the previous branch, new branch, transfer date, and authorizing space owner.

The branch management module implements role-based access controls that ensure staff members can only access data and perform operations within their assigned branches. This security boundary prevents unauthorized cross-branch data access while enabling space owners to maintain centralized oversight of all branch operations. The system also supports branch-specific inventory management, where staff members can only view and modify inventory items associated with their assigned branch location.

Technical implementation of the branch management feature required modifications to the user authentication middleware to incorporate branch context into session tokens. The JWT token structure was extended to include branch identifiers, enabling the system to enforce branch-scoped permissions at the API level. Database schema changes included the addition of a branches collection and modifications to the users collection to support branch associations. The front-end implementation utilized React context providers to propagate branch selection state throughout the component tree, ensuring consistent branch context across all dashboard views.

#### 3.9.1.2 Purchase Price and Selling Price Tracking for Inventory Profit Calculation

The inventory management system was enhanced to support dual-price tracking, capturing both the purchase price (cost of goods acquired) and the selling price (revenue from goods sold) for all inventory items. This enhancement enables comprehensive profit analysis and financial reporting capabilities that were previously unavailable in the system. The dual-price tracking system addresses a fundamental business requirement for space owners who operate on-site retail services, such as snack bars, merchandise sales, and equipment rentals.

For each inventory item, the system now records the purchase price at the time of stock acquisition, enabling calculation of cost of goods sold and gross profit margins. The profit per unit is computed automatically as the difference between the selling price and the purchase price. The system also calculates the profit margin percentage using the formula: ((Selling Price - Purchase Price) / Purchase Price) multiplied by 100, providing space owners with immediate visibility into product profitability.

The potential revenue analysis feature compares current inventory stock levels against selling prices to project potential revenue, while simultaneously calculating total inventory cost based on purchase prices. This analysis enables space owners to make informed decisions about inventory investment, identifying high-margin products that warrant increased stock levels and low-margin items that may require pricing adjustment or discontinuation. The profit preview functionality

displays real-time profit projections within the inventory management interface, updating dynamically as stock quantities change through sales and replenishment activities.

#### 3.9.1.3 Twenty-Four Seven Schedule Management with Per-Day Toggle

The scheduling system underwent significant enhancement to support twenty-four seven operating schedules with granular per-day configuration options. This feature addresses the diverse operating requirements of co-working spaces in Iloilo City, where some facilities operate continuously while others maintain restricted hours based on location, security considerations, and target user demographics.

The enhanced schedule management interface provides space owners with a quick-apply option that sets all seven days of the week to twenty-four hour operation with a single action, eliminating the repetitive data entry previously required for continuous operations. Each day of the week can be individually configured with its own toggle switch, enabling mixed schedules where weekdays operate continuously while weekends maintain restricted hours, or any other combination that matches operational requirements.

The per-day twenty-four seven toggle includes a cancel option that allows space owners to revert specific days to custom time ranges without affecting the twenty-four hour setting on other days. This granular control ensures that scheduling configurations can precisely match the operational reality of each co-working space. The system validates schedule configurations to prevent logical inconsistencies, such as closing times that precede opening times, and provides visual indicators to highlight days with non-standard or overnight schedules.

#### 3.9.1.4 Real-Time Availability Verification Before Booking Confirmation

A critical enhancement to the reservation engine implements real-time availability verification immediately before booking confirmation, preventing overbooking scenarios that could occur due to race conditions in high-demand periods. The enhanced availability check executes a database query at the moment of booking submission, verifying that the requested resource remains available after accounting for all confirmed bookings that may have been created since the user initiated the booking process.

The real-time availability verification integrates with the dynamic status system, which automatically computes space availability based on current schedule configurations and capacity utilization. When a user initiates the booking process, the system captures a timestamp and validates that the availability state has not changed significantly since the user began their

selection. If another user has confirmed a booking for the same resource during this window, the system presents a clear message indicating the unavailability and suggests alternative time slots or resources.

#### 3.9.1.5 Dynamic Status Computation Based on Schedule and Capacity

The space status system was enhanced to support dynamic status computation that automatically determines a space's operational status based on its schedule configuration and current capacity utilization. Rather than relying solely on manual status updates from space owners, the system now evaluates multiple data points to derive an accurate real-time status indicator. The status computation considers the current time relative to the space's operating hours, the number of active bookings versus total capacity, and any special closures or maintenance periods that have been configured.

The dynamic status system implements three primary status categories: Open and Available (operating hours active with available capacity), Open but Limited (operating hours active but capacity near exhaustion), and Closed (outside operating hours or explicitly closed by the owner). These status indicators are displayed to users through color-coded badges and filtering options, enabling informed decision-making when selecting co-working spaces. The status computation runs as a background process with periodic refresh intervals to ensure displayed information remains current.

#### 3.9.1.6 Enhanced Image Upload Capacity

The image upload system was upgraded to support file sizes up to fifty megabytes, a substantial increase from the previous five megabyte limit. This enhancement accommodates high-resolution photographs of co-working spaces, which are essential for accurate visual representation of facilities. The increased upload limit supports modern camera resolutions and enables space owners to showcase their facilities with professional-quality imagery that improves user engagement and booking conversion rates.

The enhanced upload system implements chunked transfer encoding for large file uploads, preventing timeout errors that could occur with previous single-transfer methods. Client-side image compression is applied prior to upload to optimize file sizes without significant quality degradation. Server-side processing includes automatic image resizing to generate thumbnail and preview variants, reducing bandwidth requirements for listing pages while preserving full-

resolution images for detail views. The upload interface provides real-time progress indicators for large file transfers, improving user experience during the upload process.

## 3.9.2 User Interface and User Experience Improvements

The post-deployment enhancement cycle placed significant emphasis on user interface refinements and user experience optimization. These improvements were informed by usability testing results, user feedback analytics, and industry best practices in human-computer interaction. The following subsections document the specific interface modifications and their rationale within the context of the FlexSpace system's user-centered design philosophy.

### 3.9.2.1 Dedicated Page Interface for Space Listing Creation

The space listing creation workflow was fundamentally redesigned, transitioning from a modal-based interface to a dedicated full-page experience. The previous modal implementation, while functional for basic space registration, presented limitations as the complexity of space configurations increased with the introduction of branch management, advanced scheduling, and inventory tracking features. Modal interfaces imposed spatial constraints that necessitated scrolling within a confined area and limited the ability to present comprehensive contextual information during the listing creation process.

The dedicated page implementation opens the listing creation interface in a new browser window, providing the full viewport for form elements, contextual help, and preview functionality. This redesign enables a multi-column layout that presents the space information form alongside a live preview of how the listing will appear to users, facilitating immediate visual feedback during content creation. The dedicated page approach also eliminates the risk of accidental modal closure, which could result in lost form data, by implementing explicit save and cancel actions with unsaved change detection and confirmation dialogs.

The new listing creation page implements a step-by-step wizard interface that guides space owners through the configuration process in logical stages: basic information, location and mapping, amenities and features, schedule configuration, pricing setup, and media uploads. Each step includes validation before progression, ensuring that required information is captured in the correct sequence. The wizard interface reduces cognitive load by presenting focused subsets of configuration options rather than overwhelming users with all fields simultaneously.

#### 3.9.2.2 Enhanced Status Badges with Interactive Tooltips

The status badge components throughout the administrative interface were enhanced with hover-triggered tooltip functionality that provides contextual information about each status state. Status badges serve as critical visual indicators for booking states, space availability, staff assignments, and inventory levels, but the previous implementation conveyed only the status label without explanatory context. The enhanced badges now display descriptive tooltips on hover, explaining the meaning of each status and providing relevant action guidance.

The tooltip implementation utilizes CSS hover states with a fade-in animation for smooth appearance transitions. Each tooltip is positioned dynamically to remain within the viewport boundaries, preventing truncation on smaller screens or near viewport edges. The tooltip content is customizable per status type, enabling space owners to configure status descriptions that match their operational terminology. For booking statuses, tooltips include additional context such as the timestamp of the last status change and the user who initiated the change, providing a mini audit trail within the hover interaction.

#### 3.9.2.3 Character Counter for Space Name Input Fields

Input fields for space names and similar text entries now include real-time character counters that display the current character count alongside the maximum allowed length. The character counter implements the format "current/maximum" (for example, "0/50") positioned adjacent to the input field. This enhancement addresses user feedback regarding uncertainty about name length restrictions and prevents submission errors caused by exceeding the maximum character limit.

The character counter updates in real-time as the user types, with visual feedback transitioning from a neutral color to a warning color when the input approaches the maximum limit, and to an error color when the limit is exceeded. The counter is integrated with the form validation system, preventing form submission when the character count exceeds the allowed maximum and displaying a clear error message indicating the excess. For space names, the fifty-character maximum was established based on analysis of optimal display widths across the user interface, ensuring that space names render correctly in list views, map popups, and booking confirmations without truncation.

#### 3.9.2.4 Enhanced Error Messaging with Visual Indicators

The form validation error messaging system was comprehensively enhanced to provide clearer, more actionable feedback when user input fails validation checks. Error messages are now

accompanied by red asterisk indicators on required field labels, providing immediate visual identification of mandatory fields before submission attempts. When validation errors occur, the system highlights the specific fields in error with red border styling and displays contextually positioned error messages that explain the validation failure and provide guidance for correction.

The enhanced error messaging implements a three-tier severity system: Information (blue) for suggestions and optional improvements, Warning (yellow) for values that are acceptable but may not produce optimal results, and Error (red) for values that prevent form submission. Each error message includes a specific description of the validation failure rather than generic rejection notices. For example, instead of "Invalid input," the system displays messages such as "Hourly rate must be a positive number between 10 and 10,000 Philippine pesos" or "Space name must contain between 3 and 50 characters." This specificity reduces user confusion and accelerates error resolution.

#### 3.9.2.5 Weekly Schedule Interface with Improved Time Input Visibility

The weekly schedule configuration interface underwent significant redesign to improve the visibility and usability of time input fields. The previous implementation used compact time selectors that were difficult to interact with on touch devices and provided limited visual feedback regarding the configured schedule. The enhanced interface presents time inputs in a more prominent format with increased field sizes, improved contrast, and embedded clock icons that provide clear visual identification of time-related controls.

Each time input field now includes a clock icon positioned inside the input boundary, providing immediate visual identification of the field's purpose without requiring the user to read the associated label. The time picker dropdown interface was enhanced with larger touch targets, making it easier to select precise times on mobile devices. The schedule grid now displays configured hours in a visual timeline format, enabling space owners to quickly verify their schedule configuration at a glance without reading individual time values.

#### 3.9.2.6 Helpful Tips Section for Schedule Management

A contextual tips section was added to the schedule management interface, providing space owners with guidance on effective schedule configuration strategies. The tips section appears as a collapsible panel adjacent to the schedule configuration form, offering best practice recommendations based on the specific type of co-working space being configured. Tips cover

topics such as optimal operating hours for different districts, peak demand time recommendations based on historical booking data, and suggestions for managing twenty-four hour operations.

The tips system implements contextual variation, displaying different guidance based on the space type (study hub, professional co-working, meeting room facility) and the district where the space is located. For study hubs in the Jaro district, tips may recommend extended evening hours during examination periods. For professional co-working spaces in Mandurriao, tips may suggest early morning opening hours to accommodate the business park workforce. This contextual approach ensures that guidance is relevant and actionable for each specific use case.

#### 3.9.2.7 Interactive Status Buttons with Scale Animation

The status toggle buttons throughout the administrative interface were enhanced with interactive scale animations that provide visual feedback when clicked. When a user clicks a status button to change the state of a booking, space, or inventory item, the button briefly scales down and then returns to normal size, creating a satisfying tactile response that confirms the interaction has been registered. This micro-interaction follows principles of motion design that enhance perceived system responsiveness and user engagement.

The scale animation is implemented using CSS transforms with hardware acceleration for smooth performance across devices. The animation duration is calibrated to 150 milliseconds, providing perceptible feedback without introducing interaction delay. Status buttons also implement hover state changes that indicate clickability, transitioning from a static appearance to an elevated state with subtle shadow effects. These combined interactions make status controls more discoverable and provide clear affordances for their interactive nature.

#### 3.9.2.8 Unified Location and District Fields with Map Auto-Detection

The location input system was redesigned to unify the address and district fields, implementing automatic district detection based on map coordinate selection. Previously, space owners manually entered their address and selected their district from a dropdown, creating the potential for mismatches between the geographic coordinates and the administrative district assignment. The enhanced system automatically determines the district based on the geographic coordinates selected through the map interface, ensuring consistency between spatial data and administrative categorization.

When a space owner places the map marker at their location, the system performs a reverse geocoding operation and cross-references the coordinates against the district boundary data stored

in the Districts collection. The district field is automatically populated with the correct district name and is displayed as a read-only value to prevent manual modification that could introduce inconsistencies. If a location falls near a district boundary, the system displays a visual indicator on the map showing the boundary proximity and confirms the district assignment before finalizing the location configuration.

### 3.9.3 Validation and Security Enhancements

The post-deployment enhancement cycle included comprehensive improvements to input validation and security measures. These enhancements address both user experience concerns related to data entry errors and security requirements for protecting system integrity and user data. The following subsections document the specific validation and security improvements implemented during the enhancement phase.

#### 3.9.3.1 Negative Number Prevention for Financial and Capacity Fields

The input validation system was enhanced to prevent the entry of negative values in fields where such values are logically invalid. This improvement addresses a previously identified issue where users could inadvertently or maliciously enter negative numbers in fields such as hourly rate, daily rate, capacity, and inventory quantities. The validation now enforces a minimum value of zero for all financial fields and a minimum value of one for capacity-related fields, ensuring that submitted data conforms to business logic constraints.

The negative number prevention is implemented at multiple layers of the application architecture. Client-side validation provides immediate feedback by blocking the entry of negative symbols in numeric input fields and displaying a tooltip explaining the restriction. Server-side validation through the express-validator middleware rejects any requests containing negative values for protected fields, returning a 400 Bad Request response with a descriptive error message. Database-level validation through Mongoose schema constraints provides a final safeguard against invalid data persistence. This multi-layer approach ensures that negative values cannot reach the database even if client-side validation is bypassed.

#### 3.9.3.2 Positive Number Validation for All Numeric Input Fields

Building on the negative number prevention, a comprehensive positive number validation system was implemented across all numeric input fields throughout the application. This system ensures

that values such as pricing, capacity, room counts, and inventory quantities are always positive numbers within acceptable business ranges. Each numeric field now has defined minimum and maximum values based on domain-specific constraints. Hourly rates must be between ten and ten thousand Philippine pesos. Capacity values must be between one and one thousand seats. Inventory quantities must be non-negative integers not exceeding ten thousand units.

The positive number validation system implements progressive validation that provides increasingly specific feedback as the user refines their input. If a user enters a value below the minimum, the system displays a message indicating the minimum acceptable value. If a user enters a value above the maximum, the system displays a message indicating the maximum acceptable value and suggesting alternative approaches if the user's legitimate needs exceed the standard limits. For values that are not valid numbers at all, the system displays a format validation message requesting a numeric entry.

## 3.9.3.3 Maximum Length Validation for Text Input Fields

Maximum length constraints were defined and enforced for all text input fields to prevent database storage issues and ensure optimal display rendering. The space name field is limited to fifty characters, balancing the need for descriptive names with the display constraints of list views and mobile interfaces. The amenities description field is limited to twenty characters per item, encouraging concise amenity labels that display cleanly in filter interfaces and space detail cards.

The maximum length validation system implements real-time character counting with visual feedback as users approach the character limit. When the user reaches eighty percent of the maximum length, the character counter transitions to a warning color. When the maximum length is reached, further text input is blocked and a message is displayed explaining the limit. The system also implements a graceful truncation strategy for legacy data that may exceed the new length limits, displaying truncated values with an ellipsis indicator and a tooltip showing the full text on hover.

## 3.9.3.4 Duplicate Amenity Prevention

The amenities management system was enhanced with validation that prevents the entry of duplicate amenities within a single space listing. Previously, users could inadvertently add the same amenity multiple times, resulting in redundant display and potential confusion for users viewing the space details. The duplicate prevention system normalizes amenity names by converting to

lowercase and removing extra whitespace before comparison, ensuring that variations in capitalization or spacing do not circumvent the duplicate detection.

When a user attempts to add an amenity that already exists in the space's amenity list, the system displays a notification indicating the duplication and prevents the addition. The notification includes the existing amenity name as stored in the system, helping users understand why their entry was rejected. For batch amenity imports or edits, the system deduplicates the submitted list before processing, silently removing duplicates while preserving the first occurrence of each unique amenity.

### 3.9.3.5 Staff Permission Filtering by Branch Assignment

The authorization system was enhanced to implement branch-scoped permission filtering for staff members. Staff users can now only view and manage data associated with branches to which they have been explicitly assigned. This security enhancement prevents staff members from accessing information about other branches within the same organization, protecting sensitive operational data and ensuring that staff actions are limited to their authorized scope.

The permission filtering is implemented at the database query level, where all queries executed by staff users are automatically augmented with branch filter conditions based on their assigned branches. This approach ensures that unauthorized data is never retrieved from the database, providing security that is independent of front-end implementation. The system also filters navigation elements and dashboard widgets to show only branches and data relevant to the staff member's assignments, creating a personalized interface that focuses on their operational responsibilities.

### 3.9.3.6 Anti-DDoS Enhancement with Localhost Whitelist

The distributed denial of service protection system was improved with the addition of a localhost whitelist that exempts local development and testing environments from rate limiting restrictions. During development and quality assurance testing, the rate limiting middleware could interfere with rapid request sequences needed for testing, such as automated test suites that execute multiple authentication or booking operations in quick succession. The localhost whitelist identifies requests originating from the loopback address (127.0.0.1) and the local hostname, bypassing rate limit checks for these trusted sources.

In production environments, the whitelist provides protection against misconfigured rate limiting that could inadvertently block legitimate local monitoring or health check requests from the server infrastructure. The whitelist implementation includes safeguards to prevent exploitation through header spoofing, verifying the actual connection origin rather than relying solely on declared headers that could be forged. All whitelist bypass events are logged for security auditing, enabling detection of any attempts to exploit the whitelist mechanism.

### 3.9.4 Bug Fixes and Issue Resolution

The post-deployment enhancement cycle included systematic identification and resolution of software defects discovered through user reports, automated testing, and operational monitoring. Each bug was documented, prioritized, and resolved following a structured bug tracking process. The following subsections describe the significant bug fixes implemented during this cycle.

## 3.9.4.1 Booking on Closed Days Resolution

A critical bug was identified and resolved that allowed users to create bookings for days when the co-working space was explicitly closed. The bug occurred because the availability checking logic validated capacity constraints without verifying whether the space was open on the requested date. This oversight enabled bookings to be created for holidays, special closure days, and days outside the regular weekly schedule, resulting in user confusion and operational disruption when arriving at closed facilities.

The resolution involved modifying the booking availability check to include schedule verification as a prerequisite step before capacity validation. The enhanced logic first checks whether the requested date falls within the space's operating schedule, including evaluation of the weekly schedule, special closure dates, and temporary suspensions. Only if the space is confirmed to be open on the requested date does the system proceed to check capacity availability. This sequential validation ensures that closed days are properly respected regardless of capacity status.

## 3.9.4.2 Real-Time Booking Re-render Fix

An issue was resolved where the booking management interface failed to automatically re-render when new bookings were created or existing bookings changed status. Space owners and administrators needed to manually refresh the page to see updated booking information, creating a suboptimal experience and potentially causing missed bookings or delayed responses. The root

cause was identified as a missing dependency in the React useEffect hook that controlled real-time data updates through the Socket.io connection.

The fix involved adding the missing dependency to the effect hook and implementing a more robust event listener pattern for Socket.io booking events. The enhanced implementation uses a dedicated booking update channel that broadcasts changes to all connected clients, with the front-end state management system incorporating incoming updates through a reducer pattern that maintains data consistency. The fix also includes reconnection logic that ensures real-time updates resume after temporary network interruptions, and a fallback polling mechanism that activates if the WebSocket connection remains unavailable for an extended period.

### 3.9.4.3 Notification Permission Bug Resolution

A bug affecting browser notification permissions was identified and resolved. The issue manifested when users who had previously denied notification permissions attempted to enable notifications through the application settings. The system would attempt to request permission again without checking the current permission state, resulting in a silent failure because browsers automatically reject repeated permission requests without presenting the permission dialog to users who have already denied access.

The fix implements a permission state check before attempting to request notification access. If the permission state is "denied," the system displays guidance explaining how users can manually enable notifications through their browser settings, with browser-specific instructions for Chrome, Firefox, Safari, and Edge. If the permission state is "default" or "prompt," the system proceeds with the permission request. The fix also adds a permission status indicator in the user settings page, showing the current notification permission state and providing contextually appropriate actions based on that state.

### 3.9.4.4 Amenity Text Overlap Resolution

A display issue was fixed where long amenity names would overlap adjacent elements in the space detail view and filter panels. The bug occurred because the amenity display components used fixed-width containers without text overflow handling, causing amenity names exceeding the container width to visually encroach on neighboring elements. This issue was particularly prevalent on mobile devices where the reduced viewport width compressed the available space for amenity display.

The resolution implemented a multi-faceted approach to text overflow handling. Amenity containers now use flexible sizing with minimum and maximum width constraints. Text overflow is handled through intelligent truncation with ellipsis, preserving the first and last portions of long amenity names while truncating the middle. Hover interactions reveal the full amenity name through a tooltip. The amenity grid layout was converted to a responsive flexbox arrangement that automatically adjusts column count based on available width, preventing the fixed-column layout that contributed to the overlap issue.

## 3.9.4.5 Staff Space Assignment and Branch Data Corrections

Two related data integrity issues were identified and resolved in the staff management system. The first issue involved incorrect population of the space identifier field when creating staff assignments, resulting in staff records being associated with the wrong space or lacking a space association entirely. The second issue involved branch assignment data not being properly persisted when saving staff configuration changes, causing branch assignments to revert to default values after page refreshes.

The root cause for both issues was traced to inconsistent data handling between the front-end form state and the API request payload. The front-end was sending nested object structures that the API endpoint was not properly destructuring, resulting in missing or incorrectly mapped fields. The fix involved standardizing the data format for staff management API requests and implementing server-side validation that verifies all required fields are present and correctly typed before processing. A data migration script was also executed to correct existing records affected by the bug, ensuring that all current staff assignments reflect the intended space and branch associations.

## 3.9.4.6 Controller Route Conflict Resolution

A route conflict was identified between the space controller and the staff controller, where both controllers defined overlapping route patterns that caused request routing ambiguity. Specifically, the space detail endpoint and the staff assignment endpoint shared a similar URL structure that the Express.js router could not consistently disambiguate, resulting in intermittent 404 errors or incorrect handler execution depending on the route registration order.

The resolution involved redesigning the API route structure to eliminate ambiguity through explicit path differentiation. Space-related routes were prefixed with a consistent "/spaces" path segment, while staff-related routes were prefixed with "/staff". Route parameters were renamed to be more descriptive, changing generic parameter names like ":id" to specific names like ":spaceId" and

":staffId". The route registration order was also standardized, with a centralized route configuration file that documents all available endpoints and their corresponding controller functions, preventing future conflicts through clear visibility of the complete route structure.

## 3.9.5 Inventory Management and Point-of-Sale Enhancements

The inventory management and point-of-sale capabilities of the FlexSpace system were substantially enhanced during the post-deployment iteration. These improvements address the commercial requirements of co-working spaces that operate on-site retail operations, enabling more sophisticated inventory tracking, profit analysis, and sales processing. The following subsections document the specific enhancements to the inventory and point-of-sale modules.

### 3.9.5.1 Purchase Price Integration in Product Records

The product data model was extended to include a purchase price field that captures the cost at which inventory items are acquired from suppliers. This field complements the existing selling price field to enable complete profit analysis for each product in the inventory. The purchase price is recorded at the time of stock intake and is stored historically to support cost averaging calculations when the same product is acquired at different prices across multiple purchase orders.

The purchase price field is restricted to staff members with inventory management permissions and space owners, preventing unauthorized viewing of cost data by general staff or users. The interface for entering purchase prices is presented during the stock intake workflow, where receiving staff can record the unit cost alongside the received quantity. For products without recorded purchase prices, the system displays a warning indicator in the profit analysis dashboard, prompting space owners to complete the cost information for accurate reporting.

### 3.9.5.2 Per-Unit Profit and Margin Calculation

The system now automatically calculates per-unit profit and profit margin for all inventory items that have both purchase price and selling price data recorded. The per-unit profit is computed as the selling price minus the purchase price, representing the gross profit generated by each unit sold. The profit margin percentage is calculated by dividing the per-unit profit by the purchase price and multiplying by one hundred, providing a standardized metric for comparing profitability across products with different price points.

These calculations are performed in real-time as price data is updated, ensuring that profit metrics are always current. The calculations are displayed in the inventory management dashboard within a dedicated profit analysis panel that shows the per-unit profit, margin percentage, total units sold, and cumulative profit for each product. The panel includes sorting and filtering capabilities that enable space owners to identify their most and least profitable products, supporting data-driven decisions about product mix and pricing strategy.

### 3.9.5.3 Potential Revenue versus Inventory Cost Analysis

A financial projection feature was added that compares potential revenue from current inventory stock against the total cost of acquiring that inventory. The potential revenue is calculated by multiplying the current stock quantity by the selling price for each product and summing across the entire inventory. The inventory cost is calculated by multiplying the current stock quantity by the purchase price for each product. The difference between potential revenue and inventory cost represents the total potential gross profit embedded in the current inventory.

This analysis provides space owners with valuable insights for inventory investment decisions. A high potential profit relative to inventory cost indicates efficient inventory selection, while a low ratio may suggest the need for pricing adjustments or product mix changes. The analysis is presented through a comparative visualization that displays revenue potential, inventory cost, and potential profit as grouped bars, with percentage indicators showing the margin relationship. The visualization updates automatically as inventory quantities change through sales and replenishment activities.

### 3.9.5.4 Branch-Specific Inventory Filtering

The inventory management interface was enhanced with branch-specific filtering that enables space owners to view and manage inventory separately for each branch location. When a space owner operates multiple branches, the inventory filter defaults to the currently selected branch context, showing only products and stock levels relevant to that location. The branch selector in the inventory interface synchronizes with the global branch selection, maintaining consistent context as users navigate between different sections of the dashboard.

The branch filtering system implements stock isolation, where inventory quantities are tracked independently per branch and stock transfers between branches require explicit transfer transactions. This approach prevents inventory quantity confusion that could occur if all branches shared a common stock pool. Staff members see only the inventory for branches they are assigned

to, ensuring that stock management responsibilities align with operational scope. The system also supports cross-branch inventory comparison, enabling space owners to view stock levels across all branches in a unified table for centralized replenishment planning.

### 3.9.5.5 Automated Stock Deduction on Order Creation

The order processing workflow was enhanced to automatically deduct sold quantities from inventory stock when orders are created. Previously, stock deductions required manual confirmation by staff members, creating a delay between sale and inventory update that could lead to overselling when multiple orders were processed simultaneously. The automated deduction system uses database transactions to ensure that stock quantities are updated atomically with order creation, preventing race conditions that could result in selling products that are no longer in stock.

When stock levels fall below a configurable reorder threshold after deduction, the system generates a low-stock notification alerting staff members that replenishment may be needed. If an order is cancelled, the deducted stock is automatically restored to inventory through a reversal transaction that maintains data consistency. For partial cancellations where only some items on an order are cancelled, the system calculates the appropriate quantity to restore and updates the inventory accordingly. This automated approach reduces manual inventory management workload and improves stock accuracy.

## 3.9.6 Architectural Improvements

The system architecture underwent significant refactoring during the post-deployment enhancement cycle to improve code organization, component reusability, and maintainability. These architectural improvements establish a foundation for future feature development and reduce technical debt accumulated during the initial development phase. The following subsections document the major architectural changes implemented.

### 3.9.6.1 Modal Component Extraction and Reusability

All modal dialog components throughout the application were extracted from their original inline implementations into a centralized modal component library located in the components slash modal directory. The previous implementation embedded modal logic within parent components, resulting in duplicated code for common modal behaviors such as open and close animations, backdrop clicking, escape key handling, and focus trapping. The extracted modal system provides

a reusable Modal component that encapsulates these common behaviors and accepts content through React children or specialized modal variant components.

The modal library includes specialized variants for common use cases: ConfirmationModal for delete confirmations and action verification, FormModal for forms that require modal presentation, AlertModal for important notifications that require user acknowledgment, and ReviewModal for displaying and collecting user reviews. Each variant extends the base Modal component with domain-specific behaviors while maintaining consistent visual styling and interaction patterns. The extraction reduced the total modal-related code by approximately sixty percent and ensures that modal behavior modifications propagate consistently across all modal instances in the application.

## 3.9.6.2 Review Modal and Booking Details Modal Components

Two new specialized modal components were developed as part of the modal extraction initiative. The showReviewModal component provides a comprehensive interface for viewing user reviews, displaying the reviewer information, star rating, written feedback, submission date, and any associated photographs. The component implements a scrollable layout that accommodates reviews of varying length and includes pagination controls when multiple reviews are displayed in a gallery format.

The booking details modal component presents comprehensive information about a specific booking in a structured format suitable for both customer service and administrative purposes. The modal displays booking identification information, user details, space and resource specifications, timing information, payment status, check-in and check-out timestamps, and a timeline visualization of the booking lifecycle. Staff members can access action buttons within the modal to perform common operations such as modifying the booking, processing refunds, or communicating with the user. Both modal components follow the centralized modal system patterns and are accessible throughout the application through programmatic invocation functions.

## 3.9.6.3 Enhanced Polling for Real-Time Order Updates

The real-time order update mechanism was improved through optimized polling strategies that reduce server load while maintaining timely update delivery. The previous implementation used a fixed-interval polling approach that queried the server at a constant rate regardless of order activity levels. The enhanced system implements adaptive polling that adjusts query frequency based on order state and recent activity. When an order is in a pending or processing state, the polling

frequency increases to provide rapid status updates. When an order is completed or cancelled, polling frequency decreases to conserve resources.

The polling optimization also includes a progressive backoff mechanism that reduces polling frequency when consecutive polls return no changes, and a rapid resumption feature that immediately restores frequent polling when activity is detected. Server-side optimizations include an endpoint specifically designed for polling that returns only changed data since the last query, reducing payload sizes and database query complexity. The enhanced polling system reduced average server request volume for order updates by approximately forty percent while improving the median update detection time.

## 3.9.6.4 Payment Webhook Handling Infrastructure

A webhook handling infrastructure was developed to process asynchronous payment notifications from external payment service providers. Webhooks enable the system to receive real-time notifications when payment events occur, such as successful payment completion, payment failure, refund processing, or dispute initiation. The webhook infrastructure provides a secure endpoint that validates incoming webhook signatures to verify authenticity, preventing malicious actors from sending fraudulent payment notifications.

The webhook handler processes validated notifications through a state machine that updates booking and payment records according to the notification type. Each webhook event is logged with its payload, processing result, and timestamp for audit purposes. The system implements idempotent processing to handle duplicate webhook deliveries gracefully, ensuring that repeated notifications for the same event do not result in inconsistent state changes. Failed webhook processing triggers retry logic with exponential backoff and alerts administrative users if manual intervention is required.

## 3.9.6.5 PayMongo Integration for Online Payment Processing

The simulated payment system was replaced with a production-ready integration with PayMongo, a Philippine-based payment processing platform that supports GCash, Maya, credit cards, and other local payment methods. The PayMongo integration enables real financial transactions within the FlexSpace platform, processing actual payments from users to space owners. The integration implements PayMongo's payment intent API, which creates a secure transaction context that captures payment details and processes charges through the appropriate payment method.

The payment flow begins when a user confirms a booking, at which point the system creates a PayMongo payment intent with the booking amount and currency. The user is redirected to PayMongo's secure checkout page where they complete payment using their preferred method. Upon payment completion, PayMongo sends a webhook notification to the system, which updates the booking status to confirmed and notifies the user of successful payment. For failed payments, the system provides clear error messaging and allows users to retry with a different payment method. The integration includes comprehensive logging and reconciliation features that enable space owners to track payment status and resolve discrepancies.

## 3.9.7 Staff Management System Enhancements

The staff management capabilities of the FlexSpace system were comprehensively enhanced to support the operational requirements of multi-branch co-working space operators. These enhancements establish a robust framework for staff authorization, branch assignment, and operational oversight that scales from single-location operators to enterprise-level space networks. The following subsections document the specific staff management improvements implemented during the post-deployment enhancement cycle.

### 3.9.7.1 Branch-Restricted Staff Access Control

The authorization system was enhanced to enforce branch-restricted access for all staff users. Staff members can now only access data, perform operations, and view analytics for branches to which they have been explicitly assigned by the space owner. This restriction applies to all system functions including booking management, inventory control, user review responses, and financial reporting. The branch restriction is enforced at the API middleware level, ensuring that staff users cannot circumvent access controls through direct API requests.

The branch access control system implements a hierarchical permission model where space owners have unrestricted access to all branches within their organization, while staff members operate within the boundaries of their branch assignments. When a staff member logs into the system, their branch assignments are loaded into the session context and applied to all subsequent data queries. The dashboard interface adapts to show only the branches and data relevant to the staff member's assignments, creating a focused work environment that reduces information overload and prevents accidental operations on unauthorized branches.

### 3.9.7.2 Staff Branch Transfer Functionality

A branch transfer system was developed to enable space owners to move staff members between branches as operational needs change. The transfer workflow is initiated from the staff management interface, where the space owner selects the staff member to transfer and the destination branch. The system validates that the destination branch has capacity for additional staff assignments and that the transfer does not violate any role constraints, such as requiring at least one manager to remain at the source branch.

The transfer process maintains a complete history of all branch assignments, recording the start and end dates for each branch association. This historical record enables space owners to track staff mobility patterns and analyze tenure at different locations. When a transfer is executed, the system updates all active permissions and refreshes the staff member's session to reflect their new branch context. Pending tasks or active bookings associated with the staff member at the source branch are flagged for reassignment, ensuring that no operational items are orphaned during the transfer process.

### 3.9.7.3 Branch Selector for Space Owner Dashboard

A global branch selector was added to the space owner dashboard, providing a consistent mechanism for switching between branches across all dashboard views. The branch selector is persistently visible in the dashboard header and displays the currently active branch with a dropdown menu for selecting alternative branches. When a space owner switches branches, all dashboard data updates to reflect the selected branch context, including booking counts, revenue figures, occupancy rates, and inventory levels.

The branch selector implements a quick-switch feature that allows space owners to navigate between recently accessed branches without opening the dropdown menu. For space owners with many branches, the selector includes a search function that filters the branch list based on typed input. The selected branch is persisted in the user's session and local storage, ensuring that the branch context is maintained across page refreshes and browser sessions. Aggregate views that display data across all branches are accessible through an "All Branches" option in the selector.

### 3.9.7.4 Staff Listing with Branch Name Display

The staff listing interface was enhanced to display the branch names associated with each staff member, providing immediate visual context about staff distribution across locations. Each staff

entry in the listing shows the staff member's name, role, primary branch, additional branch assignments, and account status. The branch information is presented as labeled badges that use consistent color coding with the branch selector, enabling rapid visual association between staff members and their operational locations.

The staff listing supports filtering by branch, enabling space owners to view only staff members assigned to a specific location. The listing also supports sorting by branch name, grouping staff members by their primary branch for organizational clarity. When a staff member is assigned to multiple branches, all associated branch names are displayed with the primary branch highlighted. Clicking on a branch name in the staff listing navigates to the branch-specific view, streamlining the workflow for branch-level staff management.

### 3.9.7.5 Interactive Status Toggle with Tooltip Guidance

The staff status controls were enhanced with interactive toggle functionality that enables space owners to change staff account status through a single click. Each staff member entry includes a status toggle that switches between active and inactive states. The toggle implements a confirmation dialog for deactivation actions to prevent accidental status changes that could lock staff members out of the system. When activated, the toggle displays a tooltip explaining the implications of each status state.

The active status enables full system access according to the staff member's role and branch assignments. The inactive status prevents login and suspends all system access while preserving the staff member's historical data and configuration for potential reactivation. The tooltip for active status explains that the staff member can access all authorized functions, while the inactive status tooltip explains that the account is suspended and provides guidance on when deactivation is appropriate, such as for temporary leave or employment termination.

## 3.9.8 Performance and User Experience Optimization

Beyond feature development and bug resolution, the post-deployment enhancement cycle included focused efforts on system performance optimization and user experience refinement. These improvements target the responsiveness, efficiency, and satisfaction dimensions of the user experience, ensuring that the FlexSpace system performs reliably under real-world usage conditions. The following subsections document the performance and user experience optimizations implemented.

### 3.9.8.1 Polling Frequency Optimization

The real-time data polling mechanisms throughout the application were optimized to reduce unnecessary server requests while maintaining timely data updates. The previous implementation used uniform polling intervals that queried the server at fixed rates regardless of data change frequency or user activity. The optimized system implements contextual polling that adjusts query frequency based on multiple factors: the page's visibility state (reducing polls when the browser tab is not active), the recency of data changes (increasing polls after detected changes), and the user's current activity (reducing polls during periods of inactivity).

The polling optimization includes a tab visibility awareness feature that uses the Page Visibility API to detect when the browser tab is hidden from view. When a tab is hidden, polling frequency is reduced to a minimum level since the user is not actively viewing the interface. When the tab becomes visible again, polling frequency is temporarily increased to rapidly refresh any data that changed during the hidden period. These optimizations reduced overall polling request volume by approximately fifty-five percent during typical usage patterns, significantly decreasing server load and improving battery life for mobile users.

### 3.9.8.2 Mobile Responsiveness Improvements

The mobile user experience was enhanced through targeted improvements to responsive layouts, touch interactions, and mobile-specific navigation patterns. The dashboard interface was redesigned with a collapsible sidebar that transforms into a bottom navigation bar on mobile devices, maximizing the available screen area for content display. Data tables were converted to card-based layouts on narrow viewports, presenting tabular information in a vertically scrollable format that is more readable on mobile screens.

Touch target sizes were increased throughout the interface to meet the minimum forty-eight pixel recommendation for mobile accessibility. Form inputs were optimized for mobile interaction by implementing appropriate input types (numeric keyboards for number fields, date pickers for calendar inputs) and autocomplete attributes for common fields. The booking workflow was streamlined for mobile users with a simplified step progression and larger touch-friendly controls for time selection and date picking. These improvements increased the mobile task completion rate by approximately thirty percent compared to the previous interface.

### 3.9.8.3 Error Handling and User Feedback Enhancement

The error handling system was improved to provide more informative and actionable feedback when operations fail. Error messages were rewritten to use plain language that explains what went wrong and what the user can do to resolve the issue. Technical error codes and stack traces were removed from user-facing messages and replaced with user-friendly explanations. Error notifications now include suggested next steps and, where applicable, direct links to take corrective action.

A retry mechanism was implemented for transient errors such as network timeouts, automatically attempting failed operations up to three times with progressive delay intervals before presenting an error to the user. For network connectivity loss, the system displays a non-intrusive banner indicating the connection status and queues user actions for automatic submission when connectivity is restored. Loading states were standardized across the application with skeleton screens that preview the content layout while data is being fetched, reducing perceived wait times and preventing layout shift when content loads.

### 3.9.8.4 Modal Component Optimization

The centralized modal components were further optimized for performance and accessibility. Modal content is now lazily loaded, with the modal component rendered in the DOM only when activated rather than being persistently present. This change reduced the initial DOM size by approximately twenty percent on pages with multiple modal options. Modal animations were optimized using hardware-accelerated CSS transforms instead of JavaScript-driven animations, resulting in smoother transitions on devices with limited processing capacity.

Accessibility improvements included full keyboard navigation support within modals, with tab cycling confined to the modal content, escape key dismissal, and focus restoration to the triggering element upon close. Screen reader announcements were added for modal open and close events, using ARIA live regions to communicate modal state changes. Focus trapping within active modals prevents keyboard users from inadvertently navigating to background content while a modal is open, ensuring that interactive elements outside the modal cannot be accessed until the modal is dismissed.

### 3.9.8.5 Form Validation Strengthening

The form validation system was comprehensively strengthened with additional validation rules, improved error messaging, and real-time validation feedback. All forms now implement client-side validation that provides immediate feedback as users complete fields, preventing submission of invalid data and reducing server load from rejected requests. Validation rules were expanded to cover business logic constraints such as date range validity, price consistency, and cross-field dependencies.

Real-time validation triggers when a user completes a field and moves to the next input, providing immediate feedback without waiting for form submission. Complex validations that require server verification, such as email uniqueness checks, implement debounced validation that queries the server after the user pauses typing, preventing excessive validation requests. Validation summary panels were added to long forms, displaying all validation errors in a consolidated location that can be accessed without scrolling through the entire form to find individual error messages.

### 3.9.8.6 Loading State Enhancement

Loading states throughout the application were enhanced to provide clearer progress indication and reduce perceived wait times. Skeleton screens replaced traditional spinner loading indicators on data-driven pages, displaying a placeholder layout that mirrors the expected content structure. As data loads, the skeleton placeholders are progressively replaced with actual content, creating a smooth visual transition that reduces the cognitive disruption of content appearing abruptly.

For operations with predictable duration, such as file uploads and report generation, progress indicators display percentage completion and estimated time remaining. For operations with unpredictable duration, indeterminate progress indicators are accompanied by contextual messages that explain what the system is doing and why it may take time. Button loading states prevent duplicate submission by disabling the action button and displaying a loading indicator within the button itself, providing clear feedback that the action is being processed without requiring the user to look elsewhere on the page for confirmation.

## 3.9.9 Additional Use Cases for Enhanced Features

The post-deployment enhancements introduced new system capabilities that require additional use case documentation to fully describe the interactions between users and the enhanced system. The following use cases extend the original twelve use cases documented in Section 3.6, providing

comprehensive narratives for the branch management, inventory profit tracking, schedule configuration, staff management, and payment processing features.

**Use Case UC-013: Space Owner Assigns Staff to Branch**

Actor: Space Owner

Pre-conditions: Space owner is authenticated and has at least one registered branch. The target staff member has an active account in the system.

Primary Success Scenario:

1. Space owner navigates to the Staff Management section of the dashboard and clicks on the Branch Assignment tab.

2. The system displays a list of all staff members with their current branch assignments indicated by branch name badges.

3. Space owner clicks the "Assign Branch" button on the target staff member's entry.

4. The system displays a branch selection modal showing all available branches with checkboxes for selection and a radio button for primary branch designation.

5. Space owner selects one or more branches and designates the primary branch for the staff member.

6. Space owner clicks "Save Assignment." The system validates the selections, updates the staff member's branch associations in the database, and refreshes the staff listing to reflect the new assignments.

7. The system sends a notification email to the staff member informing them of their new branch assignments and login instructions.

Post-conditions: Staff member is associated with the selected branches and can access data for those branches upon next login.

Alternative/Exception Handler A1: Staff already assigned to selected branch. At step 6, if the staff member is already assigned to one of the selected branches, the system displays a notification and removes the duplicate from the update operation.

**Use Case UC-014: Space Owner Tracks Inventory Profit**

Actor: Space Owner

Pre-conditions: Space owner is authenticated and has inventory items with both purchase price and selling price recorded.

Primary Success Scenario:

1. Space owner navigates to the Inventory Management section and clicks on the Profit Analysis tab.

2. The system displays a profit analysis dashboard showing all inventory items with per-unit profit, profit margin percentage, total units sold, and cumulative profit.

3. Space owner clicks on a specific product to view detailed profit information.

4. The system displays a detailed view showing the purchase price history, selling price, current stock level, potential revenue from current stock, total inventory cost, and potential profit.

5. Space owner can export the profit analysis data as a CSV file for external analysis by clicking the Export button.

Post-conditions: Space owner has viewed profit information and optionally exported the data.

Alternative/Exception Handler A1: Missing purchase price. If a product does not have a recorded purchase price, the system displays a warning indicator and excludes that product from margin calculations, prompting the space owner to complete the purchase price information.

**Use Case UC-015: Space Owner Configures Twenty-Four Seven Schedule**

Actor: Space Owner

Pre-conditions: Space owner is authenticated and has an active space listing.

Primary Success Scenario:

1. Space owner navigates to the Space Management section and selects the Schedule Configuration option for the target space.

2. The system displays the weekly schedule interface with seven day rows, each showing current operating hours.

3. Space owner clicks the "Set All Days to 24/7" button at the top of the schedule grid.

4. The system updates all seven days to display "Open 24 Hours" with a green indicator showing the twenty-four seven status.

5. Space owner clicks on Saturday's row to modify it to limited hours.

6. The system displays a toggle switch for twenty-four seven operation. Space owner turns off the toggle.

7. The system reveals time input fields for Saturday's opening and closing times. Space owner enters 08:00 and 18:00.

8. Space owner clicks "Save Schedule." The system validates the configuration and updates the space's operating hours.

Post-conditions: Space schedule reflects twenty-four seven operation for six days and limited hours on Saturday.

**Use Case UC-016: Staff Member Processes Point-of-Sale Order**

Actor: Staff Member

Pre-conditions: Staff member is authenticated and assigned to a branch with inventory items.

Primary Success Scenario:

1. Staff member navigates to the Point of Sale section of the dashboard.

2. The system displays the product catalog for the staff member's assigned branch, showing available items, prices, and current stock levels.

3. Staff member clicks on products to add them to the order cart, specifying quantities for each item.

4. The system validates that requested quantities do not exceed available stock and updates the cart total.

5. Staff member clicks "Process Payment" and selects the payment method (GCash, Maya, or cash).

6. For digital payments, the system generates a PayMongo payment link. For cash payments, the system records the cash received and calculates change due.

7. Upon payment confirmation, the system creates the order record, deducts sold quantities from inventory stock, and prints a receipt.

Post-conditions: Order is recorded, inventory stock is updated, and payment is processed.

Alternative/Exception Handler A1: Insufficient stock. At step 4, if the requested quantity exceeds available stock, the system displays a warning and adjusts the cart quantity to the available amount.

**Use Case UC-017: Administrator Reviews System Logs**

Actor: System Administrator

Pre-conditions: Administrator is authenticated with system-level privileges.

Primary Success Scenario:

1. Administrator navigates to the System Monitoring section and selects Activity Logs.

2. The system displays a filterable, sortable table of system activity including user actions, API requests, error events, and security alerts.

3. Administrator applies filters to view staff permission changes within the last thirty days.

4. The system displays filtered results showing all branch assignments, role changes, and status modifications with timestamps and initiating user information.

5. Administrator clicks on a specific log entry to view detailed information including the request payload, response status, and IP address.

Post-conditions: Administrator has reviewed activity logs and can take corrective action if anomalies are detected.

# 3.9.10 Completed Enhancement Tasks Summary

The post-deployment enhancement cycle resulted in the successful completion of numerous development tasks spanning all functional areas of the FlexSpace system. This section provides a consolidated summary of completed enhancements, organized by functional category, demonstrating the scope and depth of the post-deployment improvement effort.

**Branch and Staff Management.** The branch staff assignment and transfer functionality was fully implemented, enabling space owners to manage staff across multiple locations. Staff branch

filtering was deployed, ensuring that staff members can only access data for their assigned branches. The branch selector was integrated into the space owner dashboard, providing seamless branch context switching. Staff listing improvements added branch name display to the staff roster view. Clickable status toggles with tooltips were implemented for rapid account status management.

**Inventory and Profit Tracking.** Inventory profit tracking was completed with the integration of purchase price and selling price fields in the product data model. Per-unit profit and margin percentage calculations are now automatically computed and displayed. Potential revenue versus inventory cost analysis provides space owners with financial projections for current stock. Branch-specific inventory filtering enables independent stock management across multiple locations. Stock deduction automation ensures that inventory quantities are accurately updated upon order creation.

**Schedule Management.** The twenty-four seven schedule configuration with per-day toggle was fully implemented, providing granular control over operating hours. Dynamic status computation based on schedule and capacity eliminates the need for manual status updates. The quick-apply feature enables rapid configuration of uniform schedules across all days. The schedule interface was enhanced with embedded clock icons and contextual tips for improved usability.

**User Interface and Experience.** The space listing creation workflow was successfully transitioned from a modal to a dedicated page interface, providing an improved configuration experience. Status badges were enhanced with hover-triggered tooltips that provide contextual status information. Character counters were added to text input fields, preventing submission errors from length violations. Error messages were improved with red asterisk indicators and descriptive validation feedback. The weekly schedule interface received visual improvements with better time input visibility and helpful tips. Status buttons were enhanced with interactive scale animations that provide tactile feedback on click.

**Validation and Security.** Negative number validation was implemented across all financial and capacity fields, preventing logically invalid data entry. Positive number validation enforces minimum values appropriate to each field type. Maximum length validation ensures that text inputs conform to display constraints. Duplicate amenity prevention eliminates redundant entries in space listings. Staff permission filtering enforces branch-scoped data access. Anti-DDoS protection was enhanced with a localhost whitelist for development environments.

**Technical Architecture.** All modal components were extracted to a centralized component library, improving code reusability and maintainability. The showReviewModal and booking details modal components were developed as reusable elements. Real-time order update polling was optimized with adaptive frequency algorithms. Payment webhook handling infrastructure was established for asynchronous payment notifications. PayMongo integration was completed, enabling real financial transactions through GCash, Maya, and credit card processing.

**System Reliability.** The booking on closed days bug was resolved through enhanced schedule validation in the availability checking workflow. Real-time booking re-render issues were fixed by correcting React effect dependencies and implementing robust Socket.io event handling. Notification permission bugs were resolved with improved browser permission state management. Amenity text overlap display issues were corrected with responsive layout adjustments and intelligent text truncation. Staff data population and branch assignment persistence issues were resolved through standardized API data formats. Controller route conflicts were eliminated through route path redesign and centralized route configuration.

**Performance Optimization.** Unnecessary polling was reduced through adaptive polling strategies that respond to data change frequency and page visibility. Mobile responsiveness was improved with redesigned layouts, optimized touch targets, and mobile-specific interaction patterns. Error handling was enhanced with informative messaging, retry mechanisms, and offline operation support. Modal components were optimized with lazy loading and hardware-accelerated animations. Form validation was strengthened with real-time feedback and comprehensive business logic checks. Loading states were enhanced with skeleton screens and progress indicators that improve perceived performance.

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